

Engineer On a Disk

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Reference Information

2. MATHEMATICAL TOOLS

***** This contains additions and sections by Dr. Andrew Sterian.

- We use math in almost every problem we solve. As a result the more relevant topics of mathematics are summarized here.
- This is not intended for learning, but for reference.

2.1 INTRODUCTION

- This section has been greatly enhanced, and tailored to meet our engineering requirements.
- The section outlined here is not intended to teach the elements of mathematics, but it is designed to be a quick reference guide to support the engineer required to use techniques that may not have been used recently.
- For those planning to write the first ABET Fundamentals of Engineering exam, the following topics are commonly on the exam.
 - quadratic equation
 - straight line equations - slope and perpendicular
 - conics, circles, ellipses, etc.
 - matrices, determinants, adjoint, inverse, cofactors, multiplication
 - limits, L'Hospital's rule, small angle approximation
 - integration of areas
 - complex numbers, polar form, conjugate, addition of polar forms
 - maxima, minima and inflection points
 - first order differential equations - guessing and separation
 - second order differential equation - linear, homogeneous, non-homogeneous, second order
 - triangles, sine, cosine, etc.
 - integration - by parts and separation
 - solving equations using inverse matrices, Cramer's rule, substitution
 - eigenvalues, eigenvectors
 - dot and cross products, areas of parallelograms, angles and triple product
 - divergence and curl - solenoidal and conservative fields
 - centroids
 - integration of volumes
 - integration using Laplace transforms
 - probability - permutations and combinations
 - mean, standard deviation, mode, etc.
 - log properties

- taylor series
- partial fractions
- basic coordinate transformations - cartesian, cylindrical, spherical
- trig identities
- derivative - basics, natural log, small angles approx., chain rule, partial fractions

2.1.1 Constants and Other Stuff

- A good place to start a short list of mathematical relationships is with greek letters

lower case	upper case	name
α	A	alpha
β	B	beta
γ	Γ	gamma
δ	Δ	delta
ϵ	E	epsilon
ζ	Z	zeta
η	H	eta
θ	Θ	theta
ι	I	iota
κ	K	kappa
λ	Λ	lambda
μ	M	mu
ν	N	nu
ξ	Ξ	xi
$\omicron$	O	omicron
π	Π	pi
ρ	P	rho
σ	Σ	sigma
τ	T	tau
υ	Y	upsilon
ϕ	Φ	phi
χ	X	chi
ψ	Ψ	psi
ω	Ω	omega

- The constants listed are amount some of the main ones, other values can be derived through calculation using modern calculators or computers. The values are typically given with more than 15 places of accuracy so that they can be used for double precision calculations.

$$e = 2.7182818 = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \text{natural logarithm base}$$

$$\pi = 3.1415927 = \text{pi}$$

$$\gamma = 0.57721566 = \text{Eulers constant}$$

$$1 \text{radian} = 57.29578^\circ$$

2.1.2 Basic Operations

- These operations are generally universal, and are described in sufficient detail for our use.
- Basic properties include,

$$\text{commutative} \quad a + b = b + a$$

$$\text{distributive} \quad a(b + c) = ab + ac$$

$$\text{associative} \quad a(bc) = (ab)c \quad a + (b + c) = (a + b) + c$$

2.1.2.1 - Factorial

- A compact representation of a series of increasing multiples.

$$n! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot n$$

$$0! = 1$$

2.1.3 Exponents and Logarithms

- The basic properties of exponents are so important they demand some sort of mention

$$\begin{array}{lll}
 (x^n)(x^m) = x^{n+m} & x^0 = 1, \text{ if } x \text{ is not } 0 & x^{\frac{1}{n}} = \sqrt[n]{x} \\
 \frac{(x^n)}{(x^m)} = x^{n-m} & x^{-p} = \frac{1}{x^p} & x^{\frac{m}{n}} = \sqrt[n]{x^m} \\
 (x^n)^m = x^{n \cdot m} & (xy)^n = (x^n)(y^n) & \sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}
 \end{array}$$

- Logarithms also have a few basic properties of use,

The basic base 10 logarithm:

$$\log x = y \quad x = 10^y$$

The basic base n logarithm:

$$\log_n x = y \quad x = n^y$$

The basic natural logarithm (e is a constant with a value found near the start of this section:

$$\ln x = \log_e x = y \quad x = e^y$$

- All logarithms observe a basic set of rules for their application,

$$\log_n(xy) = \log_n(x) + \log_n(y) \quad \log_n(n) = 1$$

$$\log_n\left(\frac{x}{y}\right) = \log_n(x) - \log_n(y) \quad \log_n(1) = 0$$

$$\log_n(x^y) = y \log_n(x)$$

$$\log_n(x) = \frac{\log_m(x)}{\log_m(n)}$$

$$\ln(A \angle \theta) = \ln(A) + (\theta + 2\pi k)j \quad k \in I$$

2.1.4 Polynomial Expansions

- Binomial expansion for polynomials,

$$(a + x)^n = a^n + na^{n-1}x + \frac{n(n-1)}{2!}a^{n-2}x^2 + \dots + x^n$$

2.2 FUNCTIONS

2.2.1 Discrete and Continuous Probability Distributions

- The Binomial distribution is,

$$P(m) = \sum_{t \leq m} \binom{n}{t} p^t q^{n-t} \quad q = 1 - p \quad q, p \in [0, 1]$$

- The Poisson distribution is,

$$P(m) = \sum_{t \leq m} \frac{\lambda^t e^{-\lambda}}{t!} \quad \lambda > 0$$

- The Hypergeometric distribution is,

$$P(m) = \sum_{t \leq m} \frac{\binom{r}{t} \binom{s}{n-t}}{\binom{r+s}{n}}$$

- The Normal distribution is,

$$P(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2} dt$$

2.2.2 Basic Polynomials

- The quadratic equation appears in almost every engineering discipline, therefore is of great importance.

$$ax^2 + bx + c = 0 = a(x - r_1)(x - r_2) \quad r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Cubic equations also appear on a regular basis, and as a result should also be considered.

$$x^3 + ax^2 + bx + c = 0 = (x - r_1)(x - r_2)(x - r_3)$$

First, calculate,

$$Q = \frac{3b - a^2}{9} \quad R = \frac{9ab - 27c - 2a^3}{54} \quad S = \sqrt[3]{R + \sqrt{Q^3 + R^2}} \quad T = \sqrt[3]{R - \sqrt{Q^3 + R^2}}$$

Then the roots,

$$r_1 = S + T - \frac{a}{3} \quad r_2 = \frac{S + T}{2} - \frac{a}{3} + \frac{j\sqrt{3}}{2}(S - T) \quad r_3 = \frac{S + T}{2} - \frac{a}{3} - \frac{j\sqrt{3}}{2}(S - T)$$

- On a few occasions a quartic equation will also have to be solved. This can be done by first reducing the equation to a quadratic,

$$x^4 + ax^3 + bx^2 + cx + d = 0 = (x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

First, solve the equation below to get a real root (call it 'y'),

$$y^3 - by^2 + (ac - 4d)y + (4bd - c^2 - a^2d) = 0$$

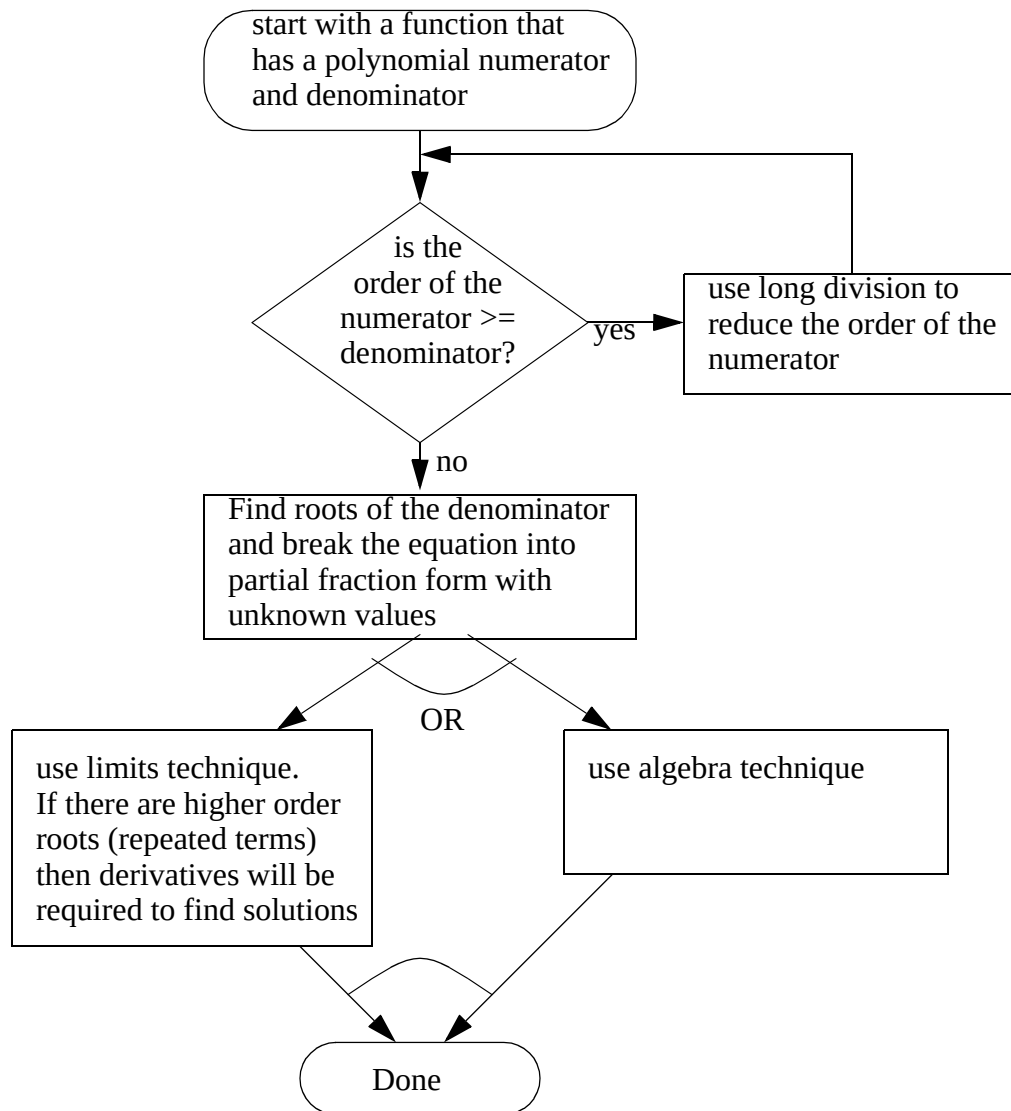
Next, find the roots of the 2 equations below,

$$r_1, r_2 = z^2 + \left(\frac{a + \sqrt{a^2 - 4b + 4y}}{2} \right)z + \left(\frac{y + \sqrt{y^2 - 4d}}{2} \right) = 0$$

$$r_3, r_4 = z^2 + \left(\frac{a - \sqrt{a^2 - 4b + 4y}}{2} \right)z + \left(\frac{y - \sqrt{y^2 - 4d}}{2} \right) = 0$$

2.2.3 Partial Fractions

- The next is a flowchart for partial fraction expansions.



- The partial fraction expansion for,

$$x(s) = \frac{1}{s^2(s+1)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+1}$$

$$C = \lim_{s \rightarrow -1} \left[(s+1) \left(\frac{1}{s^2(s+1)} \right) \right] = 1$$

$$A = \lim_{s \rightarrow 0} \left[s^2 \left(\frac{1}{s^2(s+1)} \right) \right] = \lim_{s \rightarrow 0} \left[\frac{1}{s+1} \right] = 1$$

$$B = \lim_{s \rightarrow 0} \left[\frac{d}{ds} \left[s^2 \left(\frac{1}{s^2(s+1)} \right) \right] \right] = \lim_{s \rightarrow 0} \left[\frac{d}{ds} \left(\frac{1}{s+1} \right) \right] = \lim_{s \rightarrow 0} [-(s+1)^{-2}] = -1$$

- Consider the example below where the order of the numerator is larger than the denominator.

$$x(s) = \frac{5s^3 + 3s^2 + 8s + 6}{s^2 + 4}$$

This cannot be solved using partial fractions because the numerator is 3rd order and the denominator is only 2nd order. Therefore long division can be used to reduce the order of the equation.

$$\begin{array}{r}
 5s + 3 \\
 s^2 + 4 \overline{) 5s^3 + 3s^2 + 8s + 6} \\
 \underline{5s^3 + 20s} \\
 3s^2 - 12s + 6 \\
 \underline{3s^2 + 12} \\
 -12s - 6
 \end{array}$$

This can now be used to write a new function that has a reduced portion that can be solved with partial fractions.

$$x(s) = 5s + 3 + \frac{-12s - 6}{s^2 + 4} \quad \text{solve} \quad \frac{-12s - 6}{s^2 + 4} = \frac{A}{s + 2j} + \frac{B}{s - 2j}$$

- When the order of the denominator terms is greater than 1 it requires an expanded partial fraction form, as shown below.

$$F(s) = \frac{5}{s^2(s+1)^3}$$

$$\frac{5}{s^2(s+1)^3} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{(s+1)^3} + \frac{D}{(s+1)^2} + \frac{E}{(s+1)}$$

- We can solve the previous problem using the algebra technique.

$$\begin{aligned} \frac{5}{s^2(s+1)^3} &= \frac{A}{s^2} + \frac{B}{s} + \frac{C}{(s+1)^3} + \frac{D}{(s+1)^2} + \frac{E}{(s+1)} \\ &= \frac{A(s+1)^3 + Bs(s+1)^3 + Cs^2 + Ds^2(s+1) + Es^2(s+1)^2}{s^2(s+1)^3} \\ &= \frac{s^4(B+E) + s^3(A+3B+D+2E) + s^2(3A+3B+C+D+E) + s(3A+B) + (A)}{s^2(s+1)^3} \end{aligned}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 3 & 0 & 1 & 2 \\ 3 & 3 & 1 & 1 & 1 \\ 3 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 5 \end{bmatrix} \quad \begin{bmatrix} A \\ B \\ C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 3 & 0 & 1 & 2 \\ 3 & 3 & 1 & 1 & 1 \\ 3 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ -15 \\ 5 \\ 10 \\ 15 \end{bmatrix}$$

$$\frac{5}{s^2(s+1)^3} = \frac{5}{s^2} + \frac{-15}{s} + \frac{5}{(s+1)^3} + \frac{10}{(s+1)^2} + \frac{15}{(s+1)}$$

2.2.4 Summation and Series

- The notation $\sum_{i=a}^b x_i$ is equivalent to assuming a and b are integers and $x_1 + x_2 + \dots + x_b$ and b is the index variable is a placeholder whose name does not matter.

- Operations on summations:

$$\sum_{i=a}^b x_i = \sum_{i=b}^a x_i$$

$$\sum_{i=a}^b \alpha x_i = \alpha \sum_{i=a}^b x_i$$

$$\sum_{i=a}^b x_i + \sum_{j=a}^b y_j = \sum_{i=a}^b (x_i + y_i)$$

$$\sum_{i=a}^b x_i + \sum_{i=b+1}^c x_i = \sum_{i=a}^c x_i$$

$$\left(\sum_{i=a}^b x_i \right) \left(\sum_{j=c}^d y_j \right) = \sum_{i=a}^b \sum_{j=c}^d x_i y_j$$

- Some common summations:

$$\sum_{i=1}^N i = \frac{1}{2}N(N+1)$$

$$\sum_{i=0}^{N-1} \alpha^i = \begin{cases} \frac{1-\alpha^N}{1-\alpha}, & \alpha \neq 1 \\ N, & \alpha = 1 \end{cases} \text{ for both real and complex } \alpha.$$

$$\sum_{i=0}^{\infty} \alpha^i = \frac{1}{1-\alpha}, |\alpha| < 1 \text{ for both real and complex } \alpha \text{ For } |\alpha| \geq 1, \text{ the summation does not converge.}$$

2.3 SPATIAL RELATIONSHIPS

2.3.1 Trigonometry

- The basic trigonometry functions are,

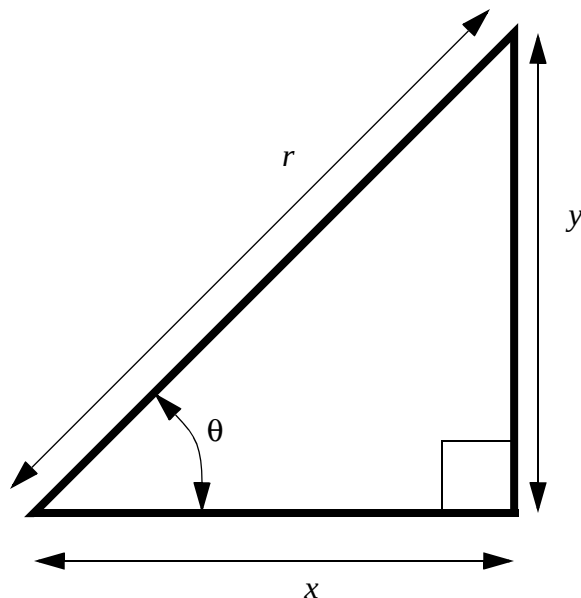
$$\sin \theta = \frac{y}{r} = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{x}{r} = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{y}{x} = \frac{1}{\cot \theta} = \frac{\sin \theta}{\cos \theta}$$

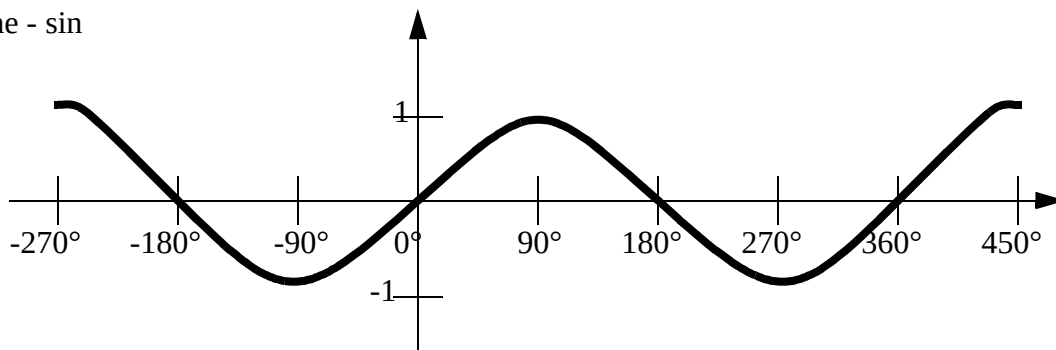
Pythagorean Formula:

$$r^2 = x^2 + y^2$$

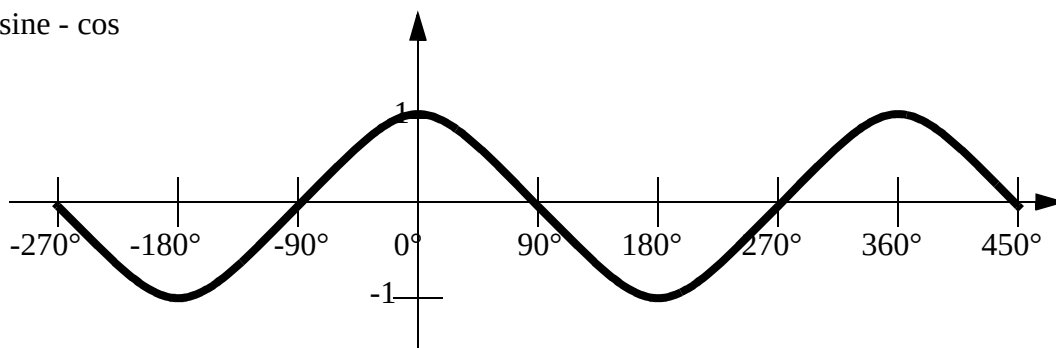


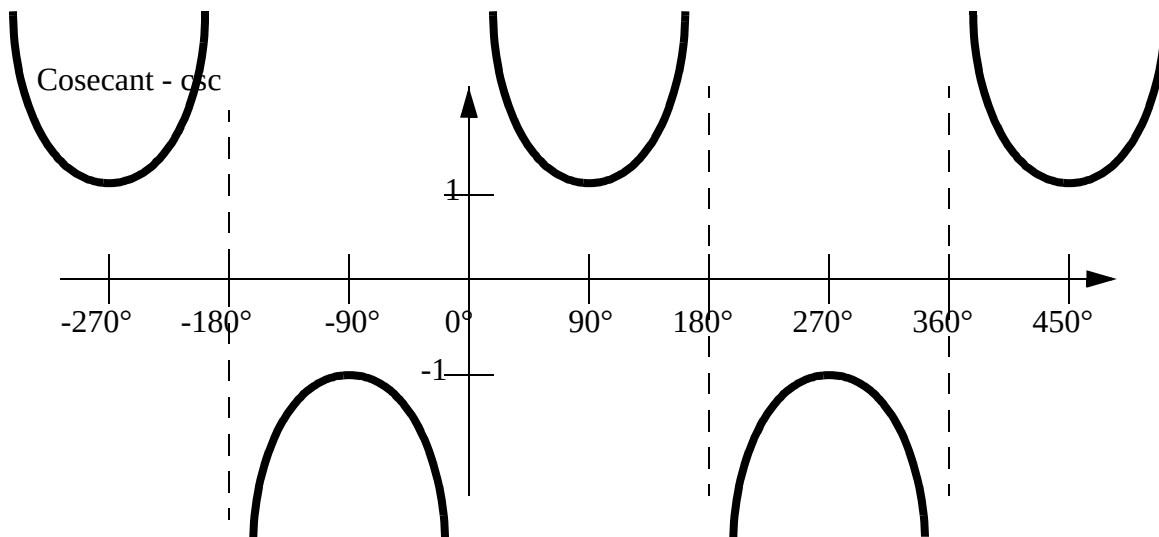
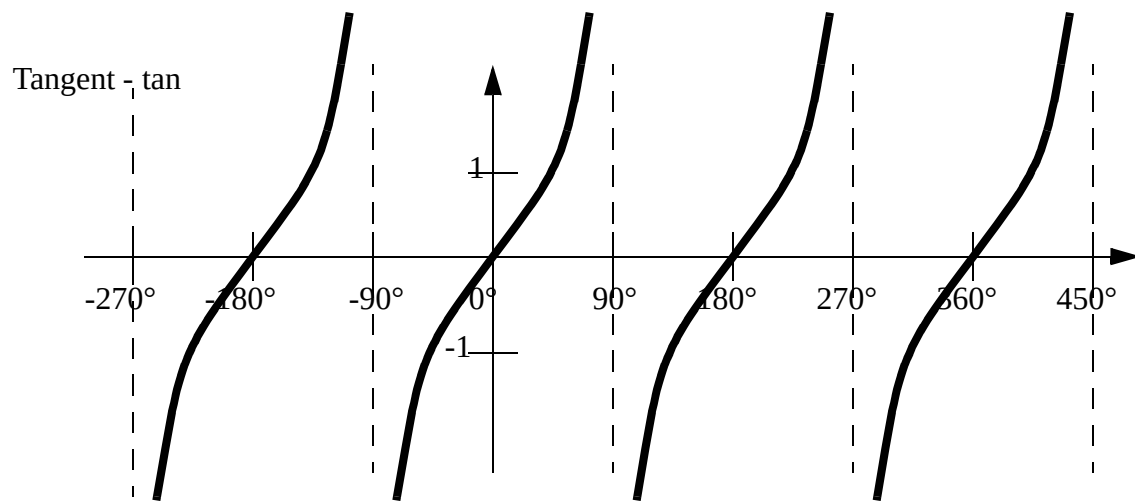
- Graphs of these functions are given below,

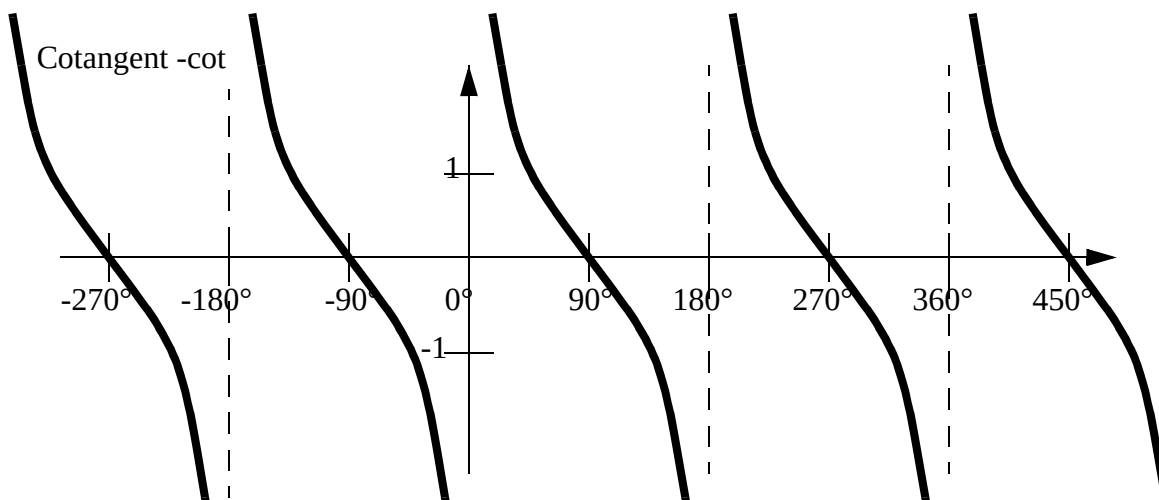
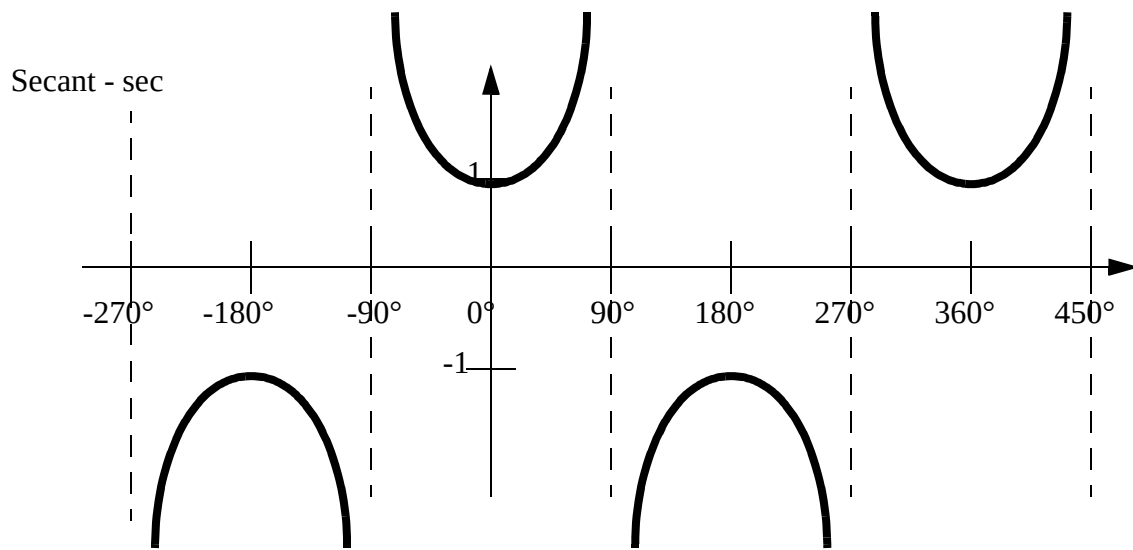
Sine - sin



Cosine - cos







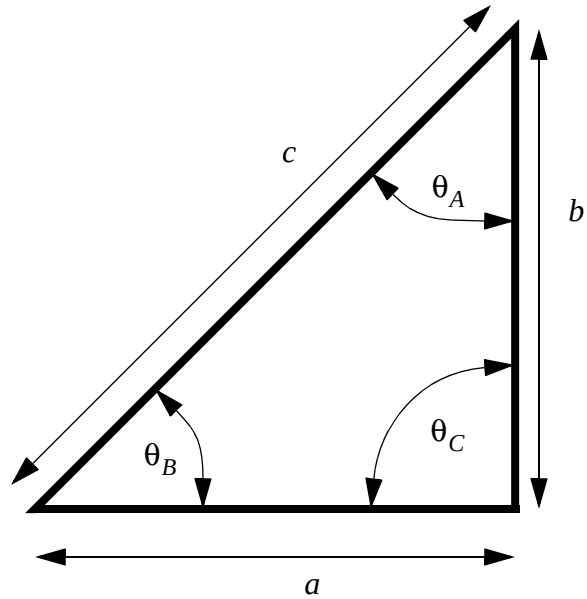
- NOTE: Keep in mind when finding these trig values, that any value that does not lie in the right hand quadrants of cartesian space, may need additions of $\pm 90^\circ$ or $\pm 180^\circ$.

Cosine Law:

$$c^2 = a^2 + b^2 - 2ab \cos \theta_c$$

Sine Law:

$$\frac{a}{\sin \theta_A} = \frac{b}{\sin \theta_B} = \frac{c}{\sin \theta_C}$$



- Now a group of trigonometric relationships will be given. These are often best used when attempting to manipulate equations.

$$\sin(-\theta) = -\sin\theta \qquad \cos(-\theta) = \cos\theta \qquad \tan(-\theta) = -\tan\theta$$

$$\sin\theta = \cos(\theta - 90^\circ) = \cos(90^\circ - \theta) = \text{etc.}$$

$$\sin(\theta_1 \pm \theta_2) = \sin\theta_1 \cos\theta_2 \pm \cos\theta_1 \sin\theta_2 \quad \text{OR} \quad \sin(2\theta) = 2\sin\theta \cos\theta$$

$$\cos(\theta_1 \pm \theta_2) = \cos\theta_1 \cos\theta_2 \mp \sin\theta_1 \sin\theta_2 \quad \text{OR} \quad \cos(2\theta) = (\cos\theta)^2 - (\sin\theta)^2$$

$$\tan(\theta_1 \pm \theta_2) = \frac{\tan\theta_1 \pm \tan\theta_2}{1 \mp \tan\theta_1 \tan\theta_2}$$

$$\cot(\theta_1 \pm \theta_2) = \frac{\cot\theta_1 \cot\theta_2 \mp 1}{\tan\theta_2 \pm \tan\theta_1}$$

$$\sin\frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos\theta}{2}} \quad \swarrow \text{-ve if in left hand quadrants}$$

$$\cos\frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos\theta}{2}} \quad \nwarrow$$

$$\tan\frac{\theta}{2} = \frac{\sin\theta}{1 + \cos\theta} = \frac{1 - \cos\theta}{\sin\theta}$$

$$(\cos\theta)^2 + (\sin\theta)^2 = 1$$

- These can also be related to complex exponents,

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2} \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

2.3.2 Hyperbolic Functions

- The basic definitions are given below,

$$\sinh(x) = \frac{e^x - e^{-x}}{2} = \text{hyperbolic sine of } x$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} = \text{hyperbolic cosine of } x$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \text{hyperbolic tangent of } x$$

$$\operatorname{csch}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^x - e^{-x}} = \text{hyperbolic cosecant of } x$$

$$\operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^x + e^{-x}} = \text{hyperbolic secant of } x$$

$$\operatorname{coth}(x) = \frac{\cosh(x)}{\sinh(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \text{hyperbolic cotangent of } x$$

- some of the basic relationships are,

$$\sinh(-x) = -\sinh(x)$$

$$\cosh(-x) = \cosh(x)$$

$$\tanh(-x) = -\tanh(x)$$

$$\operatorname{csch}(-x) = -\operatorname{csch}(x)$$

$$\operatorname{sech}(-x) = \operatorname{sech}(x)$$

$$\operatorname{coth}(-x) = -\operatorname{coth}(x)$$

- Some of the more advanced relationships are,

$$(\cosh x)^2 - (\sinh x)^2 = (\operatorname{sech} x)^2 + (\tanh x)^2 = (\coth x)^2 - (\operatorname{csch} x)^2 = 1$$

$$\sinh(x \pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y)$$

$$\cosh(x \pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh(y)$$

$$\tanh(x \pm y) = \frac{\tanh(x) \pm \tanh(y)}{1 \pm \tanh(x)\tanh(y)}$$

- Some of the relationships between the hyperbolic, and normal trigonometry functions are,

$$\sin(jx) = j \sinh(x)$$

$$j \sin(x) = \sinh(jx)$$

$$\cos(jx) = \cosh(x)$$

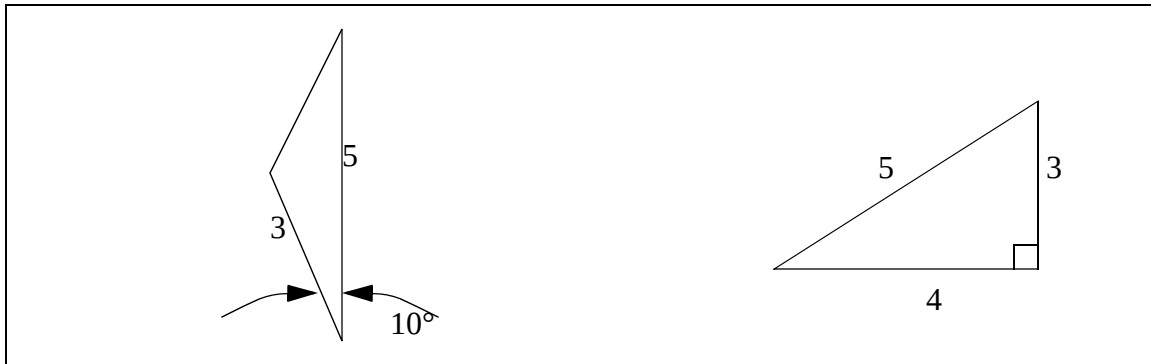
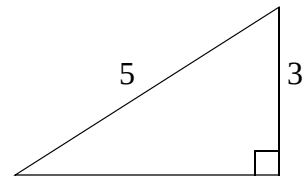
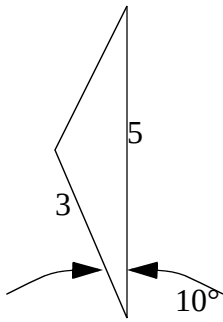
$$\cos(x) = \cosh(jx)$$

$$\tan(jx) = j \tanh(x)$$

$$j \tan(x) = \tanh(jx)$$

2.3.2.1 - Practice Problems

3. Find all of the missing side lengths and corner angles on the two triangles below,



2.3.3 Geometry

***** ADD IN MASS MOMENTS AND DESCRIPTIONS *****

- A set of the basic 2D and 3D geometric primitives are given, and the notation used is described below,

A = contained area

P = perimeter distance

V = contained volume

S = surface area

x, y, z = centre of mass

$\bar{x}, \bar{y}, \bar{z}$ = centroid

I_x, I_y, I_z = moment of inertia of area (or second moment of inertia)

AREA PROPERTIES:

$$I_x = \int_A y^2 dA = \text{the moment of inertia about the y-axis}$$

$$I_y = \int_A x^2 dA = \text{the moment of inertia about the x-axis}$$

$$I_{xy} = \int_A xy dA = \text{the product of inertia}$$

$$J_O = \int_A r^2 dA = \int_A (x^2 + y^2) dA = I_x + I_y = \text{The polar moment of inertia}$$

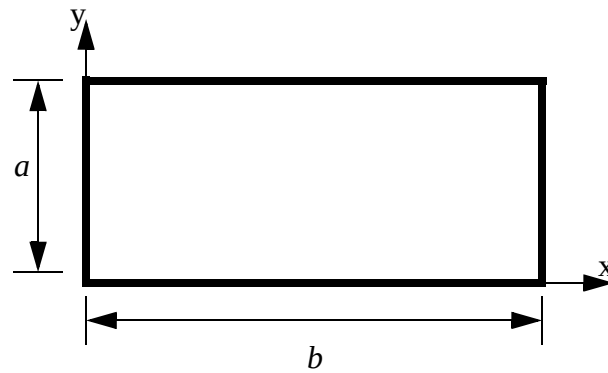
$$\bar{x} = \frac{\int_A x dA}{\int_A dA} = \text{centroid location along the x-axis}$$

$$\bar{y} = \frac{\int_A y dA}{\int_A dA} = \text{centroid location along the y-axis}$$

Rectangle/Square:

$$A = ab$$

$$P = 2a + 2b$$



Centroid:

$$\bar{x} = \frac{b}{2}$$

$$\bar{y} = \frac{a}{2}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{ba^3}{12}$$

$$\bar{I}_y = \frac{b^3a}{12}$$

$$\bar{I}_{xy} = 0$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{ba^3}{3}$$

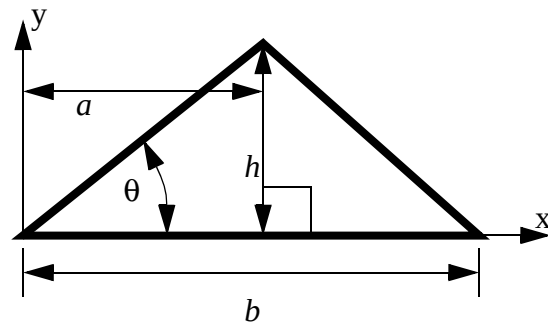
$$I_y = \frac{b^3a}{3}$$

$$I_{xy} = \frac{b^2a^2}{4}$$

Triangle:

$$A = \frac{bh}{2}$$

$$P =$$



Centroid:

$$\bar{x} = \frac{a+b}{3}$$

$$\bar{y} = \frac{h}{3}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{bh^3}{36}$$

$$\bar{I}_y = \frac{bh}{36}(a^2 + b^2 - ab)$$

$$\bar{I}_{xy} = \frac{bh^2}{72}(2a - b)$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{bh^3}{12}$$

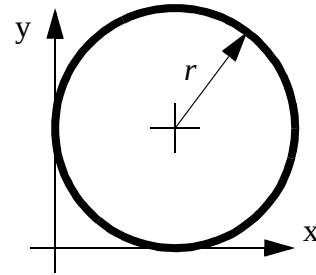
$$I_y = \frac{bh}{12}(a^2 + b^2 - ab)$$

$$I_{xy} = \frac{bh^2}{24}(2a - b)$$

Circle:

$$A = \pi r^2$$

$$P = 2\pi r$$



Centroid:

$$\bar{x} = r$$

$$\bar{y} = r$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{\pi r^4}{4}$$

$$\bar{I}_y = \frac{\pi r^4}{4}$$

$$\bar{I}_{xy} = 0$$

Moment of Inertia
(about origin axes):

$$I_x =$$

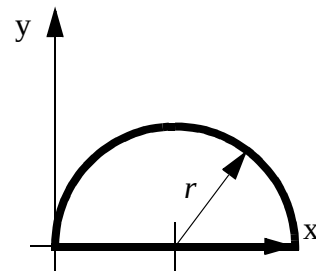
$$I_y =$$

$$I_{xy} =$$

Half Circle:

$$A = \frac{\pi r^2}{2}$$

$$P = \pi r + 2r$$



Centroid:

$$\bar{x} = r$$

$$\bar{y} = \frac{4r}{3\pi}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \left(\frac{\pi}{8} - \frac{8}{9\pi} \right) r^4$$

$$\bar{I}_y = \frac{\pi r^4}{8}$$

$$I_{xy} = 0$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{\pi r^4}{8}$$

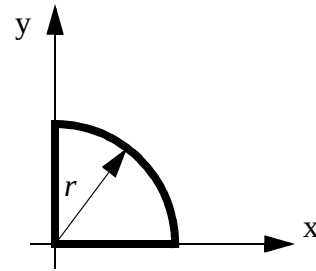
$$I_y = \frac{\pi r^4}{8}$$

$$I_{xy} = 0$$

Quarter Circle:

$$A = \frac{\pi r^2}{4}$$

$$P = \frac{\pi r}{2} + 2r$$



Centroid:

$$\bar{x} = \frac{4r}{3\pi}$$

$$\bar{y} = \frac{4r}{3\pi}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = 0.05488r^4$$

$$\bar{I}_y = 0.05488r^4$$

$$\bar{I}_{xy} = -0.01647r^4$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{\pi r^4}{16}$$

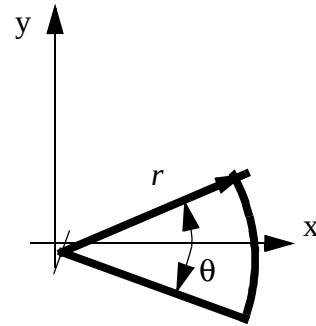
$$I_y = \frac{\pi r^4}{16}$$

$$I_{xy} = \frac{r^4}{8}$$

Circular Arc:

$$A = \frac{\theta r^2}{2}$$

$$P = \theta r + 2r$$



Centroid:

$$\bar{x} = \frac{2r \sin \frac{\theta}{2}}{3\theta}$$

$$\bar{y} = 0$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x =$$

$$\bar{I}_y =$$

$$\bar{I}_{xy} =$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{r^4}{8}(\theta - \sin \theta)$$

$$I_y = \frac{r^4}{8}(\theta + \sin \theta)$$

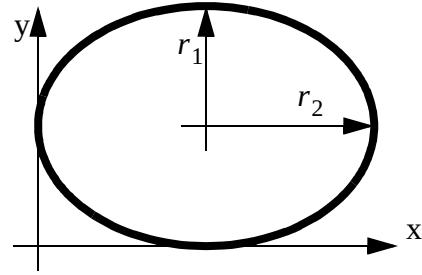
$$I_{xy} = 0$$

Ellipse:

$$A = \pi r_1 r_2$$

$$P = 4r_1 \int_0^{\frac{\pi}{2}} \sqrt{1 - \frac{\sqrt{r_1^2 + r_2^2}}{a} (\sin \theta)^2} d\theta$$

$$P \approx 2\pi \sqrt{\frac{r_1^2 + r_2^2}{2}}$$



Centroid:

$$\bar{x} = r_2$$

$$\bar{y} = r_1$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{\pi r_1^3 r_2}{4}$$

$$\bar{I}_y = \frac{\pi r_1 r_2^3}{4}$$

$$\bar{I}_{xy} =$$

Moment of Inertia
(about origin axes):

$$I_x =$$

$$I_y =$$

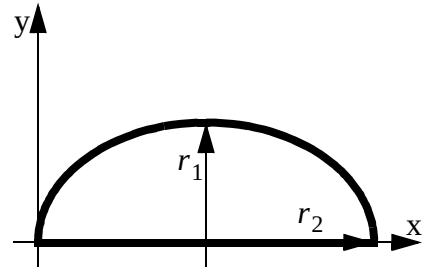
$$I_{xy} =$$

Half Ellipse:

$$A = \frac{\pi r_1 r_2}{2}$$

$$P = 2r_1 \int_0^{\frac{\pi}{2}} \sqrt{1 - \frac{\sqrt{r_1^2 + r_2^2}}{a} (\sin \theta)^2} d\theta + 2r_2$$

$$P \approx \pi \sqrt{\frac{r_1^2 + r_2^2}{2}} + 2r_2$$



Centroid:

$$\bar{x} = r_2$$

$$\bar{y} = \frac{4r_1}{3\pi}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = 0.05488 r_2 r_1^3$$

$$\bar{I}_y = 0.05488 r_2^3 r_1$$

$$\bar{I}_{xy} = -0.01647 r_1^2 r_2^2$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{\pi r_2 r_1^3}{16}$$

$$I_y = \frac{\pi r_2^3 r_1}{16}$$

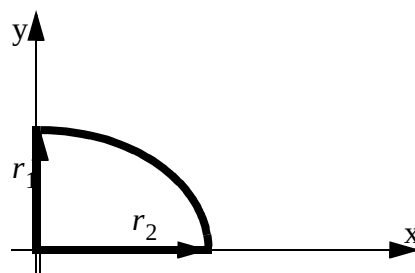
$$I_{xy} = \frac{r_1^2 r_2^2}{8}$$

Quarter Ellipse:

$$A = \frac{\pi r_1 r_2}{4}$$

$$P = r_1 \int_0^{\frac{\pi}{2}} \sqrt{1 - \frac{\sqrt{r_1^2 + r_2^2}}{a} (\sin \theta)^2} d\theta + 2r_2$$

$$P \approx \frac{\pi}{2} \sqrt{\frac{r_1^2 + r_2^2}{2}} + 2r_2$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = \frac{4r_2}{3\pi}$$

$$\bar{I}_x =$$

$$I_x = \pi r_2^3 r_1$$

$$\bar{y} = \frac{4r_1}{3\pi}$$

$$\bar{I}_y =$$

$$I_y = \pi r_2^3 r_1$$

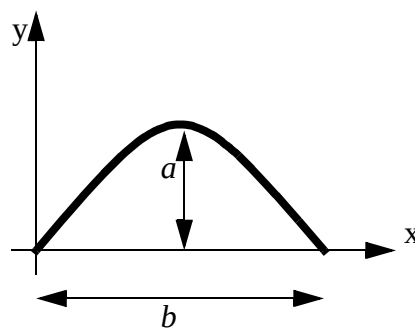
$$\bar{I}_{xy} =$$

$$I_{xy} = \frac{r_2^2 r_1^2}{8}$$

Parabola:

$$A = \frac{2}{3}ab$$

$$P = \frac{\sqrt{b^2 + 16a^2}}{2} + \frac{b^2}{8a} \ln \left(\frac{4a + \sqrt{b^2 + 16a^2}}{b} \right)$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = \frac{b}{2}$$

$$\bar{I}_x =$$

$$I_x =$$

$$\bar{y} = \frac{2a}{5}$$

$$\bar{I}_y =$$

$$I_y =$$

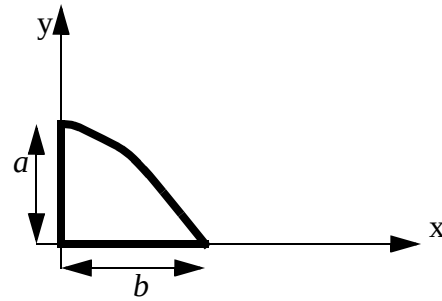
$$\bar{I}_{xy} =$$

$$I_{xy} =$$

Half Parabola:

$$A = \frac{ab}{3}$$

$$P = \frac{\sqrt{b^2 + 16a^2}}{4} + \frac{b^2}{16a} \ln \left(\frac{4a + \sqrt{b^2 + 16a^2}}{b} \right)$$



Centroid:

$$\bar{x} = \frac{3b}{8}$$

$$\bar{y} = \frac{2a}{5}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{8ba^3}{175}$$

$$\bar{I}_y = \frac{19b^3a}{480}$$

$$\bar{I}_{xy} = \frac{b^2a^2}{60}$$

Moment of Inertia
(about origin axes):

$$I_x = \frac{2ba^3}{7}$$

$$I_y = \frac{2b^3a}{15}$$

$$I_{xy} = \frac{b^2a^2}{6}$$

- A general class of geometries are conics. This form is shown below, and can be used to represent many of the simple shapes represented by a polynomial.

$$Ax^2 + 2Bxy + Cy^2 + 2Dx + 2Ey + F = 0$$

Conditions

$A = B = C = 0$ straight line

$B = 0, A = C$ circle

$B^2 - AC < 0$ ellipse

$B^2 - AC = 0$ parabola

$B^2 - AC > 0$ hyperbola

VOLUME PROPERTIES:

$$I_x = \int_V r_x^2 dV = \text{the moment of inertia about the x-axis}$$

$$I_y = \int_V r_y^2 dV = \text{the moment of inertia about the y-axis}$$

$$I_z = \int_V r_z^2 dV = \text{the moment of inertia about the z-axis}$$

$$\bar{x} = \frac{\int_V x dV}{\int_V dV} = \text{centroid location along the x-axis}$$

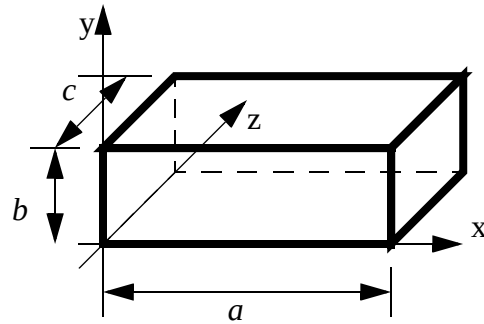
$$\bar{y} = \frac{\int_V y dV}{\int_V dV} = \text{centroid location along the y-axis}$$

$$\bar{z} = \frac{\int_V z dV}{\int_V dV} = \text{centroid location along the z-axis}$$

Parallelepiped (box):

$$V = abc$$

$$S = 2(ab + ac + bc)$$



Centroid:

$$\bar{x} = \frac{a}{2}$$

$$\bar{y} = \frac{b}{2}$$

$$\bar{z} = \frac{c}{2}$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{M(a^2 + b^2)}{12}$$

$$\bar{I}_y = \frac{M(a^2 + c^2)}{12}$$

$$\bar{I}_z = \frac{M(b^2 + a^2)}{12}$$

Moment of Inertia
(about origin axes):

$$I_x =$$

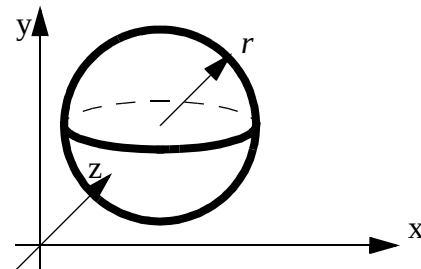
$$I_y =$$

$$I_z =$$

Sphere:

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$



Centroid:

$$\bar{x} = r$$

$$\bar{y} = r$$

$$\bar{z} = r$$

Moment of Inertia
(about centroid axes):

$$\bar{I}_x = \frac{2Mr^2}{5}$$

$$\bar{I}_y = \frac{2Mr^2}{5}$$

$$\bar{I}_z = \frac{2Mr^2}{5}$$

Moment of Inertia
(about origin axes):

$$I_x =$$

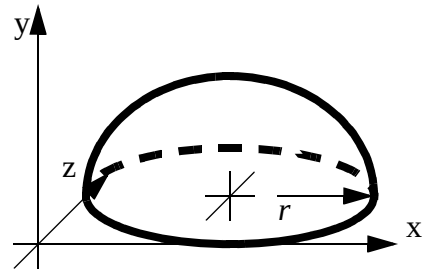
$$I_y =$$

$$I_z =$$

Hemisphere:

$$V = \frac{2}{3}\pi r^3$$

$$S =$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = r$$

$$\bar{I}_x = \frac{83}{320}Mr^2$$

$$I_x =$$

$$\bar{y} = \frac{3r}{8}$$

$$\bar{I}_y = \frac{2Mr^2}{5}$$

$$I_y =$$

$$\bar{z} = r$$

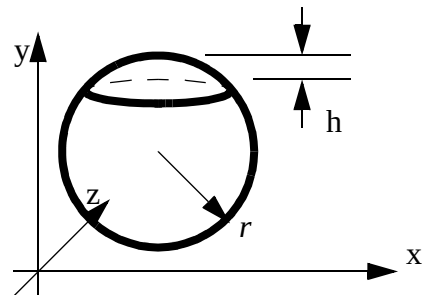
$$\bar{I}_z = \frac{83}{320}Mr^2$$

$$I_z =$$

Cap of a Sphere:

$$V = \frac{1}{3}\pi h^2(3r - h)$$

$$S = 2\pi rh$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = r$$

$$\bar{I}_x =$$

$$I_x =$$

$$\bar{y} =$$

$$\bar{I}_y =$$

$$I_y =$$

$$\bar{z} = r$$

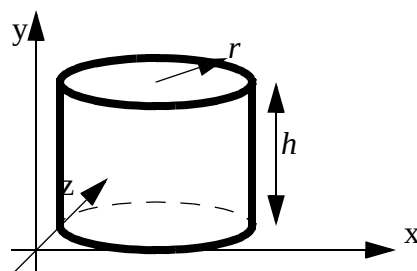
$$\bar{I}_z =$$

$$I_z =$$

Cylinder:

$$V = h\pi r^2$$

$$S = 2\pi rh + 2\pi r^2$$



Centroid:

Moment of Inertia
(about centroid axis):

Moment of Inertia
(about origin axis):

$$\bar{x} = r$$

$$\bar{I}_x = M\left(\frac{h^2}{12} + \frac{r^2}{4}\right)$$

$$I_x = M\left(\frac{h^2}{3} + \frac{r^2}{4}\right)$$

$$\bar{y} = \frac{h}{2}$$

$$\bar{I}_y = \frac{Mr^2}{2}$$

$$I_y =$$

$$\bar{z} = r$$

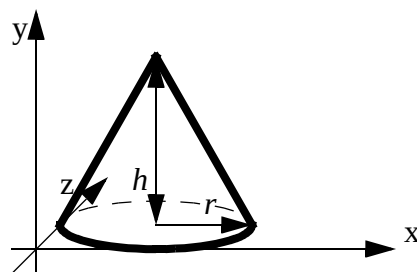
$$\bar{I}_z = M\left(\frac{h^2}{12} + \frac{r^2}{4}\right)$$

$$I_z = M\left(\frac{h^2}{3} + \frac{r^2}{4}\right)$$

Cone:

$$V = \frac{1}{3}\pi r^2 h$$

$$S = \pi r \sqrt{r^2 + h^2}$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = r$$

$$\bar{I}_x = M\left(\frac{3h^3}{80} + \frac{3r^2}{20}\right)$$

$$I_x =$$

$$\bar{y} = \frac{h}{4}$$

$$\bar{I}_y = \frac{3Mr^2}{10}$$

$$I_y =$$

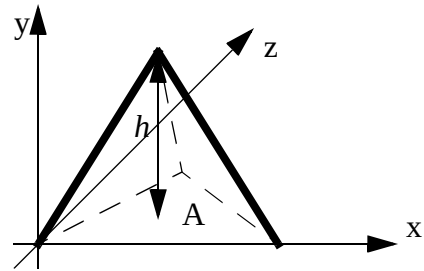
$$\bar{z} = r$$

$$\bar{I}_z = M\left(\frac{3h^3}{80} + \frac{3r^2}{20}\right)$$

$$I_z =$$

Tetrahedron:

$$V = \frac{1}{3}Ah$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} =$$

$$\bar{I}_x =$$

$$I_x =$$

$$\bar{y} = \frac{h}{4}$$

$$\bar{I}_y =$$

$$I_y =$$

$$\bar{z} =$$

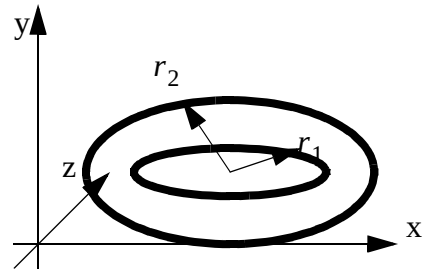
$$\bar{I}_z =$$

$$I_z =$$

Torus:

$$V = \frac{1}{4}\pi^2(r_1 + r_2)(r_2 - r_1)^2$$

$$S = \pi^2(r_2^2 - r_1^2)$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = r_2$$

$$\bar{I}_x =$$

$$I_x =$$

$$\bar{y} = \left(\frac{r_2 - r_1}{2}\right)$$

$$\bar{I}_y =$$

$$I_y =$$

$$\bar{z} = r_2$$

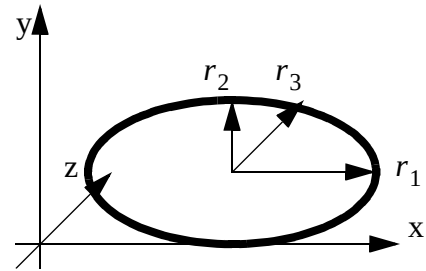
$$\bar{I}_z =$$

$$I_z =$$

Ellipsoid:

$$V = \frac{4}{3}\pi r_1 r_2 r_3$$

$$S =$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = r_1$$

$$\bar{I}_x =$$

$$I_x =$$

$$\bar{y} = r_2$$

$$\bar{I}_y =$$

$$I_y =$$

$$\bar{z} = r_3$$

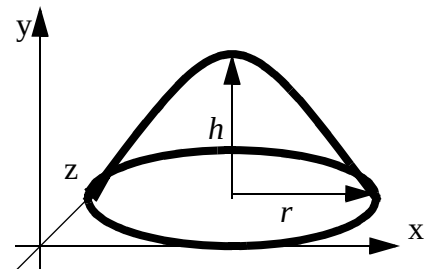
$$\bar{I}_z =$$

$$I_z =$$

Paraboloid:

$$V = \frac{1}{2}\pi r^2 h$$

$$S =$$



Centroid:

Moment of Inertia
(about centroid axes):

Moment of Inertia
(about origin axes):

$$\bar{x} = r$$

$$\bar{I}_x =$$

$$I_x =$$

$$\bar{y} =$$

$$\bar{I}_y =$$

$$I_y =$$

$$\bar{z} = r$$

$$\bar{I}_z =$$

$$I_z =$$

2.3.4 Planes, Lines, etc.

- The most fundamental mathematical geometry is a line. The basic relationships are given below,

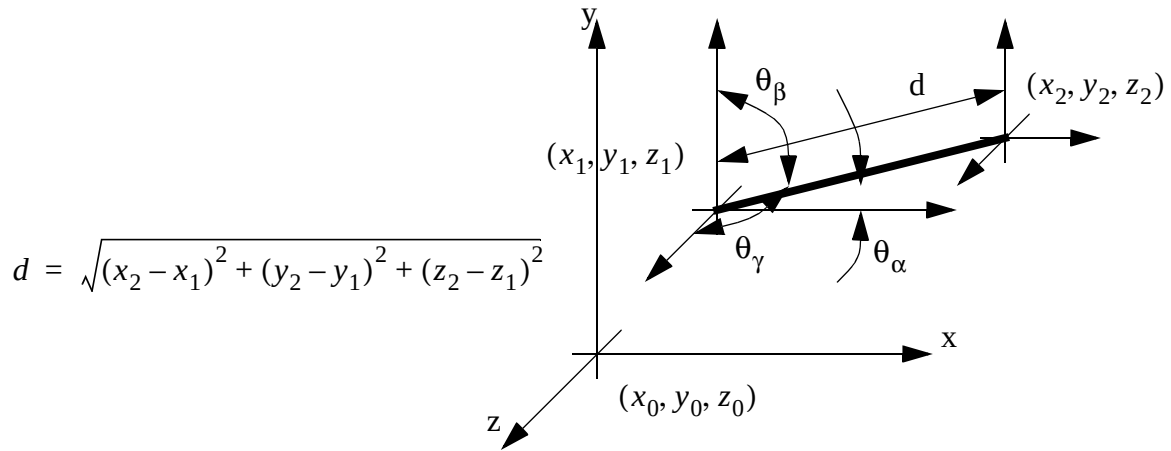
$$y = mx + b \quad \text{defined with a slope and intercept}$$

$$m_{\text{perpendicular}} = \frac{1}{m} \quad \text{a slope perpendicular to a line}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{the slope using two points}$$

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{as defined by two intercepts}$$

- If we assume a line is between two points in space, and that at one end we have a local reference frame, there are some basic relationships that can be derived.



The direction cosines of the angles are,

$$\theta_\alpha = \arccos\left(\frac{x_2 - x_1}{d}\right) \quad \theta_\beta = \arccos\left(\frac{y_2 - y_1}{d}\right) \quad \theta_\gamma = \arccos\left(\frac{z_2 - z_1}{d}\right)$$

$$(\cos\theta_\alpha)^2 + (\cos\theta_\beta)^2 + (\cos\theta_\gamma)^2 = 1$$

The equation of the line is,

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Explicit

$$(x, y, z) = (x_1, y_1, z_1) + t((x_2, y_2, z_2) - (x_1, y_1, z_1))$$

Parametric $t \in [0, 1]$

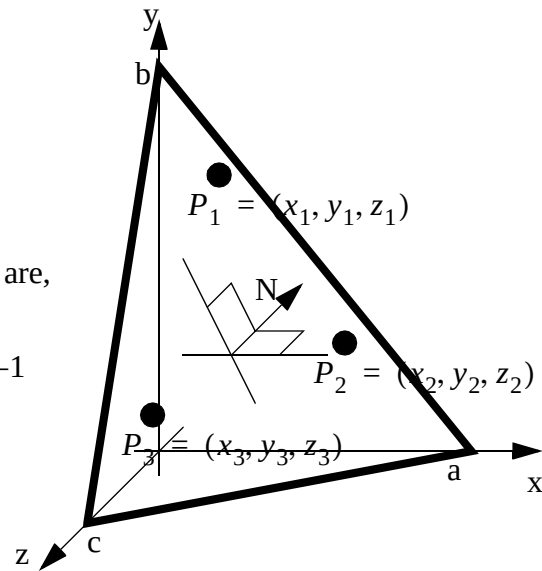
- The relationships for a plane are,

The explicit equation for a plane is,

$$Ax + By + Cz + D = 0$$

where the coefficients defined by the intercepts are,

$$A = \frac{1}{a} \quad B = \frac{1}{b} \quad C = \frac{1}{c} \quad D = -1$$



The determinant can also be used,

$$\det \begin{bmatrix} x-x_1 & y-y_1 & z-z_1 \\ x-x_2 & y-y_2 & z-z_2 \\ x-x_3 & y-y_3 & z-z_3 \end{bmatrix} = 0$$

$$\therefore \det \begin{bmatrix} y_2-y_1 & z_2-z_1 \\ y_3-y_1 & z_3-z_1 \end{bmatrix} (x-x_1) + \det \begin{bmatrix} z_2-z_1 & x_2-x_1 \\ z_3-z_1 & x_3-x_1 \end{bmatrix} (y-y_1)$$

$$+ \det \begin{bmatrix} x_2-x_1 & y_2-y_1 \\ x_3-x_1 & y_3-y_1 \end{bmatrix} (z-z_1) = 0$$

The normal to the plane (through the origin) is,

$$(x, y, z) = t(A, B, C)$$

2.4 COORDINATE SYSTEMS

2.4.1 Complex Numbers

- In this section, as in all others, ‘j’ will be the preferred notation for the complex number, this is to help minimize confusion with the ‘i’ used for current in electrical engineering.
- The basic algebraic properties of these numbers are,

The Complex Number:

$$j = \sqrt{-1} \qquad j^2 = -1$$

Complex Numbers:

$$a + bj \qquad \text{where,} \qquad a \text{ and } b \text{ are both real numbers}$$

Complex Conjugates (denoted by adding an asterisk ‘*’ the variable):

$$N = a + bj \qquad N^* = a - bj$$

Basic Properties:

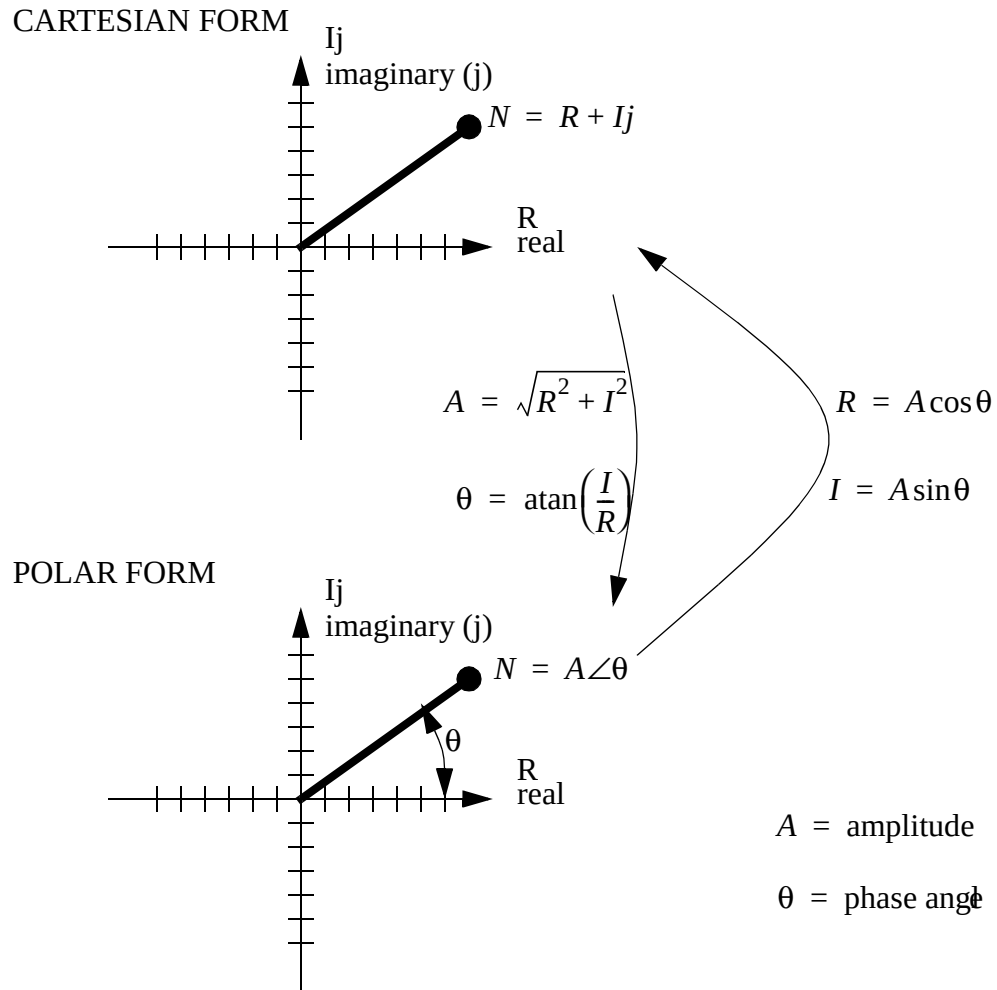
$$(a + bj) + (c + dj) = (a + c) + (b + d)j$$

$$(a + bj) - (c + dj) = (a - c) + (b - d)j$$

$$(a + bj) \cdot (c + dj) = (ac - bd) + (ad + bc)j$$

$$\frac{N}{M} = \frac{a + bj}{c + dj} = \frac{N(N^*)}{M(N^*)} = \left(\frac{a + bj}{c + dj} \right) \left(\frac{c - dj}{c - dj} \right) = \frac{ac + bd}{c^2 + d^2} + \left(\frac{bc - ad}{c^2 + d^2} \right)j$$

- We can also show complex numbers graphically. These representations lead to alternative representations. If it is not obvious above, please consider the notation above uses a cartesian notation, but a polar notation can also be very useful when doing large calculations.



- We can also do calculations using polar notation (this is well suited to multiplication and division, whereas cartesian notation is easier for addition and subtraction),

$$A \angle \theta = A(\cos \theta + j \sin \theta) = Ae^{j\theta}$$

$$(A_1 \angle \theta_1)(A_2 \angle \theta_2) = (A_1 A_2) \angle (\theta_1 + \theta_2)$$

$$\frac{(A_1 \angle \theta_1)}{(A_2 \angle \theta_2)} = \left(\frac{A_1}{A_2}\right) \angle (\theta_1 - \theta_2)$$

$$(A \angle \theta)^n = (A^n) \angle (n\theta) \quad (\text{DeMoivre's theorem})$$

- Note that DeMoivre's theorem can be used to find exponents (including roots) of complex numbers

- Euler's formula: $e^{j\theta} = \cos\theta + j\sin\theta$

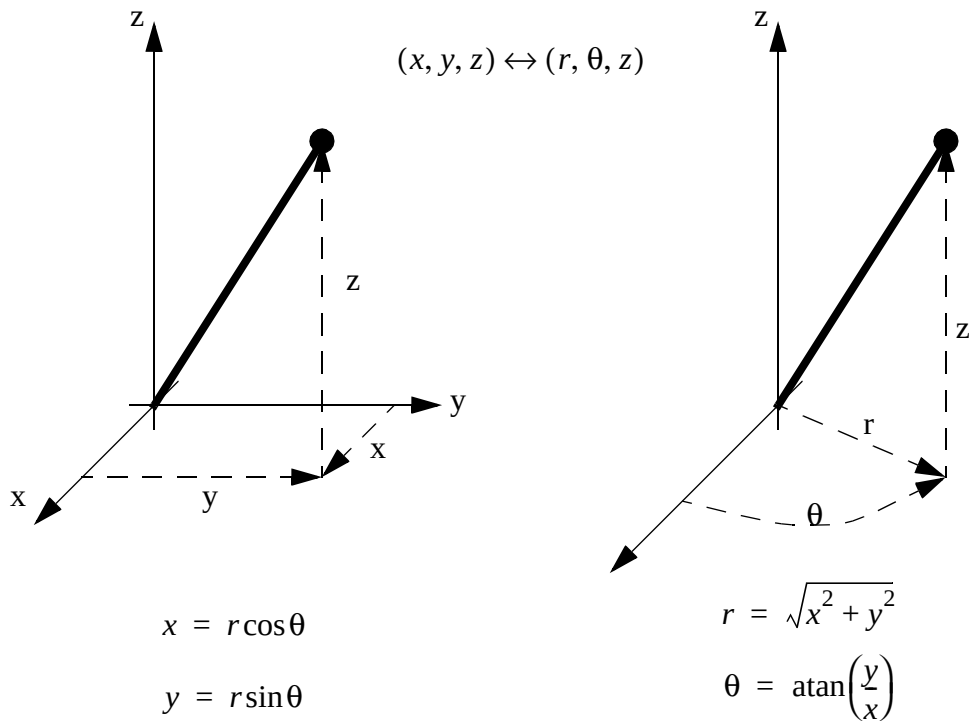
- From the above, the following useful identities arise:

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

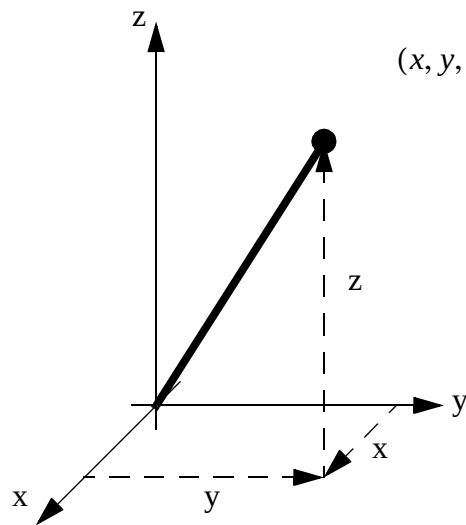
2.4.2 Cylindrical Coordinates

- Basically, these coordinates appear as if the cartesian box has been replaced with a cylinder,



2.4.3 Spherical Coordinates

- This system replaces the cartesian box with a sphere,

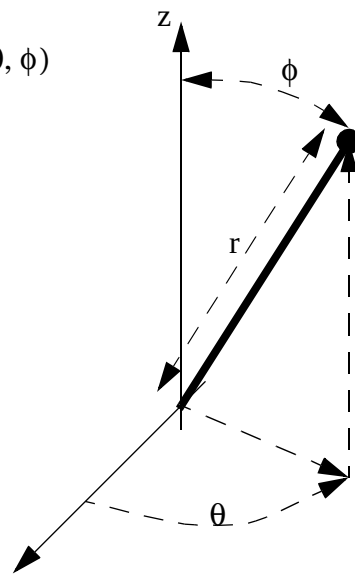


$$(x, y, z) \leftrightarrow (r, \theta, \phi)$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$



$$r = \sqrt{x^2 + y^2 + z^2}$$

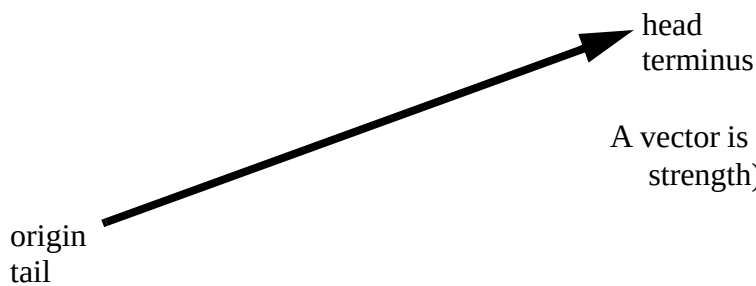
$$\theta = \text{atan}\left(\frac{y}{x}\right)$$

$$\phi = \text{acos}\left(\frac{z}{r}\right)$$

2.5 MATRICES AND VECTORS

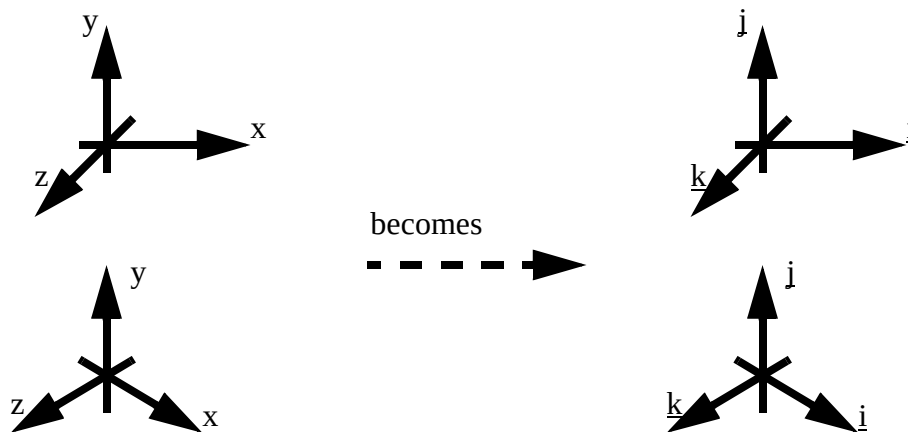
2.5.1 Vectors

- Vectors are often drawn with arrows, as shown below,



A vector is said to have magnitude (length or strength) and direction.

- Cartesian notation is also a common form of usage.



- Vectors can be added and subtracted, numerically and graphically,

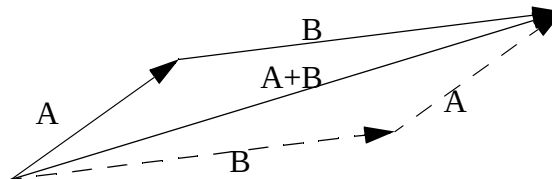
$$A = (2, 3, 4)$$

$$A + B = (2 + 7, 3 + 8, 4 + 9)$$

$$B = (7, 8, 9)$$

$$A - B = (2 - 7, 3 - 8, 4 - 9)$$

Parallelogram Law



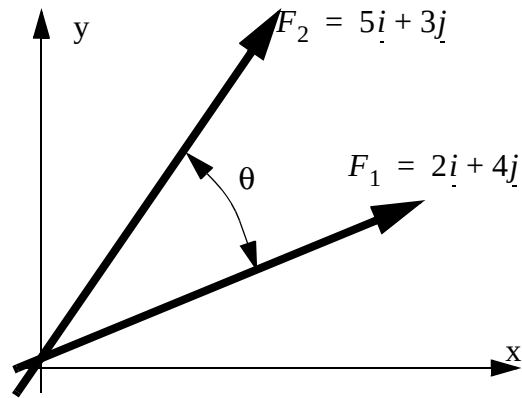
2.5.2 Dot (Scalar) Product

- We can use a dot product to find the angle between two vectors

$$\cos \theta = \frac{\mathbf{F}_1 \cdot \mathbf{F}_2}{|\mathbf{F}_1||\mathbf{F}_2|}$$

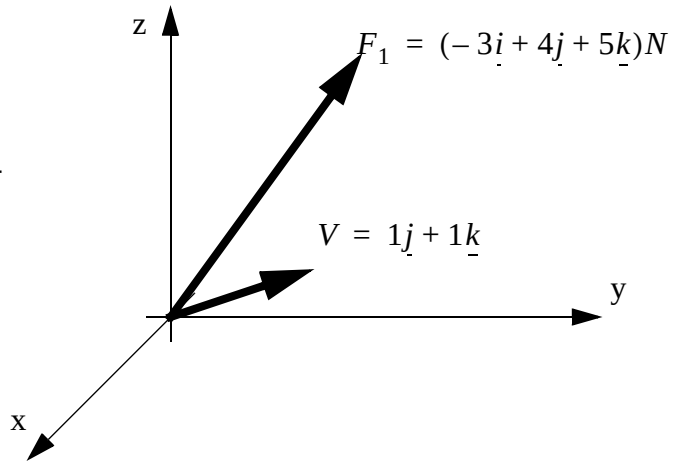
$$\therefore \theta = \arccos \left(\frac{(2)(5) + (4)(3)}{\sqrt{2^2 + 4^2} \sqrt{5^2 + 3^2}} \right)$$

$$\therefore \theta = \arccos \left(\frac{22}{(4.47)(6)} \right) = 32.5^\circ$$



- We can use a dot product to project one vector onto another vector.

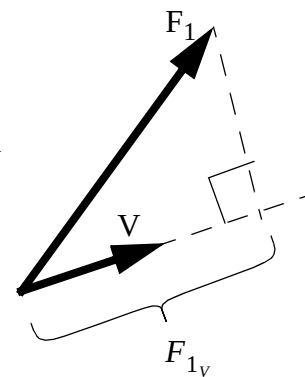
We want to find the component of force $\mathbf{F}_1$ that projects onto the vector $\mathbf{V}$. To do this we first convert $\mathbf{V}$ to a unit vector, if we do not, the component we find will be multiplied by the magnitude of $\mathbf{V}$.



$$\lambda_V = \frac{\mathbf{V}}{|\mathbf{V}|} = \frac{1\mathbf{j} + 1\mathbf{k}}{\sqrt{1^2 + 1^2}} = 0.707\mathbf{j} + 0.707\mathbf{k}$$

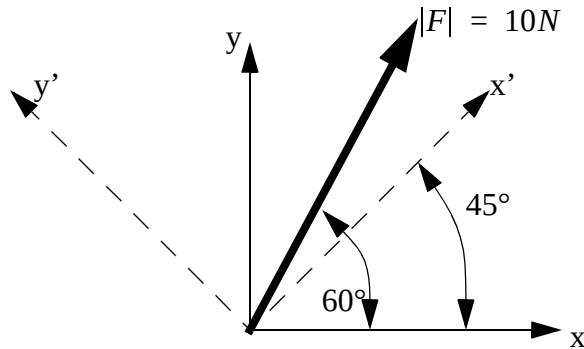
$$\mathbf{F}_{1_V} = \lambda_V \cdot \mathbf{F}_1 = (0.707\mathbf{j} + 0.707\mathbf{k}) \cdot (-3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})N$$

$$\therefore \mathbf{F}_{1_V} = (0)(-3) + (0.707)(4) + (0.707)(5) = 6N$$



- We can consider the basic properties of the dot product and units vectors.

Unit vectors are useful when breaking up vector magnitudes and direction. As an example consider the vector, and the displaced x-y axes shown below as x'-y'.



We could write out 5 vectors here, relative to the x-y axis,

$$\underline{x \text{ axis}} = 2\underline{i}$$

$$\underline{y \text{ axis}} = 3\underline{j}$$

$$\underline{x' \text{ axis}} = 1\underline{i} + 1\underline{j}$$

$$\underline{y' \text{ axis}} = -1\underline{i} + 1\underline{j}$$

$$\underline{F} = 10N \angle 60^\circ = (10 \cos 60^\circ)\underline{i} + (10 \sin 60^\circ)\underline{j}$$

None of these vectors has a magnitude of 1, and hence they are not unit vectors. But, if we find the equivalent vectors with a magnitude of one we can simplify many tasks. In particular if we want to find the x and y components of F relative to the x-y axis we can use the dot product.

$$\underline{\lambda_x} = 1\underline{i} + 0\underline{j} \quad (\text{unit vector for the x-axis})$$

$$\underline{F_x} = \underline{\lambda_x} \bullet \underline{F} = (1\underline{i} + 0\underline{j}) \bullet [(10 \cos 60^\circ)\underline{i} + (10 \sin 60^\circ)\underline{j}]$$

$$\therefore = (1)(10 \cos 60^\circ) + (0)(10 \sin 60^\circ) = 10N \cos 60^\circ$$

This result is obvious, but consider the other obvious case where we want to project a vector onto itself,

$$\lambda_F = \frac{F}{|F|} = \frac{10 \cos 60^\circ \underline{i} + 10 \sin 60^\circ \underline{j}}{10} = \cos 60^\circ \underline{i} + \sin 60^\circ \underline{j}$$

Incorrect - Not using a unit vector

$$\begin{aligned} F_F &= F \bullet F \\ &= ((10 \cos 60^\circ) \underline{i} + (10 \sin 60^\circ) \underline{j}) \bullet ((10 \cos 60^\circ) \underline{i} + (10 \sin 60^\circ) \underline{j}) \\ &= (10 \cos 60^\circ)(10 \cos 60^\circ) + (10 \sin 60^\circ)(10 \sin 60^\circ) \\ &= 100((\cos 60^\circ)^2 + (\sin 60^\circ)^2) = \cancel{100} \end{aligned}$$

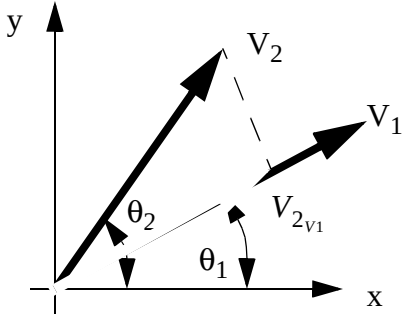
Using a unit vector

$$\begin{aligned} F_F &= F \bullet \lambda_F \\ &= ((10 \cos 60^\circ) \underline{i} + (10 \sin 60^\circ) \underline{j}) \bullet ((\cos 60^\circ) \underline{i} + (\sin 60^\circ) \underline{j}) \\ &= (10 \cos 60^\circ)(\cos 60^\circ) + (10 \sin 60^\circ)(\sin 60^\circ) \\ &= 10((\cos 60^\circ)^2 + (\sin 60^\circ)^2) = 10 \quad \text{Correct} \end{aligned}$$

Now consider the case where we find the component of F in the x' direction. Again, this can be done using the dot product to project F onto a unit vector.

$$\begin{aligned} u_{x'} &= \cos 45^\circ \underline{i} + \sin 45^\circ \underline{j} \\ F_{x'} &= F \bullet \lambda_{x'} = ((10 \cos 60^\circ) \underline{i} + (10 \sin 60^\circ) \underline{j}) \bullet ((\cos 45^\circ) \underline{i} + (\sin 45^\circ) \underline{j}) \\ &= (10 \cos 60^\circ)(\cos 45^\circ) + (10 \sin 60^\circ)(\sin 45^\circ) \\ &= 10(\cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ) = 10(\cos(60^\circ - 45^\circ)) \end{aligned}$$

Here we see a few cases where the dot product has been applied to find the vector projected onto a unit vector. Now finally consider the more general case,



First, by inspection, we can see that the component of V_2 (projected) in the direction of V_1 will be,

$$|V_{2v1}| = |V_2| \cos(\theta_2 - \theta_1)$$

Next, we can manipulate this expression into the dot product form,

$$\begin{aligned} &= |V_2|(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) \\ &= |V_2|[(\cos \theta_1 i + \sin \theta_1 j) \bullet (\cos \theta_2 i + \sin \theta_2 j)] \\ &= |V_2| \left[\frac{V_1}{|V_1|} \bullet \frac{V_2}{|V_2|} \right] = |V_2| \left[\frac{V_1 \bullet V_2}{|V_1||V_2|} \right] = \frac{V_1 \bullet V_2}{|V_1|} = V_2 \bullet \lambda_{V_1} \end{aligned}$$

Or more generally,

$$\begin{aligned} |V_{2v1}| &= |V_2| \cos(\theta_2 - \theta_1) = |V_2| \left[\frac{V_1 \bullet V_2}{|V_1||V_2|} \right] \\ \therefore |V_2| \cos(\theta_2 - \theta_1) &= |V_2| \left[\frac{V_1 \bullet V_2}{|V_1||V_2|} \right] \\ \therefore \cos(\theta_2 - \theta_1) &= \left[\frac{V_1 \bullet V_2}{|V_1||V_2|} \right] \end{aligned}$$

*Note that the dot product also works in 3D, and similar proofs are used.

2.5.3 Cross Product

- First, consider an example,

$$\underline{F} = (-6.43\underline{i} + 7.66\underline{j} + 0\underline{k})N$$

$$\underline{d} = (2\underline{i} + 0\underline{j} + 0\underline{k})m$$

$$\underline{M} = \underline{d} \times \underline{F} = \begin{bmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2m & 0m & 0m \\ -6.43N & 7.66N & 0N \end{bmatrix}$$

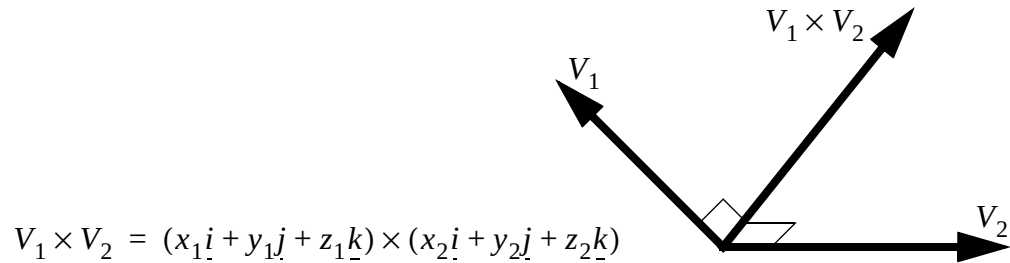
NOTE: note that the cross product here is for the right hand rule coordinates. If the left handed coordinate system is used $\underline{F}$ and $\underline{d}$ should be reversed.

$$\therefore \underline{M} = (0m0N - 0m(7.66N))\underline{i} - (2m0N - 0m(-6.43N))\underline{j} + (2m(7.66N) - 0m(-6.43N))\underline{k} = 15.3\underline{k}(mN)$$

NOTE: there are two things to note about the solution. First, it is a vector. Here there is only a z component because this vector points out of the page, and a rotation about this vector would rotate on the plane of the page. Second, this result is positive, because the positive sense is defined by the vector system. In this right handed system find the positive rotation by pointing your right hand thumb towards the positive axis (the 'k' means that the vector is about the z-axis here), and curl your fingers, that is the positive direction.

- The basic properties of the cross product are,

The cross (or vector) product of two vectors will yield a new vector perpendicular to both vectors, with a magnitude that is a product of the two magnitudes.



$$\underline{V_1} \times \underline{V_2} = (x_1 \underline{i} + y_1 \underline{j} + z_1 \underline{k}) \times (x_2 \underline{i} + y_2 \underline{j} + z_2 \underline{k})$$

$$\underline{V_1} \times \underline{V_2} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

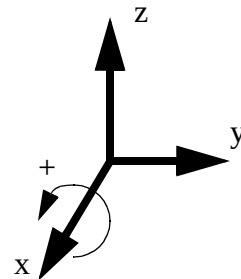
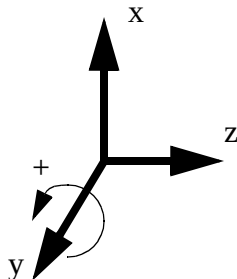
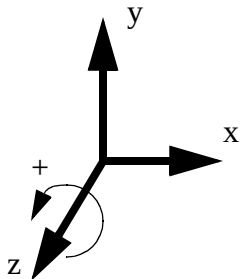
$$\underline{V_1} \times \underline{V_2} = (y_1 z_2 - z_1 y_2) \underline{i} + (z_1 x_2 - x_1 z_2) \underline{j} + (x_1 y_2 - y_1 x_2) \underline{k}$$

We can also find a unit vector normal 'n' to the vectors 'V1' and 'V2' using a cross product, divided by the magnitude.

$$\lambda_n = \frac{\underline{V_1} \times \underline{V_2}}{|\underline{V_1} \times \underline{V_2}|}$$

- When using a left/right handed coordinate system,

The positive orientation of angles and moments about an axis can be determined by pointing the thumb of the right hand along the axis of rotation. The fingers curl in the positive direction.



- The properties of the cross products are,

The cross product is distributive, but not associative. This allows us to collect terms in a cross product operation, but we cannot change the order of the cross product.

$r_1 \times F + r_2 \times F = (r_1 + r_2) \times F$	DISTRIBUTIVE
$r \times F \neq F \times r$ but	NOT ASSOCIATIVE
$r \times F = -(F \times r)$	

2.5.4 Triple Product

When we want to do a cross product, followed by a dot product (called the mixed triple product), we can do both steps in one operation by finding the determinant of the following. An example of a problem that would use this shortcut is when a moment is found about one point on a pipe, and then the moment component twisting the pipe is found using the dot product.

$$(d \times F) \cdot u = \begin{vmatrix} u_x & u_y & u_z \\ d_x & d_y & d_z \\ F_x & F_y & F_z \end{vmatrix}$$

2.5.5 Matrices

- Matrices allow simple equations that drive a large number of repetitive calculations - as a result they are found in many computer applications.
- A matrix has the form seen below,

$$\begin{array}{c}
 \longleftrightarrow \text{n columns} \\
 \begin{bmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \dots & \dots & \dots & \dots \\ a_{1m} & a_{2m} & \dots & a_{nm} \end{bmatrix} \\
 \begin{array}{c} \updownarrow \\ \text{m rows} \end{array}
 \end{array}$$

If $n=m$ then the matrix is said to be square.

Many applications require square matrices.
We may also represent a matrix as a 1-by-3
for a vector.

- Matrix operations are available for many of the basic algebraic expressions, examples are given below. There are also many restrictions - many of these are indicated.

$$A = 2 \quad B = \begin{bmatrix} 3 & 4 & 5 \\ 6 & 7 & 8 \\ 9 & 10 & 11 \end{bmatrix} \quad C = \begin{bmatrix} 12 & 13 & 14 \\ 15 & 16 & 17 \\ 18 & 19 & 20 \end{bmatrix} \quad D = \begin{bmatrix} 21 \\ 22 \\ 23 \end{bmatrix} \quad E = [24 \ 25 \ 26]$$

Addition/Subtraction

$$A + B = \begin{bmatrix} 3+2 & 4+2 & 5+2 \\ 6+2 & 7+2 & 8+2 \\ 9+2 & 10+2 & 11+2 \end{bmatrix} \quad B + C = \begin{bmatrix} 3+12 & 4+13 & 5+14 \\ 6+15 & 7+16 & 8+17 \\ 9+18 & 10+19 & 11+20 \end{bmatrix}$$

$$B + D = \text{not valid}$$

$$B - A = \begin{bmatrix} 3-2 & 4-2 & 5-2 \\ 6-2 & 7-2 & 8-2 \\ 9-2 & 10-2 & 11-2 \end{bmatrix} \quad B - C = \begin{bmatrix} 3-12 & 4-13 & 5-14 \\ 6-15 & 7-16 & 8-17 \\ 9-18 & 10-19 & 11-20 \end{bmatrix}$$

$$B - D = \text{not valid}$$

Multiplication/Division

$$A \cdot B = \begin{bmatrix} 3(2) & 4(2) & 5(2) \\ 6(2) & 7(2) & 8(2) \\ 9(2) & 10(2) & 11(2) \end{bmatrix} \quad \frac{B}{A} = \begin{bmatrix} \frac{3}{2} & \frac{4}{2} & \frac{5}{2} \\ \frac{6}{2} & \frac{7}{2} & \frac{8}{2} \\ \frac{9}{2} & \frac{10}{2} & \frac{11}{2} \end{bmatrix}$$

$$B \cdot D = \begin{bmatrix} (3 \cdot 21 + 4 \cdot 22 + 5 \cdot 23) \\ (6 \cdot 21 + 7 \cdot 22 + 8 \cdot 23) \\ (9 \cdot 21 + 10 \cdot 22 + 11 \cdot 23) \end{bmatrix} \quad D \cdot E = 21 \cdot 24 + 22 \cdot 25 + 23 \cdot 26$$

$$B \cdot C = \begin{bmatrix} (3 \cdot 12 + 4 \cdot 15 + 5 \cdot 18) & (3 \cdot 13 + 4 \cdot 16 + 5 \cdot 19) & (3 \cdot 14 + 4 \cdot 17 + 5 \cdot 20) \\ (6 \cdot 12 + 7 \cdot 15 + 8 \cdot 18) & (6 \cdot 13 + 7 \cdot 16 + 8 \cdot 19) & (6 \cdot 14 + 7 \cdot 17 + 8 \cdot 20) \\ (9 \cdot 12 + 10 \cdot 15 + 11 \cdot 18) & (9 \cdot 13 + 10 \cdot 16 + 11 \cdot 19) & (9 \cdot 14 + 10 \cdot 17 + 11 \cdot 20) \end{bmatrix}$$

$$\frac{B}{C}, \frac{B}{D}, \frac{D}{B}, \text{ etc } = \text{ not allowed (see inverse) }$$

Note: To multiply matrices, the first matrix must have the same number of columns as the second matrix has rows.

Determinant

$$|B| = 3 \cdot \begin{vmatrix} 7 & 8 \\ 10 & 11 \end{vmatrix} - 4 \cdot \begin{vmatrix} 6 & 8 \\ 9 & 11 \end{vmatrix} + 5 \cdot \begin{vmatrix} 6 & 7 \\ 9 & 10 \end{vmatrix} = 3 \cdot (-3) - 4 \cdot (-6) + 5 \cdot (-3) = 0$$

$$\begin{vmatrix} 7 & 8 \\ 10 & 11 \end{vmatrix} = (7 \cdot 11) - (8 \cdot 10) = -3$$

$$\begin{vmatrix} 6 & 8 \\ 9 & 11 \end{vmatrix} = (6 \cdot 11) - (8 \cdot 9) = -6$$

$$\begin{vmatrix} 6 & 7 \\ 9 & 10 \end{vmatrix} = (6 \cdot 10) - (7 \cdot 9) = -3$$

$$|D|, |E| = \text{ not valid (matrices not square) }$$

Transpose

$$B^T = \begin{bmatrix} 3 & 6 & 9 \\ 4 & 7 & 10 \\ 5 & 8 & 11 \end{bmatrix} \quad D^T = \begin{bmatrix} 21 & 22 & 23 \end{bmatrix} \quad E^T = \begin{bmatrix} 24 \\ 25 \\ 26 \end{bmatrix}$$

Adjoint

$$||B|| = \begin{bmatrix} \left| \begin{array}{cc} 7 & 8 \\ 10 & 11 \end{array} \right| & - \left| \begin{array}{cc} 6 & 8 \\ 10 & 11 \end{array} \right| & \left| \begin{array}{cc} 6 & 7 \\ 9 & 10 \end{array} \right| \\ - \left| \begin{array}{cc} 4 & 5 \\ 10 & 11 \end{array} \right| & \left| \begin{array}{cc} 3 & 5 \\ 9 & 11 \end{array} \right| & - \left| \begin{array}{cc} 3 & 4 \\ 9 & 10 \end{array} \right| \\ \left| \begin{array}{cc} 4 & 5 \\ 7 & 8 \end{array} \right| & - \left| \begin{array}{cc} 3 & 5 \\ 6 & 8 \end{array} \right| & \left| \begin{array}{cc} 3 & 4 \\ 6 & 7 \end{array} \right| \end{bmatrix}^T$$

The matrix of determinant to the left is made up by getting rid of the row and column of the element, and then finding the determinant of what is left. Note the sign changes on alternating elements.

$$||D|| = \text{invalid (must be square)}$$

Inverse

$$D = B \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

To solve this equation for x,y,z we need to move B to the left hand side. To do this we use the inverse.

$$B^{-1}D = B^{-1} \cdot B \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B^{-1}D = I \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B^{-1} = \frac{||B||}{|B|}$$

In this case B is singular, so the inverse is undetermined, and the matrix is indeterminate.

$$D^{-1} = \text{invalid (must be square)}$$

Identity Matrix

This is a square matrix that is the matrix equivalent to '1'.

$$B \cdot I = I \cdot B = B$$

$$D \cdot I = I \cdot D = D$$

$$B^{-1} \cdot B = I$$

$$\begin{bmatrix} 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{etc} = I$$

- The eigenvalue of a matrix is found using,

$$|A - \lambda I| = 0$$

2.5.6 Solving Linear Equations with Matrices

- We can solve systems of equations using the inverse matrix,

Given,

$$2 \cdot x + 4 \cdot y + 3 \cdot z = 5$$

$$9 \cdot x + 6 \cdot y + 8 \cdot z = 7$$

$$11 \cdot x + 13 \cdot y + 10 \cdot z = 12$$

Write down the matrix, then rearrange, and solve.

$$\begin{bmatrix} 2 & 4 & 3 \\ 9 & 6 & 8 \\ 11 & 13 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 12 \end{bmatrix} \quad \therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 4 & 3 \\ 9 & 6 & 8 \\ 11 & 13 & 10 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 7 \\ 12 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

- We can solve systems of equations using Cramer's rule (with determinants),

Given,

$$2 \cdot x + 4 \cdot y + 3 \cdot z = 5$$

$$9 \cdot x + 6 \cdot y + 8 \cdot z = 7$$

$$11 \cdot x + 13 \cdot y + 10 \cdot z = 12$$

Write down the coefficient and parameter matrices,

$$A = \begin{bmatrix} 2 & 4 & 3 \\ 9 & 6 & 8 \\ 11 & 13 & 10 \end{bmatrix} \quad B = \begin{bmatrix} 5 \\ 7 \\ 12 \end{bmatrix}$$

Calculate the determinant for A (this will be reused), and calculate the determinants for matrices below. Note: when trying to find the first parameter 'x' we replace the first column in A with B.

$$x = \frac{\begin{vmatrix} 5 & 4 & 3 \\ 7 & 6 & 8 \\ 12 & 13 & 10 \end{vmatrix}}{|A|} =$$

$$y = \frac{\begin{vmatrix} 2 & 5 & 3 \\ 9 & 7 & 8 \\ 11 & 12 & 10 \end{vmatrix}}{|A|} =$$

$$z = \frac{\begin{vmatrix} 2 & 4 & 5 \\ 9 & 6 & 7 \\ 11 & 13 & 12 \end{vmatrix}}{|A|} =$$

2.5.7 Practice Problems

1. Perform the matrix operations below.

Multiply

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 10 \\ 11 \\ 12 \end{bmatrix} =$$

Determinant

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 2 & 6 \\ 7 & 8 & 9 \end{vmatrix} =$$

Inverse

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 6 \\ 7 & 8 & 9 \end{bmatrix}^{-1} =$$

ANS.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 10 \\ 11 \\ 12 \end{bmatrix} = \begin{bmatrix} 68 \\ 167 \\ 266 \end{bmatrix}$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 2 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 36$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 6 \\ 7 & 8 & 9 \end{bmatrix}^{-1} = \begin{bmatrix} -0.833 & 0.167 & 0.167 \\ 0.167 & -0.333 & 0.167 \\ 0.5 & 0.167 & -0.167 \end{bmatrix}$$

2. Perform the vector operations below,

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Cross Product $A \times B =$ Dot Product $A \bullet B =$

ANS.

$$A \times B = (-4, 17, -10)$$

$$A \bullet B = 13$$

4. Solve the following equations using any technique,

$$5x - 2y + 4z = -1$$

$$6x + 7y + 5z = -2$$

$$2x - 3y + 6z = -3$$

ANS.

$$x = 0.273$$

$$y = -0.072$$

$$z = -0.627$$

2.6 CALCULUS

- NOTE: Calculus is very useful when looking at real systems. Many students are turned off by the topic because they "don't get it". But, the secret to calculus is to remember that there is no single "truth" - it is more a loose collection of tricks and techniques. Each one has to be learned separately, and when needed you must remember it, or know where to look.

2.6.1 Single Variable Functions

2.6.1.1 - Differentiation

- The basic principles of differentiation are,

Both u , v and w are functions of x , but this is not shown for brevity.

Also note that C is used as a constant, and all angles are in radians.

$$\frac{d}{dx}(C) = 0$$

$$\frac{d}{dx}(Cu) = (C)\frac{d}{dx}(u)$$

$$\frac{d}{dx}(u + v + \dots) = \frac{d}{dx}(u) + \frac{d}{dx}(v) + \dots$$

$$\frac{d}{dx}(u^n) = (nu^{n-1})\frac{d}{dx}(u)$$

$$\frac{d}{dx}(uv) = (u)\frac{d}{dx}(v) + (v)\frac{d}{dx}(u)$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \left(\frac{v}{v^2}\right)\frac{d}{dx}(u) - \left(\frac{u}{v^2}\right)\frac{d}{dx}(v)$$

$$\frac{d}{dx}(uvw) = (uv)\frac{d}{dx}(w) + (uw)\frac{d}{dx}(v) + (vw)\frac{d}{dx}(u)$$

$$\frac{d}{dx}(y) = \frac{d}{du}(y)\frac{d}{dx}(u) = \text{chain rule}$$

$$\frac{d}{dx}(u) = \frac{1}{\frac{d}{du}(x)}$$

$$\frac{d}{dx}(y) = \frac{\frac{d}{du}(y)}{\frac{d}{du}(x)}$$

- Differentiation rules specific to basic trigonometry and logarithm functions

$$\frac{d}{dx}(\sin u) = (\cos u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\cot u) = (-\csc u)^2 \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\cos u) = (-\sin u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\sec u) = (\tan u \sec u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\tan u) = \left(\frac{1}{\cos u}\right)^2 \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\csc u) = (-\csc u \cot u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(e^u) = (e^u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\sinh u) = (\cosh u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\cosh u) = (\sinh u) \frac{d}{dx}(u)$$

$$\frac{d}{dx}(\tanh u) = (\operatorname{sech} u)^2 \frac{d}{dx}(u)$$

- L'Hospital's rule can be used when evaluating limits that go to infinity.

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \lim_{x \rightarrow a} \left(\frac{\left(\frac{d}{dt} \right) f(x)}{\left(\frac{d}{dt} \right) g(x)} \right) = \lim_{x \rightarrow a} \left(\frac{\left(\frac{d}{dt} \right)^2 f(x)}{\left(\frac{d}{dt} \right)^2 g(x)} \right) = \dots$$

- Some techniques used for finding derivatives are,

Leibnitz's Rule, (notice the form is similar to the binomial equation) can be used for finding the derivatives of multiplied functions.

$$\left(\frac{d}{dx} \right)^n (uv) = \left(\frac{d}{dx} \right)^0 (u) \left(\frac{d}{dx} \right)^n (v) + \binom{n}{1} \left(\frac{d}{dx} \right)^1 (u) \left(\frac{d}{dx} \right)^{n-1} (v)$$

$$\binom{n}{2} \left(\frac{d}{dx} \right)^2 (u) \left(\frac{d}{dx} \right)^{n-2} (v) + \dots + \binom{n}{n} \left(\frac{d}{dx} \right)^n (u) \left(\frac{d}{dx} \right)^0 (v)$$

2.6.1.2 - Integration

- Some basic properties of integrals include,

In the following expressions, u , v , and w are functions of x . in addition to this, C is a constant. and all angles are radians.

$$\int C dx = ax + C$$

$$\int C f(x) dx = C \int f(x) dx$$

$$\int (u + v + w + \dots) dx = \int u dx + \int v dx + \int w dx + \dots$$

$$\int u dv = uv - \int v du = \text{integration by parts}$$

$$\int f(Cx) dx = \frac{1}{C} \int f(u) du \quad u = Cx$$

$$\int F(f(x)) dx = \int F(u) \frac{d}{du}(x) du = \int \frac{F(u)}{f'(x)} du \quad u = f(x)$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad \int \frac{1}{x} dx = \ln|x| + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C \quad \int e^x dx = e^x + C$$

- Some of the trigonometric integrals are,

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int (\sin x)^2 dx = -\frac{\sin x \cos x + x}{2} + C$$

$$\int (\cos x)^2 dx = \frac{\sin x \cos x + x}{2} + C$$

$$\int (\sin x)^3 dx = -\frac{\cos x((\sin x)^2 + 2)}{3} + C$$

$$\int (\cos x)^3 dx = \frac{\sin x((\cos x)^2 + 2)}{3} + C$$

$$\int x \cos(ax) dx = \frac{\cos(ax)}{a^2} + \frac{x}{a} \sin(ax) + C$$

$$\int x^2 \cos(ax) dx = \frac{2x \cos(ax)}{a^2} + \frac{a^2 x^2 - 2}{a^3} \sin(ax) + C$$

$$\int (\cos x)^4 dx = \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$$

$$\int \cos x (\sin x)^n dx = \frac{(\sin x)^{n+1}}{n+1} + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$

$$\int \tanh x dx = \ln(\cosh x) + C$$

- Some other integrals of use that are basically functions of x are,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int (a + bx)^{-1} dx = \frac{\ln(a + bx)}{b} + C$$

$$\int (a + bx^2)^{-1} dx = \frac{1}{2\sqrt{(-b)a}} \ln\left(\frac{\sqrt{a} + 2\sqrt{-b}}{\sqrt{a} - x\sqrt{-b}}\right) + C, a > 0, b < 0$$

$$\int x(a + bx^2)^{-1} dx = \frac{\ln(bx^2 + a)}{2b} + C$$

$$\int x^2(a + bx^2)^{-1} dx = \frac{x}{b} - \frac{a}{b\sqrt{ab}} \operatorname{atan}\left(\frac{x\sqrt{ab}}{a}\right) + C$$

$$\int (a^2 - x^2)^{-1} dx = \frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right) + C, a^2 > x^2$$

$$\int (a + bx)^{-1} dx = \frac{2\sqrt{a+bx}}{b} + C$$

$$\int x(x^2 \pm a^2)^{-\frac{1}{2}} dx = \sqrt{x^2 \pm a^2} + C$$

$$\int (a + bx + cx^2)^{-1} dx = \frac{1}{\sqrt{c}} \ln\left[\sqrt{a + bx + cx^2} + x\sqrt{c} + \frac{b}{2\sqrt{c}}\right] + C, c > 0$$

$$\int (a + bx + cx^2)^{-1} dx = \frac{1}{\sqrt{-c}} \operatorname{asin}\left[\frac{-2cx - b}{\sqrt{b^2 - 4ac}}\right] + C, c < 0$$

$$\int (a + bx)^{\frac{1}{2}} dx = \frac{2}{3b} (a + bx)^{\frac{3}{2}}$$

$$\int (a + bx)^{\frac{1}{2}} dx = \frac{2}{3b} (a + bx)^{\frac{3}{2}}$$

$$\int x(a + bx)^{\frac{1}{2}} dx = -\frac{2(2a - 3bx)(a + bx)^{\frac{3}{2}}}{15b^2}$$

$$\int (1 + a^2 x^2)^{\frac{1}{2}} dx = \frac{x(1 + a^2 x^2)^{\frac{1}{2}} + \frac{\ln\left(x + \left(\frac{1}{a^2} + x^2\right)^{\frac{1}{2}}\right)}{a}}{2}$$

$$\int x(1 + a^2 x^2)^{\frac{1}{2}} dx = \frac{a\left(\frac{1}{a^2} + x^2\right)^{\frac{3}{2}}}{3}$$

$$\int x^2(1 + a^2 x^2)^{\frac{1}{2}} dx = \frac{ax\left(\frac{1}{a^2} + x^2\right)^{\frac{3}{2}}}{4} - \frac{8}{8a^2} x(1 + a^2 x^2)^{\frac{1}{2}} - \frac{\ln\left(x + \left(\frac{1}{a^2} + x^2\right)^{\frac{1}{2}}\right)}{8a^3}$$

$$\int (1 - a^2 x^2)^{\frac{1}{2}} dx = \frac{1}{2} \left[x(1 - a^2 x^2)^{\frac{1}{2}} + \frac{\operatorname{asin}(ax)}{a} \right]$$

$$\int x(1 - a^2 x^2)^{\frac{1}{2}} dx = -\frac{a}{3} \left(\frac{1}{a^2} - x^2\right)^{\frac{3}{2}}$$

$$\int x^2(a^2 - x^2)^{\frac{1}{2}} dx = -\frac{x}{4}(a^2 - x^2)^{\frac{3}{2}} + \frac{1}{8} \left[x(a^2 - x^2)^{\frac{1}{2}} + a^2 \operatorname{asin}\left(\frac{x}{a}\right) \right]$$

$$\int (1 + a^2 x^2)^{-\frac{1}{2}} dx = \frac{1}{a} \ln \left[x + \left(\frac{1}{a^2} + x^2\right)^{\frac{1}{2}} \right]$$

$$\int (1 - a^2 x^2)^{-\frac{1}{2}} dx = \frac{1}{a} \operatorname{asin}(ax) = -\frac{1}{a} \operatorname{acos}(ax)$$

- Integrals using the natural logarithm base 'e',

$$\int e^{ax} dx = \frac{e^{ax}}{a} + C$$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1) + C$$

2.6.2 Vector Calculus

- When dealing with large and/or time varying objects or phenomenon we must be able to describe the state at locations, and as a whole. To do this vectors are a very useful tool.
- Consider a basic function and how it may be represented with partial derivatives.

$$y = f(x, y, z)$$

We can write this in differential form, but the right hand side must contain partial derivatives. If we separate the operators from the function, we get a simpler form. We can then look at them as the result of a dot product, and divide it into two vectors.

$$(d)y = \left(\left(\frac{\partial}{\partial x} \right) f(x, y, z) \right) dx + \left(\left(\frac{\partial}{\partial y} \right) f(x, y, z) \right) dy + \left(\left(\frac{\partial}{\partial z} \right) f(x, y, z) \right) dz$$

$$(d)y = \left[\left(\frac{\partial}{\partial x} \right) dx + \left(\frac{\partial}{\partial y} \right) dy + \left(\frac{\partial}{\partial z} \right) dz \right] f(x, y, z)$$

$$(d)y = \left[\left(\frac{\partial}{\partial x} \right) \underline{i} + \left(\frac{\partial}{\partial y} \right) \underline{j} + \left(\frac{\partial}{\partial z} \right) \underline{k} \right] \bullet (dx \underline{i} + dy \underline{j} + dz \underline{k}) f(x, y, z)$$

We then replace these vectors with the operators below. In this form we can manipulate the equation easily (whereas the previous form was very awkward).

$$(d)y = [\nabla \bullet dX] f(x, y, z)$$

$$(d)y = \nabla f(x, y, z) \bullet dX$$

$$(d)y = |\nabla f(x, y, z)| |dX| \cos \theta$$

In summary,

$$\nabla = \frac{\partial}{\partial x} \underline{i} + \frac{\partial}{\partial y} \underline{j} + \frac{\partial}{\partial z} \underline{k}$$

$$\nabla \bullet F = \text{the divergence of function } F$$

$$F = F_x \underline{i} + F_y \underline{j} + F_z \underline{k}$$

$$\nabla \times F = \text{the curl of function } F$$

- Gauss's or Green's or divergence theorem is given below. Both sides give the flux across a surface, or out of a volume. This is very useful for dealing with magnetic fields.

$$\int_V (\nabla \bullet F) dV = \oint_A F dA$$

where,

V, A = a volume V enclosed by a surface A

F = a field or vector value over a volume

- Stoke's theorem is given below. Both sides give the flux across a surface, or out of a volume. This is very useful for dealing with magnetic fields.

$$\int_A (\nabla \times F) dA = \oint_L F dL$$

where,

A, L = A surface area A, with a bounding parimeter of length L

F = a field or vector value over a volume

2.6.3 Differential Equations

- Solving differential equations is not very challenging, but there are a number of forms that need to be remembered.
- Another complication that often occurs is that the solution of the equations may vary depending upon boundary or initial conditions. An example of this is a mass spring combination. If they are initially at rest then they will stay at rest, but if there is some disturbance, then they will oscillate indefinitely.
- We can judge the order of these equations by the highest order derivative in the equation.
- Note: These equations are typically shown with derivatives only, when integrals occur they are typically eliminated by taking derivatives of the entire equation.
- Some of the terms used when describing differential equations are,
ordinary differential equations - if all the derivatives are of a single variable. In the example below 'x' is the variable with derivatives.

$$\text{e.g., } \left(\frac{d}{dt}\right)^2 x + \left(\frac{d}{dt}\right)x = y$$

first order differential equations - have only first order derivatives,

$$\text{e.g., } \left(\frac{d}{dt}\right)x + \left(\frac{d}{dt}\right)y = 2$$

second order differential equations - have at least on second derivative,

$$\text{e.g., } \left(\frac{d}{dt}\right)^2 x + \left(\frac{d}{dt}\right)y = 2$$

higher order differential equations - have at least one derivative that is higher than second order.

partial differential equations - these equations have partial derivatives

- Note: when solving these equations it is common to hit blocks. In these cases backtrack and try another approach.
- linearity of a differential equation is determined by looking at the dependant variables in the equation. The equation is linear if they appear with an exponent other than 1.

eg.	$y'' + y' + 2 = 5x$	linear
	$(y'')^2 + y' + 2 = 5x$	non-linear
	$y'' + (y')^3 + 2 = 5x$	non-linear
	$y'' + \sin(y') + 2 = 5x$	non-linear

2.6.3.1 - First Order Differential Equations

- These systems tend to have a relaxed or passive nature in real applications.
- Examples of these equations are given below,

$$y' + 2xy^2 - 4x^3 = 0$$

$$y' - 2y = 0$$

- Typical methods for solving these equations include,
 guessing then testing
 separation
 homogeneous

2.6.3.1.1 - Guessing

- In this technique we guess at a function that will satisfy the equation, and test it to see if it works.

$$y' + y = 0 \quad \text{the given equation}$$

$$y = Ce^{-t} \quad \text{the guess}$$

now try to substitute into the equation

$$y' = -Ce^{-t}$$

$$y' + y = -Ce^{-t} + Ce^{-t} = 0 \quad \text{therefore the guess worked - it is correct}$$

$y = Ce^{-t}$

- The previous example showed a general solution (i.e., the value of 'C' was not found). We can also find a particular solution.

$$y = Ce^{-t} \quad \text{a general solution}$$

$$y = 5e^{-t} \quad \text{a particular solution}$$

2.6.3.1.2 - Separable Equations

- In a separable equation the differential can be split so that it is on both sides of the equation. We then integrate to get the solution. This typically means there is only a single derivative term.

e.g., $\frac{dx}{dy} + y^2 + 2y + 3 = 0$

$$\therefore dx = (-y^2 - 2y - 3)dy$$

$$\therefore x = \frac{-y^3}{3} - y^2 - 3y + C$$

e.g., $\frac{dx}{dy} + x = 0$

$$\therefore \left(-\frac{1}{x}\right)dx = dy$$

$$\therefore \ln(-x) = y$$

2.6.3.1.3 - Homogeneous Equations and Substitution

- These techniques depend upon finding some combination of the variables in the equation that can be replaced with another variable to simplify the equation. This technique requires a bit of guessing about what to substitute for, and when it is to be applied.

e.g., $\frac{dy}{dx} = \frac{y}{x} - 1$ the equation given

$u = \frac{y}{x}$ the substitution chosen

Put the substitution in and solve the differential equation,

$$\frac{dy}{dx} = u - 1$$

$$\therefore u + x \frac{du}{dx} = u - 1$$

$$\therefore \frac{du}{dx} = \frac{-1}{x}$$

$$\therefore -\frac{du}{dx} = \frac{1}{x}$$

$$\therefore -u = \ln(x) + C$$

Substitute the results back into the original substitution equation to get rid of 'u',

$$-\frac{y}{x} = \ln(x) + C$$

$$\therefore y = -x \ln(x) - Cx$$

2.6.3.2 - Second Order Differential Equations

- These equations have at least one second order derivative.
- In engineering we will encounter a number of forms,
 - homogeneous
 - nonhomogeneous

2.6.3.2.1 - Linear Homogeneous

- These equations will have a standard form,

$$\left(\frac{d}{dt}\right)^2 y + A\left(\frac{d}{dt}\right)y + B = 0$$

- An example of a solution is,

e.g., $\left(\frac{d}{dt}\right)^2 y + 6\left(\frac{d}{dt}\right)y + 3 = 0$

Guess,

$$y = e^{Bt}$$

$$\left(\frac{d}{dt}\right)y = Be^{Bt}$$

$$\left(\frac{d}{dt}\right)^2 y = B^2 e^{Bt}$$

substitute and solve for B,

$$B^2 e^{Bt} + 6Be^{Bt} + 3e^{Bt} = 0$$

$$B^2 + 6B + 3 = 0$$

$$B = -3 + 2.449j, -3 - 2.449j$$

substitute and solve for B,

$$y = e^{(-3 + 2.449j)t}$$

$$y = e^{-3t} e^{2.449jt}$$

$$y = e^{-3t} (\cos(2.449t) + j \sin(2.449t))$$

Note: if both the roots are the same,

$$y = C_1 e^{Bt} + C_2 t e^{Bt}$$

2.6.3.2.2 - Nonhomogeneous Linear Equations

- These equations have the general form,

$$\left(\frac{d}{dt}\right)^2 y + A\left(\frac{d}{dt}\right)y + By = Cx$$

- to solve these equations we need to find the homogeneous and particular solutions and then add the two solutions.

$$y = y_h + y_p$$

to find y_h solve,

$$\left(\frac{d}{dt}\right)^2 y + A\left(\frac{d}{dt}\right)y + B = 0$$

to find y_p guess at a value of y and then test for validity, A good table of guesses is,

Cx form	Guess
A	C
$Ax + B$	$Cx + D$
e^{Ax}	$Ce^{Ax} \quad Cxe^{Ax}$
$B\sin(Ax) \quad \text{or} \quad B\cos(Ax)$	$C\sin(Ax) + D\cos(Ax)$ or $Cx\sin(Ax) + xD\cos(Ax)$

- Consider the example below,

$$\left(\frac{d}{dt}\right)^2 y + \left(\frac{d}{dt}\right)y - 6y = e^{-2x}$$

First solve for the homogeneous part,

$$\left(\frac{d}{dt}\right)^2 y + \left(\frac{d}{dt}\right)y - 6y = 0 \quad \text{try} \quad y = e^{Bx}$$

$$\left(\frac{d}{dt}\right)y = Be^{Bx}$$

$$\left(\frac{d}{dt}\right)^2 y = B^2 e^{Bx}$$

$$B^2 e^{Bx} + Be^{Bx} - 6e^{Bx} = 0$$

$$B^2 + B - 6 = 0$$

$$B = -3, 2$$

$$y_h = e^{-3x} + e^{2x}$$

Next, solve for the particular part. We will guess the function below.

$$y = Ce^{-2x}$$

$$\left(\frac{d}{dt}\right)y = -2Ce^{-2x}$$

$$\left(\frac{d}{dt}\right)^2 y = 4Ce^{-2x}$$

$$4Ce^{-2x} + -2Ce^{-2x} - 6Ce^{-2x} = e^{-2x}$$

$$4C - 2C - 6C = 1$$

$$C = 0.25$$

$$y_p = 0.25e^{-2x}$$

Finally,

$$y = e^{-3x} + e^{2x} + 0.25e^{-2x}$$

2.6.3.3 - Higher Order Differential Equations

2.6.3.4 - Partial Differential Equations

- Partial difference equations become critical with many engineering applications involving flows, etc.

2.6.4 Other Calculus Stuff

- The Taylor series expansion can be used to find polynomial approximations of functions.

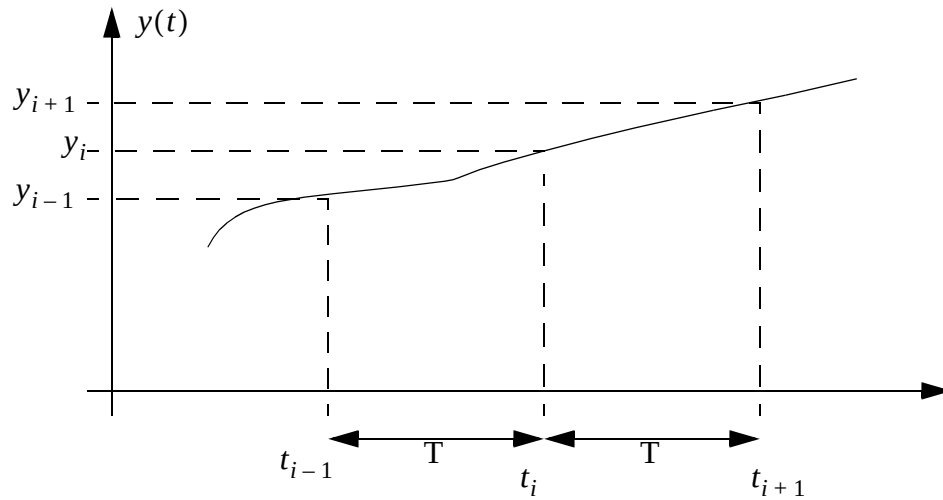
$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n-1)}(a)(x-a)^{n-1}}{(n-1)!}$$

2.7 NUMERICAL METHODS

- These techniques approximate system responses without doing integrations, etc.

2.7.1 Approximation of Integrals and Derivatives from Sampled Data

- This form of integration is done numerically - this means by doing repeated calculations to solve the equation. Numerical techniques are not as elegant as solving differential equations, and will result in small errors. But these techniques make it possible to solve complex problems much faster.
- This method uses forward/backward differences to estimate derivatives or integrals from measured data.



$$\int_{t_{i-1}}^{t_i} y(t) dt \approx \left(\frac{y_i + y_{i-1}}{2} \right) (t_i - t_{i-1}) = \frac{T}{2} (y_i + y_{i-1})$$

$$\frac{d}{dt} y(t_i) \approx \left(\frac{y_i - y_{i-1}}{t_i - t_{i-1}} \right) = \left(\frac{y_{i+1} - y_i}{t_{i+1} - t_i} \right) = \frac{1}{T} (y_i - y_{i-1}) = \frac{1}{T} (y_{i+1} - y_i)$$

$$\left(\frac{d}{dt} \right)^2 y(t_i) \approx \frac{\frac{1}{T} (y_{i+1} - y_i) - \frac{1}{T} (y_i - y_{i-1})}{T} = \frac{-2y_i + y_{i-1} + y_{i+1}}{T^2}$$

2.7.2 Euler First Order Integration

- We can also estimate the change resulting from a derivative using Euler's equation for a first order difference equation.

$$y(t + h) \approx y(t) + h \frac{d}{dt} y(t)$$

2.7.3 Taylor Series Integration

- Recall the basic Taylor series,

$$x(t+h) = x(t) + h\left(\frac{d}{dt}\right)x(t) + \frac{1}{2!}h^2\left(\frac{d}{dt}\right)^2x(t) + \frac{1}{3!}h^3\left(\frac{d}{dt}\right)^3x(t) + \frac{1}{4!}h^4\left(\frac{d}{dt}\right)^4x(t) + \dots$$

- When $h=0$ this is called a MacLaurin series.
- We can integrate a function by,

$$\left(\frac{d}{dt}\right)x = 1 + x^2 + t^3$$

$$\left(\frac{d}{dt}\right)x_0 = 0$$

$$x_0 = 0$$

$$h = 0.1$$

t (s)	x(t)	d/dt x(t)
0	0	0
0.1		
0.2		
0.3		
0.4		
0.5		
0.6		
0.7		
0.8		
0.9		

2.7.4 Runge-Kutta Integration

- The equations below are for calculating a fourth order Runge-Kutta integration.

$$x(t+h) = x(t) + \frac{1}{6}(F_1 + 2F_2 + 2F_3 + F_4)$$

$$F_1 = hf(t, x)$$

$$F_2 = hf\left(t + \frac{h}{2}, x + \frac{F_1}{2}\right)$$

$$F_3 = hf\left(t + \frac{h}{2}, x + \frac{F_2}{2}\right)$$

$$F_4 = hf(t+h, x+F_3)$$

where,

x = the state variables

f = the differential function

t = current point in time

h = the time step to the next integration point

2.7.5 Newton-Raphson to Find Roots

- When given an equation where an algebraic solution is not feasible, a numerical solution may be required. One simple technique uses an instantaneous slope of the function, and takes iterative steps towards a solution.

$$x_{i+1} = x_i - \frac{f(x_i)}{\left(\frac{d}{dx}f(x_i)\right)}$$

- The function f(x) is supplied by the user.
- This method can become divergent if the function has an inflection point near the root.
- The technique is also sensitive to the initial guess.
- This calculation should be repeated until the final solution is found.

2.8 LAPLACE TRANSFORMS

- The Laplace transform allows us to reverse time. And, as you recall from before the inverse of time is frequency. Because we are normally concerned with response, the Laplace transform is much more useful in system analysis.
- The basic Laplace transform equations is shown below,

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

where,

$f(t)$ = the function in terms of time t

$F(s)$ = the function in terms of the Laplace s

2.8.1 Laplace Transform Tables

- Basic Laplace Transforms for operational transformations are given below,

TIME DOMAIN	FREQUENCY DOMAIN
$Kf(t)$	$Kf(s)$
$f_1(t) + f_2(t) - f_3(t) + \dots$	$f_1(s) + f_2(s) - f_3(s) + \dots$
$\frac{df(t)}{dt}$	$sf(s) - f(0^-)$
$\frac{d^2f(t)}{dt^2}$	$s^2f(s) - sf(0^-) - \frac{df(0^-)}{dt}$
$\frac{d^nf(t)}{dt^n}$	$s^nf(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} - \dots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}$
$\int_0^t f(t)dt$	$\frac{f(s)}{s}$
$f(t-a)u(t-a), a > 0$	$e^{-as}f(s)$
$e^{-at}f(t)$	$f(s-a)$
$f(at), a > 0$	$\frac{1}{a}f\left(\frac{s}{a}\right)$
$tf(t)$	$-\frac{df(s)}{ds}$
$t^n f(t)$	$(-1)^n \frac{d^n f(s)}{ds^n}$
$\frac{f(t)}{t}$	$\int_s^\infty f(u)du$

- A set of useful functional Laplace transforms are given below,

TIME DOMAIN	FREQUENCY DOMAIN
A	$\frac{A}{s}$
t	$\frac{1}{s^2}$
e^{-at}	$\frac{1}{s+a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
te^{-at}	$\frac{1}{(s+a)^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$
Ae^{-at}	$\frac{A}{s-a}$
Ate^{-at}	$\frac{A}{(s-a)^2}$
$2 A e^{-\alpha t} \cos(\beta t + \theta)$	$\frac{A}{s + \alpha - \beta j} + \frac{A^{\text{complex conjugate}}}{s + \alpha + \beta j}$
$2t A e^{-\alpha t} \cos(\beta t + \theta)$	$\frac{A}{(s + \alpha - \beta j)^2} + \frac{A^{\text{complex conjugate}}}{(s + \alpha + \beta j)^2}$

- Laplace transforms can be used to solve differential equations.

2.9 z-TRANSFORMS

- For a discrete-time signal $x[n]$, the two-sided z-transform is defined by $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$

The one-sided z-transform is defined by $X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$. In both cases, the z-transform is a polynomial in the complex variable z .

- The inverse z-transform is obtained by contour integration in the complex plane

$x[n] = \frac{1}{j2\pi} \oint X(z)z^{n-1}dz$. This is usually avoided by partial fraction inversion techniques, similar to the Laplace transform.

- Along with a z-transform we associate its region of convergence (or ROC). These are the values of z for which $X(z)$ is bounded (i.e., of finite magnitude).

- Some common z-transforms are shown below.

Table 1: Common z-transforms

Signal $x[n]$	z-Transform $X(z)$	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$nu[n]$	$\frac{z^{-1}}{(1 - z^{-1})^2}$	$ z > 1$
$n^2u[n]$	$\frac{z^{-1}(1 + z^{-1})}{(1 - z^{-1})^3}$	$ z > 1$
$a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$(-a^n)u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$(-na^n)u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$\cos(\omega_0 n)u[n]$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
$\sin(\omega_0 n)u[n]$	$\frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
$a^n \cos(\omega_0 n)u[n]$	$\frac{1 - az^{-1} \cos \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

Table 1: Common z-transforms

Signal $x[n]$	z-Transform $X(z)$	ROC
$a^n \sin(\omega_0 n) u[n]$	$\frac{az^{-1} \sin \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
$\frac{n!}{k!(n-k)!} u[n]$	$\frac{z^{-k}}{(1 - z^{-1})^{k+1}}$	$ z > 1$

- The z-transform also has various properties that are useful. The table below lists properties for the two-sided z-transform. The one-sided z-transform properties can be derived from the ones below by considering the signal $x[n]u[n]$ instead of simply $x[n]$.

Table 2: Two-sided z-Transform Properties

Property	Time Domain	z-Domain	ROC
Notation	$x[n]$ $x_1[n]$ $x_2[n]$	$X(z)$ $X_1(z)$ $X_2(z)$	$r_2 < z < r_1$ ROC_1 ROC_2
Linearity	$\alpha x_1[n] + \beta x_2[n]$	$\alpha X_1(z) + \beta X_2(z)$	At least the intersection of ROC_1 and ROC_2
Time Shifting	$x[n - k]$	$z^{-k} X(z)$	That of $X(z)$, except $z = 0$ if $k > 0$ and $z = \infty$ if $k < 0$
z-Domain Scaling	$a^n x[n]$	$X(a^{-1} z)$	$ a r_2 < z < a r_1$
Time Reversal	$x[-n]$	$X(z^{-1})$	$\frac{1}{r_1} < z < \frac{1}{r_2}$
z-Domain Differentiation	$nx[n]$	$-z \frac{dX(z)}{dz}$	$r_2 < z < r_1$

Table 2: Two-sided z-Transform Properties

Property	Time Domain	z-Domain	ROC
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least the intersection of ROC_1 and ROC_2
Multiplication	$x_1[n]x_2[n]$	$\frac{1}{j2\pi} \oint X_1(v)X_2\left(\frac{z}{v}\right)v^{-1}dv$	At least $r_{1l}r_{2l} < z < r_{1u}r_{2u}$
Initial value theorem	$x[n]$ causal	$x[0] = \lim_{z \rightarrow \infty} X(z)$	

2.10 FOURIER SERIES

- These series describe functions by their frequency spectrum content. For example a square wave can be approximated with a sum of a series of sine waves with varying magnitudes.
- The basic definition of the Fourier series is given below.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

2.11 TOPICS NOT COVERED (YET)

- To ensure that the omissions are obvious, I provide a list of topics not covered below. Some of these may be added later if their need becomes obvious.
 - Frequency domain - Fourier, Bessel

2.12 REFERENCES/BIBLIOGRAPHY

Spiegel, M. R., Mathematical Handbook of Formulas and Tables, Schaum's Outline Series, McGraw-Hill Book Company, 1968.

3. WRITING REPORTS

- As engineers, reports are the most common form of document we will write.
- Report writing is an art that we often overlook, but in many cases can make a dramatic impact on our progression.
- Your reports are most likely to find their way to a superiors desk than you are to meet the individual.
- Note: typically these documents are done as a long written document -this is done in point form for more mercenary reasons.

3.1 WHY WRITE REPORTS?

- Reports are written for a number of reasons,
 - (lets forget about this one) as a student you must do them to get marks
 - to let other engineers know the results of an experiment
 - to leave a record of work done so that others may continue on
 - as a record you may use yourself if you must do work again some time later
 - they are required for legal reasons (contract or legislation)
 - they bring closure to the project

3.2 THE TECHNICAL DEPTH OF THE REPORT

- This is the most common question for beginning report writers.
- Always follow the doctrine - “What happens if I am hit by a car; could another pick up my report and continue?” if the answer to this question is no the report is too short.
- Always ask the question - “If I was reading this before starting the project would I look at this section and say it is not needed?”. If the question we are probably best to condense the section or leave it out.
- Avoid more artistic sections. If you put these in, make it clear that they are an optional part of the report and can be skipped. It is somewhat arrogant to force the reader to

3.3 TYPES OF REPORTS

- We do different types of reports, including,
 - Laboratory - Theses 'Lab Reports' describe one or more experiments, the results, and the conclusions drawn from them.
 - Consulting - A summary of details, test results, observations, and a set of conclusions. Typically they will also contain a recommendation.
 - Project - A description of work done in a project to inform other engineers who may be asked to take up further work on the project.
 - Research - A summary of current advances in a topic. This should end with some comparison of alternatives.
 - Interim - A report to apprise supervisors and others as to the progress of a project or other major undertaking.
 - Executive - A brief summary of the report, and any implications for decision making at the management levels.

3.3.1 Laboratory

- Purpose: These reports should outline your procedure and results in detail. They should also contain the analysis and conclusions. The completeness of detail allows you (and others) to review these and verify the correctness of what has been done. These have been historically used for hundreds of years and are accepted as a form of scientific and legal evidence. It is completely unacceptable to make incorrect entries or leave out important steps or data.
- Standard Format:
 1. Title, Author, Date - these make it clear what the labs contain, who did the work, and when it was done.
 2. Purpose - a brief one line statement that allows a quick overview of what the experiment is about. This is best written in the form of a scientific goal using the scientific methods.
 3. Theory - a review of applicable theory and calculations necessary. Any design work is done at this stage
 4. Equipment - a list of the required equipment will help anybody trying to replicate the procedure. Specific identifying numbers should be listed when possible. If there are problems in the data, or an instrument is found to be out of calibration, we can track the problems to specific sets of data and equipment.
 5. Procedure - these are sequential operations that describe what was done during the experiment. The level of detail should be enough that somebody else could replicate the procedure. We want to use this as a scientific protocol.
 6. Results (Note: sometimes procedure and results are mixed) - the results are recorded in tables, graphs, etc. as appropriate. It will also be very helpful to note other events that occur (e.g. power loss, high humidity, etc.)

7. Discussion - At this stage the results are reviewed for trends and other observations. At this point we want to consider the scientific method.
 8. Conclusions - To conclude we will summarize the significant results, and make general statements either upholding or rejecting our purpose.
- Style: These are meant to be written AS the work is done. As a result the work should be past tense
 - Laboratory reports should have one or more hypotheses that are to be tested. If testing designs these are the specifications. Examples might be,
 - what is the thermal capacity of a material?
 - what is the bandwidth of an amplifier?
 - will the counter increment/decrement between 0 to 9?
 - NOTE: These reports are much easier to write if you prepare all of the calculations, graphs, etc. before you start to write. If you sit down and decide to do things as you write it will take twice as long and get you half the marks..... believe me, I have written many in the past and I mark them now.

3.3.1.1 - An Example First Draft of a Report

Grand Valley State University
Padnos School of Engineering
EGR 345 Dynamics Systems Modelling and Control
Laboratory Exercise 7

Title: The Cooling of Coffee
Author: I. M. Wyred
Date: Dec., 23, 1998

Purpose: To derive a theoretical model of the rate at which coffee cools and experimentally verify the model and find coefficients.

Theory:

When coffee is heated kinetic energy is added, when coffee is cooled kinetic energy is removed.

In a typical use, coffee cools as heat is lost through convection and conduction to the air and solids in contact. The factors involved in this convection/conduction can be difficult to measure directly, but we can approximate them with a simple thermal resistance. Consider the temperature difference between the coffee and the ambient temperature. The greater the temperature difference, the higher the rate of heat flow out of the coffee. This relationship can be seen formally in the equation below. We can also assume that the atmosphere is so large that the heat transfer will not change the temperature.

$$q = \frac{1}{R}(\theta_{coffee} - \theta_{air})$$

where,

q = heat flow rate from coffee to air (J/s)

R = thermal resistance between air and coffee

θ = temperatures in the coffee and air

We can also consider that coffee has a certain thermal capacity for the heat energy. As the amount of energy rises, there will be a corresponding temperature increase. This is known as the thermal capacitance, and this value is unique for every material. The basic relationships are given below. I will assume that the energy flow rate into the coffee is negligible.

$$\Delta\theta_{coffee} = \frac{1}{C_{coffee}}(q_{in} - q_{out}) = \frac{-1}{C_{coffee}}q \quad \frac{d\theta_{coffee}}{dt} = \frac{-1}{C_{coffee}}q$$

where,

C_{coffee} = thermal capacitance

$C_{coffee} = M_{coffee}\sigma_{coffee}$

where,

M_{coffee} = mass of thermal body

σ_{coffee} = specific heat of material in mas

The temperatures can be found by consider that the energy flowing out of the cup, and into the atmosphere is governed by the resistance. And, the temperature in the coffee and air are governed by the two capacitances. We will make two assumptions, that the thermal capacitance of the atmosphere is infinite, and that there is no energy flowing into the coffee.

$$\frac{d\theta_{coffee}}{dt} = \frac{-1}{C_{coffee}}q$$

$$\therefore \frac{d\theta_{coffee}}{dt} = \frac{-1}{M_{coffee}\sigma_{coffee}} \left(\frac{1}{R}(\theta_{coffee} - \theta_{air}) \right)$$

$$\therefore \frac{d\theta_{coffee}}{dt} + \left(\frac{1}{M_{coffee}\sigma_{coffee}R} \right) \theta_{coffee} = \left(\frac{1}{M_{coffee}\sigma_{coffee}R} \right) \theta_{air}$$

This differential equation can then be solved to find the temperature as a function of time.

Guess $\theta = A + Be^{Ct}$ $\frac{d}{dt}\theta = BCe^{Ct}$

$$\therefore BCe^{Ct} + \left(\frac{1}{M_{coffee}\sigma_{coffee}R}\right)(A + Be^{Ct}) = \left(\frac{1}{M_{coffee}\sigma_{coffee}R}\right)\theta_{air}$$

$$\therefore e^{Ct}\left(BC + \frac{B}{M_{coffee}\sigma_{coffee}R}\right) + \left(\frac{A}{M_{coffee}\sigma_{coffee}R} + \left(\frac{-1}{M_{coffee}\sigma_{coffee}R}\right)\theta_{air}\right) = 0$$

$$BC + \frac{B}{M_{coffee}\sigma_{coffee}R} = 0 \quad C = \frac{-1}{M_{coffee}\sigma_{coffee}R}$$

$$\frac{A}{M_{coffee}\sigma_{coffee}R} + \left(\frac{-1}{M_{coffee}\sigma_{coffee}R}\right)\theta_{air} = 0 \quad A = \theta_{air}$$

To find B, the initial temperature of the coffee should be used,

$$\theta_0 = A + Be^{C(0)} = \theta_{air} + B \quad B = \theta_0 - \theta_{air}$$

The final equation is,

$$\theta = \theta_{air} + (\theta_0 - \theta_{air})e^{\frac{-t}{M_{coffee}\sigma_{coffee}R}}$$

The time constant of this problem can be taken from the differential equation above.

$$\tau = M_{coffee}\sigma_{coffee}R$$

Equipment:

- 1 ceramic coffee cup (14 oz.)
- 2 oz. ground coffee
- 1 coffee maker - Proctor Silex Model 1234A
- 1 thermocouple (gvsu #632357)
- 1 temperature meter (gvsu #234364)
- 1 thermometer
- 2 quarts of tap water
- 1 standard #2 coffee filter
- 1 clock with second hand
- 1 small scale (gvsu# 63424)

Procedure:

1. The coffee pot was filled with water and this was put into the coffee maker. The coffee filter and grounds were put into the machine, and the machine was turned on. After five minutes approximately the coffee was done, and the pot was full.
2. The mass of the empty coffee cup was measured on the scale and found to be 214g.
3. The air temperature in the room was measured with the thermometer and found to be 24C. The temperature of the coffee in the pot was measured using the thermocouple and temperature meter and found to be 70C.
4. Coffee was poured into the cup and, after allowing 1 minute for the temperature to equalize, the temperature was measured again. The temperature was 65C. Readings of the coffee temperature were taken every 10 minutes for the next 60 minutes. These values were recorded in Table 1 below. During this period the cup was left on a table top and allowed to cool in the ambient air temperature. During this period the mass of the full coffee cup was measured and found to be 478g.

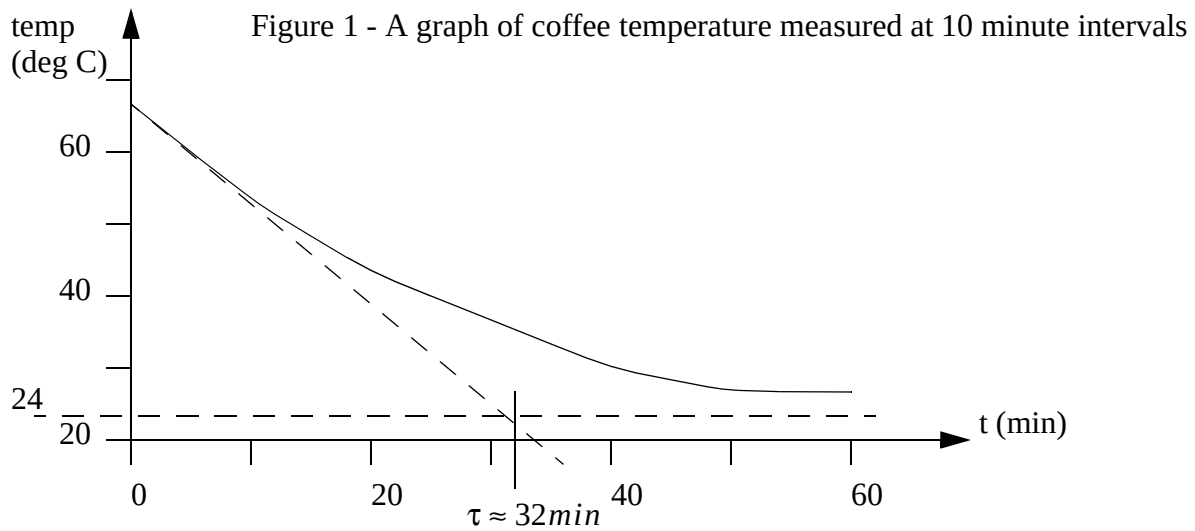
Table 1: Coffee temperatures at 10 minute intervals

time (min)	temperature (deg C)
0	65
10	53
20	43
30	35
40	30
50	28
60	26

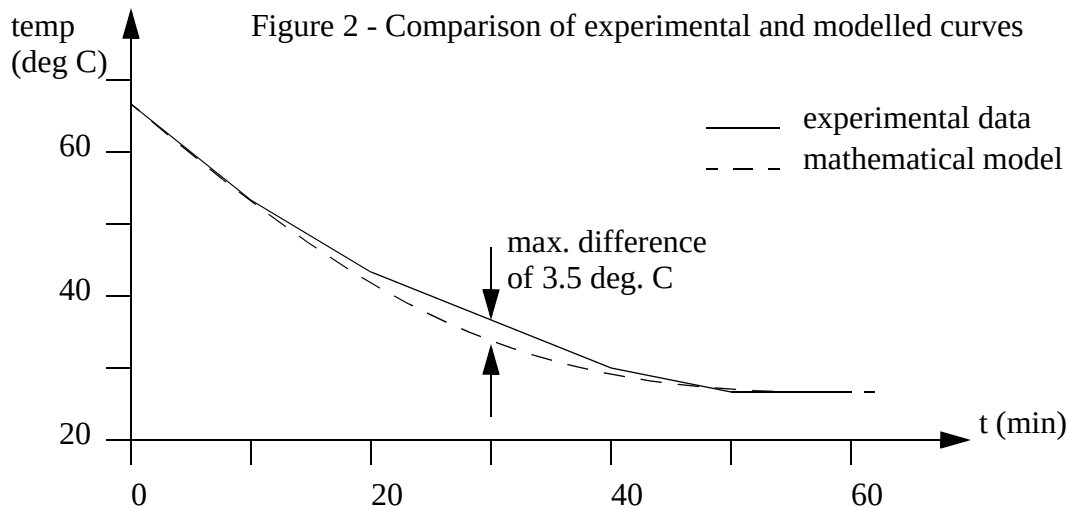
Results:

The difference between the temperature of the coffee in the pot and in the cup was 5C. This indicates that some of the heat energy in the coffee was lost to heating the cup. This change is significant, but I will assume that the heating of the cup was complete within the first minute, and this will have no effect on the data collected afterwards.

The readings for temperature over time are graphed in Figure 1 below. These show the first order response as expected, and from these we can graphically estimate the time constant at approximately 32 minutes.



We can compare the theoretical and experimental models by using plotting both on the same graph. The graph clearly shows that there is good agreement between the two curves, except for the point at 30 minutes, where there is a difference of 3.5 degrees C.



This gives an overall error of 8.5% between these two curves, compared to the total range of the data.

$$\text{error} = \frac{3.5}{65 - 24} 100 = 8.5\%$$

Finally, the results can be used to calculate a thermal resistance. If we know the mass of the coffee and assume that the coffee has the same specific heat as water, and have the time constant, the thermal resistance is found to be 1731sC/J.

$$\tau = M_{coffee} \sigma_{coffee} R$$

$$R = \frac{\tau}{M_{coffee} \sigma_{coffee}} = 1731 \frac{sC}{J}$$

$$M_{coffee} = 478g - 214g = 0.264Kg$$

$$\sigma_{coffee} = 4.2 \frac{C}{KgJ}$$

$$\tau = 32min = 1920s$$

Conclusion:

In general the models agreed well, except for a single data point. This error was relatively small, only being 8.5% of the entire data range. This error was most likely caused by a single measurement error. The error value is greater than the theoretical value, which suggests that the temperature might have been read at a "hot spot". In the procedure the temperature measuring location was not fixed, which probably resulted in a variation in measurement location.

3.3.1.2 - An Example Final Draft of a Report

- A final draft of the report is available on the course website in Mathcad format, and it will be distributed in the lab.

3.3.2 Research

- Purpose: After looking at a technical field we use these reports to condense the important details and differences. After reading a research report another reader should be able to discuss advanced topics in general terms.
- Strategy A:
 1. Clearly define the objectives for the report
 2. Outline what you know on a word processor in point form and find the 'holes'
 3. Do research to find the missing information
 4. Incorporate the new and old information (still in point form)
 5. Rearrange the points into a logical structure
 6. Convert point form into full text
 7. Proof read and edit

3.3.3 Project

- Purpose: These reports allow the developer or team to document all of the design decisions

made during the course of the project. This report should also mention avenues not taken. Quite often the projects that we start will be handed off to others after a period of time. In many cases they will not have the opportunity to talk to us, or we may not have the time. These reports serve as a well known, central document that gathers all relevant information.

- Strategy A:

1. Define the goals for the project clearly in point form
2. Examine available options and also add these in point form
3. Start to examine engineering aspects of the options
4. Make engineering decisions, and add point form to the document
5. As work continues on the project add notes and figures
6. When the project is complete, convert the point form to full text.
7. Proof read and edit

3.3.4 Executive

- Purpose: These reports condense long topics into a very brief document, typically less than one page in length. Basically these save a manager from having to read a complete report to find the details that interest him/her.

3.3.5 Consulting

- Purpose: These reports are typically commissioned by an independent third party to review a difficult problem. The consultant will review the details of the problem, do tests as required, and summarize the results. The report typically ends with conclusions, suggestions or recommendations.

3.3.6 Interim

- Purpose: This report is normally a formal report to track the progress of a project. When a project is initially planned, it will be given a timeline to follow. The interim report will indicate progress relative to the initial timeline, as well as major achievements and problems.

3.4 ELEMENTS

- In reports we must back up our opinions with data, equations, drawings, etc. As a result we use a number of common items,
 - figures
 - tables
 - equations
 -
- When these elements are included, there **MUST** be a mention of them in the written text.
- These days it is common to cut and paste figures in software. Make sure
 - the resolution is appropriate
 - the colors print properly in the final form or print well as black and white
 - the smallest features are visible
 - scanned drawings are clean and cropped to size
 - scanned photographs are clear and cropped to size
 - digital photographs should be properly lit, and cropped to size
 - screen captures are clipped to include only relevant data

3.4.1 Figures

- Figures include drawings, schematics, graphs, charts, etc.
- They should be labelled underneath sequentially and given a brief title to distinguish it from other graphs. For example “Figure 1 -Voltage and currents for 50 ohm resistor”
- In the body of the report the reference may be shortened to ‘Fig. 1’
- The figures do not need to immediately follow the reference, but they should be kept in sequence. We will often move figures to make the type setting work out better.
- If drawing graphs by computer,
 - if fitting a line/curve to the points indicate the method used (e.g. linear regression)
 - try not to use more than 5 curves on the same graph
 - use legends that can be seen in black and white
 - clearly label units and scales
 - label axes with descriptive term. For example “Hardness (RHC)” instead of “RHC”
 - scale the curve to make good use of the graph
 - avoid overly busy graphs

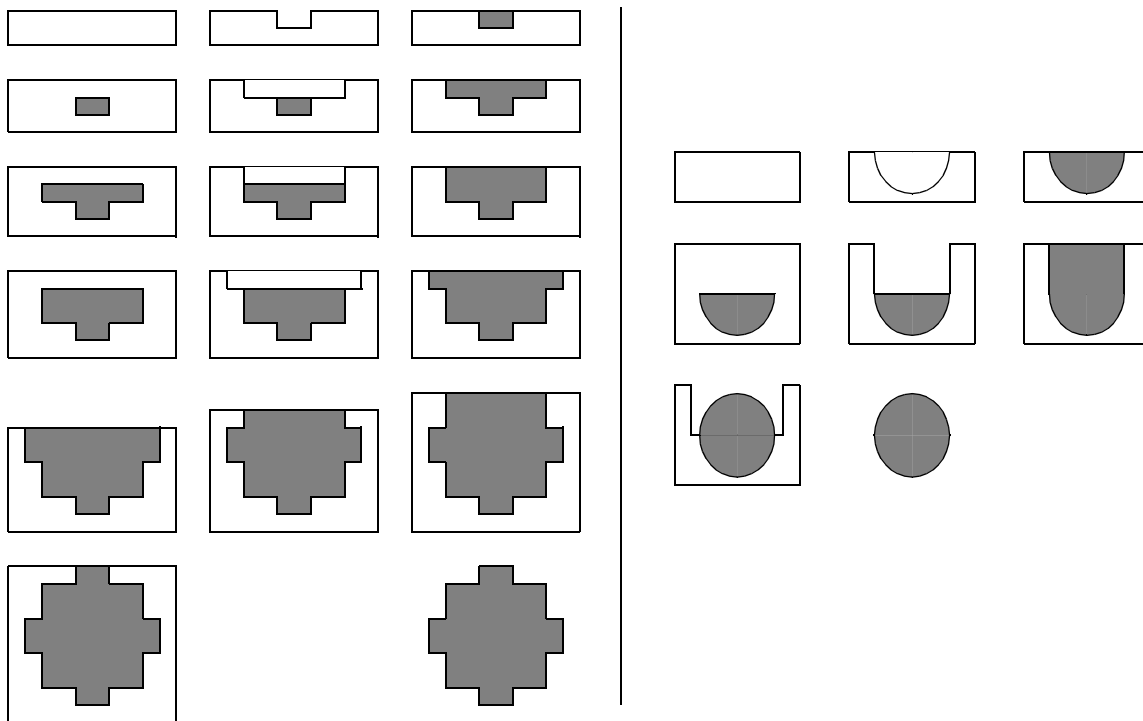


Figure 2 - Various Techniques for Making a Sphere with AMP

3.4.2 Tables

- Tables are often treated as figures.
- They allow dense information presentation, typically numerical in nature.

Table 3: A Comparison of Toy Vehicle Properties

Description	Number	Color	Shape	Material
car	3	red	rectangular	die cast
truck	6	blue	long	polyprop.
motorcycle	2	green	small	wood

- Legends can be added to tables to help condense size.

3.4.3 Equations

- When presenting equations, use a good equation editor, and watch to make sure subscripts, etc are visible.
- Number equations that are referred to in the text.
- Box in equations of great significance.

$$\overset{+}{\rightarrow} \sum F_x = -T_1 \sin 60^\circ + F_R \sin \theta_R = 0 \quad (1)$$

$$\overset{+}{\uparrow} \sum F_y = -T_1 - T_1 \cos(60^\circ) + F_R \cos \theta_R = 0 \quad (2)$$

$$\therefore F_R = \frac{T_1 \sin 60^\circ}{\sin \theta_R} = \frac{T_1 + T_1 \cos 60^\circ}{\cos \theta_R} \quad \text{sub (1) into (2)}$$

$$\therefore \frac{\sin 60^\circ}{1 + \cos 60^\circ} = \frac{\sin \theta_R}{\cos \theta_R} = \tan \theta_R$$

$$\therefore \tan \theta_R = \frac{0.866}{1 + 0.5}$$

$$\boxed{\therefore \theta_R = 30^\circ}$$

$$98 \sin 60^\circ = F_R \sin 30^\circ$$

$$\boxed{\therefore F_R = 170N}$$

3.4.4 Experimental Data

- When analyzing the results from an experiment there are a few basic methods that may be used,
 - Percent difference -
 - Mean and standard deviation -
 - Point by point -
 - Matching functions -
 - etc.....

XXXXXXXXXXXXXXXXX Add more XXXXXXXXXXXXXXXXXXXXX

3.4.5 References

- References help provide direction to the sources of information when the information may be questioned, or the reader may want to get additional detail.
- Reference formats vary between publication sources. But, the best rule is be consistent.
- One popular method for references is to number them. The numbers are used in the body of the paper (eg, [14]), and the references are listed numerically at the end.
- Another method is to list the author name and year (eg, [Yackish, 1997]) and then list the references at the end of the report.
- Footnotes are not commonly used in engineering works.

3.4.6 Acknowledgments

- When others have contributed to the work but are not listed as authors we may choose to recognize them.
- Acknowledgments are brief statements that indicate who has contributed to a work.

3.4.7 Appendices

- When we have information that is needed to support a report, but is too bulky to include, one option is to add an appendix.
- Examples of appendices include,
 - reviews of basic theory
 - sample calculations
 - long tables of materials data
 - program listings
 - long test results

3.5 GENERAL FORMATTING

- Some general formatting items are,

- number all pages sequentially,
 - roman numerals starting from 'i' on the first page
 - arabic numerals starting from '1' on the
 - or, number pages by section. This is very useful for multi part manuals
 - for example '4-7' would be the 7th page in the 4th section
 - if pages are blank label them 'this page left blank'
 - number sections sequentially with roman or Arabic numerals
 -
- For numbers,
 - use engineering notation (move exponents 3 places) so that units are always micro, milli, kilo, mega, giga, etc.
 - use significant figures to round the numbers
 - units are required always
- General English usage,
 - check spelling - note that you must read to double guess the spell checker.
 - check grammar
 - avoid informal phrases (e.g. "show me the money")
 - define acronyms and jargon the first time you use them (e.g., IBM means "Ion Beam Manufacturing")
- General style rules,
 - keep it simple (especially the introduction) - most authors trying to be eloquent end up sounding long winded and pretentious. For example, "Electronic computer based digital readings can provided a highly accurate data source to improve the quality of the ascertained data." could be replaced with "Computer based data collection is more accurate."
 - get to the point and be concise. For example, "Readings of the pressure, as the probe was ascending up the chimney towards the top, were taken." is better put "Pressure probe readings were taken as the probe was inserted".
 - it is fine to say 'I' or 'we', but don't get carried away.
 - don't be afraid to reuse terms, phrases or words if it is an exact description. For example, we could increase confusion by also describing translation as motion, movement, sliding, displacing, etc.
- General engineering rules are,
 - all statements should be justified, avoid personal opinions or 'gut feels'
 - use exact engineering terms when needed, don't try to get creative.

Title: High Tech Presentations The Easy Way

Author: Hugh Jack

1.0 PRESENTATIONS IN GENERAL

- Different purposes for presentation
 - academic lectures
 - short technical presentations
 - short non-technical presentations
 - long workshops
 -
- Main presentation types,
 - board with chalk/markers
 - overheads
 - slides
 - video
 - computer with data projector
- The main elements in a computer based presentation are
 - electronic slides
 - software demonstration
 - other media types, including sound
 - distance connection
- Typical technology presentation problems are,
 - unfamiliar with the technology
 - layout is not suited to computer projector
 - presenter stops presenting, and starts using the computer
 - the presentation is overwhelming
 -
- Some data on visual presentations¹

The Numbers on Why You Need Visuals- 1. Seeing makes the most sense. Studies show that sight is the most used human sense. A whopping 75% of all environmental stimuli is received through visual reception (Doug Malouf). So, the best presenters use visuals to maximize the impact of their presentation!

Environmental Stimuli Reception - 2. Visuals are the best way to teach your audience. According to a recent University of California at Los Angeles study, 55% percent of what an audience learns comes directly from the visual messages seen during a presentation- compared to 38% from audio messages. By combining audio and visual presentation messages, presenters can ensure their objectives are met.

Impact of Communication - 3. Visuals increase the retention of messages. A Wharton Research Center study has shown that the retention rate of verbal only presentations is approximately 10%. However, when you combine visual messages with verbal communication, you increase the retention rate to nearly 50%. A 400% increase! Why not use visual aids to help your audience remember your message?

Message Retention - 4. Visuals help you meet your audience objectives. When presenters use visual aids in their presentations, they are twice as likely (67% vs. 33%) to achieve their audience objectives, than speakers who don't use visual aids. (Decker Communications). By incorporating effective visuals in your presentation, you increase your ability to communicate your message to your audience.

Achieving Objectives - 5. Reduce the length of your meeting. A recent study from the University of Minnesota found that the average length of meetings in which visuals were used were 26.8% shorter (26.7 minutes vs. 18.6 minutes) than meetings in which no visuals were used. With executive salaries at several hundred dollars an hour, visuals can save corporate executives a lot of time and money!

Meeting Length - 6. reduces meeting time 26%??

2.0 GOOD PRESENTATION TECHNIQUES

2.1 VISUALS

- The following tips help make visuals more effective

General

- keep it simple

Content

- test your presentation for size and look - are they easy to read, can they be followed, do they convey the information

Layout

- If you plan to refer to an earlier slide, make a second copy of it - don't flip back
- Use titles to make the purpose of visuals obvious
- use bullet points - long sentences crowd the screen and are hard to read.
- do not use more than 20 words per slide, do not try to write full sentences on slides, use it as a summary

Fonts and Text

- use upper/lower case only to draw attention to words, do this sparingly
- use a large enough font so that it can be read with ease

- use white space to make the screen more readable - if the slide is too full it will overwhelm the reader
- avoid multiple fonts - these look sloppy
- avoid italics, bold and upper case for emphasis - this makes it too busy
- when doing point form punctuation can be distracting

Graphics

- do not use too many lines on graphs - more than 3 will be hard to follow
- use figures or graphs when possible to reduce the number of words
- use graphs or charts for numbers - avoid large table
- Use line drawings instead of scanned images or photographs, they are easier to make out
- avoid equations when possible
- if possible have something to pass around - don't just talk about it or show a picture

Appearance

- avoid excessive colors - use a couple to pull graphics together. Three or more is too much.
- don't use numbered steps unless identifying importance or sequence

2.2 SPEAKING TIPS

- Good elements of any presentation include,
 - eye contact with all areas of the room for a few seconds on individuals
 - interaction with audience
 - do not stand behind the podium
 - avoid wild motions, use them to keep some life in the presentation
 - move around the room, but don't continually walk
 - look at the audience much more than the screen
 - Try to act naturally - try not to 'put on a personality'
 - don't read from the visuals

3.0 PRESENTATION TECHNOLOGY

- Some technology tips include
 - limit the use of sound/video as formats are not supported universally
 - use hyperlink to files to be used for demonstrations - this can automatically start applications
 - for photographs use full screen images with good brightness
- Advantages,
 - visuals can be distributed by the web

- the presentation can be changed at the last minute
- can make presentations easy to prepare
- cut/paste is easier than with other methods
- Disadvantages,
 - more time is required to deal with the technology
 - some support is normally required for networks, projectors, etc.
 - computer support is not yet universal. Computer projectors are expensive, and network connections are uncommon and complex to connect.
 - hardware and software is not yet common, and each new piece of equipment has a learning curve.
 - The resolution of computers is well below overheads and slides
 - more time is required to get familiar with equipment
 - more equipment is needed
 - it is hard to write on the screen

3.1 COMMON HARDWARE/SOFTWARE

- The common items needed for presentations

Computer - You are likely to take your own laptop computer, or try to use one that is made available. Using your own normally reduces software problems, but getting it to work in a new place can be troublesome. If you use somebody else's computer you need to get your presentation slides there. It is a good idea to take a separate mouse to plug in to notebooks. The touchpads and other laptop 'mice' can be hard to use in presentation settings.

Data Projector - If possible get a high quality data projector that will connect to your computer. If this is not possible, use an LCD panel on an overhead projector, this will appear a bit dark. Worst case use a computer to TV converter, this will be the least expensive, but the graphic quality is very poor. Your slide fonts and images will need to be 50-100% larger.

Presentation Software - Powerpoint is one of the most common packages. It generally tends to be stable, but some of the keystrokes can get lost in a presentation.

- The following software/hardware will be sometimes used for presentations

Word Processor/Publishing Software - You may opt to convert your presentation to HTML or PDF. Most packages support these formats. I use Adobe Framemaker, although Microsoft Word is a popular choice, and free alternatives exist such as Star Office (www.stardivision.com). The word processor typically needs to be able to include equations, and figures.

Browser - If presenting in HTML you will need a browser. There are a number of excellent browsers available today, but the two best are available from Netscape and Microsoft. Both can be obtained at no charge. Either will do for the students, and this software can be used as a presentation tool in class.

Application Software - I have used packages such as Working Model, and Mathcad to support lectures. Anybody using the browsers for course notes will need copies of application software to included files. These files will allow the notes to become interactive, visual, experimental, etc. For example, in Statics I have used working model to illustrate the slip-tip problem. There are a large number o packages that offer low cost student editions, or even free demonstration versions that still be used for viewing.

Digital Camera/Scanner - A scanner can be very useful for capturing images on paper. But this should be discouraged. Scanned documents are very large, and can be very slow when downloaded for viewing. Scanned photographs also tend to have a poor quality. A very good option is to buy a digital camera that captures images directly to digital format. These cameras come in a variety of prices, but a good midrange camera can be purchased for \$600 that will give good quality photographs. Within a short period of time these costs will drop quickly, and real time video capture will be an option. Other poor options include camcorders with image capture hardware in a PC. This gives grainy pictures or low resolution.

- Computer projectors have many pitfalls,
 - the most dependable screen resolution is 640 by 480, but this is a very low resolution, most projectors support higher resolutions (800 by 600, 1024 by 768 and 1280 by 1024).
 - At higher resolutions the projector may cut the sides off your screen image.
 - cables are almost universal, most are SVGA connectors. This is not always true if you are using an Apple computer.
 - Television output is a common option on many laptops but it is not commonly used. These outputs will normally connect to an S-video connector on a normal TV projector which might not be available at the podium. A long video cable would often be needed for this option. Television projectors are poor for data projection because the pixels are set at 60 degree diagonals that make an image blurred, even when well projected.
 - each projector has unique controls. These are sometimes manual adjustments, or buttons on the unit, other times a remote control is needed. These can be annoying to set, and try to get them set ahead of time. The 'off' button is often hard to find on projectors, you can't just kill the power, it needs to cool when done. Ask somebody to show you. (Note: the buttons get harder to find when the lights are low)
 - Light from these projectors is still quite dim, so a darkened room is almost essential. Find your lighting switches before the presentation. Some rooms may be all or nothing. One trick I have used is to turn off the lights, but use an overhead projector to create ambient light

- Laptop considerations

- cables between the computer and data projector can be a nuisance. Try to position these so you don't trip over them during the presentation. A table large than the laptop, near the projector will allow you to also use a separate mouse and lay out notes on the table.
- You can often plug your laptop power supply into the projector power bar. This will help power shutdown and other problems
- Power management features in laptops will cause them to quit or go to sleep if idle for a few minutes, especially when unplugged. Turn these off before the presentation. If your computer stops during the presentation it will take a while to get it back. Windows NT is very slow rebooting and may cause your presentation to pause for a few minutes.
- Screen savers should also be turned off.
- Many laptops don't output video until you hit a key sequence such as 'FN' 'F8', look at the manual for more details. You may need to hit this a few times to get the projector and LCD screen on.

- Some items to remember

- Presentation

- Presentation on disk - in case your laptop loses it
 - Hard copy of presentation - to look ahead while presenting
 - Overheads - prepare for the worst

- Laptop

- Power cord - don't forget this
 - Battery - extra batteries can help
 - User Guide - for those little questions
 - Mouse - for ease in the dark

- Projector (if you have one)

- Projector - don't forget this one
 - Shipping case
 - Power cord
 - SVGA Video cable
 - S-Video cable
 - Mac Adapter (if applicable)
 - Remote control and extra batteries
 - Flathead/Phillips mini for cables
 - User Guide
 - Power adapters (if travelling internationally)
 - Powerstrip
 - Extension cord (25 ft)

3.2 PRESENTING WITH TECHNOLOGY

- Some techniques for success are,
 - Use the computer to present new material, then turn on the lights and do problems on the board.
 - If the computer screen is in front of a whiteboard, pull up the screen, and add notes using the screen underneath.
 - Interact with the audience - Ask questions about what was just covered, give a simple problem to solve
 - Say something ridiculous to get a response.
 - Tell a joke.
 - Walk into the seats.
 - Borrow something for an example.
 - Know the software and hardware.
 - Keep a bit of 'MTV' style in mind. Videos, sounds and other moving things help.
 - Do a 'dress rehearsal' well before - small details such as fonts can ruin all the other efforts. Ask somebody to sit through a short trial run.
 - Turn on the lights and solve problems on the board frequently, it will wake up students going to sleep in the dark.
 - use the spell checker
 - Watch out for equations
 - Set up a model for the presentation
 - Pass-arounds
 - Show up for presentations early to become familiar with equipment
 - When testing, test beginning to end, small things can halt the presentation

X.0 EXAMPLES OF PRESENTATIONS

- [REF XXXXXXXX] Top Ten Mistakes Made By Presenters!
 - No Presentation Objectives - If you don't know what your audience should do at the end of your presentation, there is no need for you to present. Knowing your objectives is the key to developing an effective presentation.
 - Poor Visual Aids - Visual aids are designed to reinforce to your audience the main points of your presentation. Without effective visuals, you are missing a key opportunity to communicate with your audience.
 - Ineffective Close - Closing your presentation is extremely important. It is when you tie up your presentation and spell out what you want your audience "to do". A weak close can kill a presentation.

- Mediocre First Impression - Audiences evaluate a presenter within the first 120 seconds of your presentation. Presenters who make a bad first impression can lose credibility with their audience and as a result diminish their ability to effectively communicate the information in the presentation.
 - No Preparation - The best presenters prepare for every presentation. Those who prepare and practice are more successful in presenting their information and anticipating audience reaction. Practice does make perfect!
 - Lack of Enthusiasm - If you aren't excited about the presentation, why should your audience be? Enthusiastic presenters are the most effective ones around!
 - Weak Eye Contact - As a presenter, you are trying to effectively communicate with your audience to get your message across. If don't make eye contact with the members in your audience, they will not take you or your message seriously.
 - No Audience Involvement - The easiest way to turn off your audience is by not getting them involved in your presentation. Use audience involvement to gain their "buy-in".
 - Lack of Facial Expressions - Don't be a zombie. Effective speakers use facial expressions to help reinforce their messages.
 - Sticky Floor Syndrome - There is nothing worse than a speaker who is glued to the floor. Be natural and don't stay in one place.
-
- Examples of presentation problems follow
 1. The demonstration - the good, the bad and the ugly - playing with your computer during the presentation
 2. All the features, sound, lights, action - too many options
 3. The screen saver - something to look at
 4. Where's the on button? - knowing your equipment
 5. Microfilm - Making the fonts too big or small
 6. Just a bad presenter - no eye contact, mumbling, etc.
 7. The cut and paste - format not right for presentation
 8. The reader - too much details on overheads, and presenter reads
 9. The constant droner - fills all gaps in sound with 'ah', 'um',
 10. The Fiddler - playing with objects and moving too much
 11. The Jedi Knight - laser pointer or weapon? (aka caffeine amplifier)
 12. The Derivation - one big equation
 13. Flipper - spends time jumping backwards in slides
 14. Where's the file? - looking for that lost file on the hard drive

4.0 OTHER TECHNOLOGY ISSUES

4.1 NETWORKS

4.1.1 Computer Addresses

- Computers are often given names, because names are easy to remember.
- In truth the computers are given numbers.

Machine Name:	claymore.engineer.gvsu.edu
Alternate Name:	www.eod.gvsu.edu
IP Number:	148.61.104.215

- When we ask for a computer by name, your computer must find the number. It does this using a DNS (Domain Name Server). On campus we have two '148.61.1.10' and '148.61.1.15'.
- The number has four parts. The first two digits '148.61' indicate to all of the internet that the computer is at 'gvsu.edu', or on campus here (we actually pay a yearly fee of about \$50 to register this internationally). The third number indicates what LAN the computer is located on (Basically each hub has its own number). Finally the last digit is specific to a machine.

4.1.2 NETWORK TYPES

- The network connection type has an impact on the effectiveness and cost of the connection.

4.1.2.1 Permanent Wires

- These networks are fast, but require a permanent connection
- For the campus network the peak data transfer rate is about (4 GB/hour)
- These types of networks include,
 - Ethernet
 - ATM
 - Fast Ethernet

4.1.2.2 Phone Lines

- The merit dialup network is a good example. It is an extension of the internet that you can reach by phone.
- The phone based connection is slower (about 5 MB/hour peak)
- There are a few main types,
 - SLIP - most common
 - PPP - also common
 - ISDN - an faster, more expensive connection, geared to permanent connections
- You need a modem in your computer, and you must dial up to another computer that has a modem and is connected to the Internet. The slower of the two modems determines the speed of the connection. Typical modem speeds are,
 - 52.4 kbps - very fast
 - 28.8/33.3 kbps - moderate speed, inexpensive
 - 14.4 kbps - a bit slow for internet access
 - 2.4, 9.6 kbps - ouch
 - 300 bps - just shoot me

4.1.3 NETWORK PROTOCOLS

- What are protocols - sequences that computers must follow when sending and receiving information. These agreed methods make sure that information is sent and received correctly.
- Why do we need protocols - Without some agreement about what information is arriving over the network, it would just seem like garbage. This would be like somebody suddenly sending stock market numbers by morse code without telling us what it is.

4.1.3.1 FTP - File Transfer Protocol

- This is a method for retrieving or sending files to remote computers.

EXERCISE: In Netscape ask for the location 'ftp://sunsite.unc.edu' This will connect you via ftp the same way as with the windows and the dos software.

4.1.3.2 HTTP - Hypertext Transfer Protocol

- This is the protocol used for talking to a web server.

4.1.3.3 Novell

- Allows us to share files stored on a server.

EXERCISE: Look at the 'my computer' icon. The drives from 'F' and up are shared by network, and files are brought to the computer as you request them.

4.1.4 DATA FORMATS

- The format of the data is important so that other programs may interpret it correctly.

4.1.4.1 HTML - Hyper Text Markup Language

- This is a format that is invisible to the user on the web. It allows documents to be formatted to fit the local screen.

EXERCISE: While looking at a home page in Netscape select 'View - Page Source'. You will see a window that includes the actual HTML file - This file was interpreted by Netscape to make the page you saw previously. Look through the file to see if you can find any text that was on the original page.

- Editors are available that allow users to update HTML documents the same way they use word processors.

EXERCISE: Find a home page in Netscape. Use the 'File - Edit Page' button to start the editor. Notice the buttons along the top for font sizes, colors, etc. Play with page and add your own name.

EXERCISE: Type in two new line of text. Name these lines 'sunsite' and 'other'. highlight 'sunsite' first, and select the small chain link at the top of the page. type in the link 'http://sunsite.unc.edu'. Accept this and then highlight the 'other' line. Enter a new link again using 'other.html'. (don't close the edit window, we will use it again shortly)

4.1.4.1.1 Publishing Web Pages

- Once a web page has been modified it is necessary to put it back on the web server.
- When publishing a page with a browser 'FTP' will be used.
- The web page called 'index.html' is the first one to be returned. If you are publishing a main page your main page should be called 'index.html'.

EXERCISE: Using the web page that you modified before, publish the results to your home page. You can do this using the 'publish' option. You will have to provide a site name 'http://www2.gvsu.edu/~YOURNAME', a user name, and a password, and call the file 'index.html'. Use Netscape to view your updated home page. Note: You may have to hit reload, as Netscape will keep old copies, and does not automatically reload web pages if it has a recent copy is stored.

EXERCISE: Edit the file again and add a link to 'other.html'. You can do this by highlighting text, and then clicking on the 'link' icon.

- Keep in mind that the website is just another computer. You have directories and files there too. To create a web site that has multiple files we need to create other files or directory names.

EXERCISE: Create a new web page, and add something to it. Publish this page as before, except call it 'other.html'. Call up the browser, and load in the 'index.html' page that you created. Click on the links and see what happens.

- Note that some web servers do not observe upper/lower case and cut the 'html' extension to 'htm'. Microsoft based computers are notorious for this, and this will be the most common source of trouble.

EXERCISE (Basic): Use the Windows ftp program to access your remote account that your web page is set up in. Look at the files and file names. Transfer the files on the web site back to your local computer.

EXERCISE (Advanced): You can open these files in Netscape, edit them, save them back to the disk, and then publish them using the ftp program.

4.1.4.2 URLs

- In HTML documents we need to refer to resources. To do this we use a label to identify the type of resource, followed by a location.
- Universal Resource Locators (URLs)
 - http:WEB_SITE_NAME
 - ftp:FTP_SITE_NAME
 - mailto:USER@MAIL_SERVER
 - news:NEWSGROUP_NAME

EXERCISE: In netscape type in 'mailto:YOUR_NAME@river.it.gvsu.edu'. After you are done try 'news:gvsu'.

4.1.4.3 Hints

- Below is a list of hints for publishing web pages
 - Windows will not allow multiple applications to open the same file at the same time. If you seem to be having trouble opening a file, make sure it is not open in another application.
 - As you add other files to your homepage, put them in the 'temp' directory. This will make all of the procedures simpler.
 - Try to make your web pages small, and link them together. This will decrease download time and make browsers happier.
 - Avoid using excessive images. Anything over 10K will make it very slow downloading over modem. Anything over 100K makes modem downloading painfully slow.
 - When putting images on the web page use 'jpg' for photographic images, and 'gif' for line images. 'jpg' images can be compressed more than 'gif', but lines will become blurred.
 - To link to other files or web pages there will be a 'link' command. If you want to add a file that is in your 'temp' directory, just put the name of the file in the 'URL' field.
 - Watch upper/lower case. This is a major cause of web page problems.

4.1.4.4 Specialized Editors

- There are a variety of editors that will allow us to edit single web pages or entire sites.
- These programs include,
 - Microsoft Word
 - Powerpoint

- WebCT
- Frontpage

EXERCISE: Start Microsoft Word and create a new document. Save this document as HTML on the hard drive. Use notepad to open the file and see how it relates to the original file.

4.1.4.5 PDF

- A format proposed by Adobe. This is not a 'standard', but is very widely accepted.
- When documents are presented in pdf format their original layout is preserved (HTML will actually change the look/layout of a document), but the files become hard/impossible to work with.
- A special plug-in is required to view these files.

EXERCISE: Point Netscape to '<http://claymore.engineer.gvsu.edu/~sahlis/214>' and look under handouts for a PDF file.

4.1.4.6 Compression

- We can make a file smaller by compressing it (unless it is already compressed, then it gets larger)
- File compression can make files harder to use in Web documents, but the smaller size makes them faster to download. A good rule of thumb is that when the file is MB in size, compression will have a large impact.
- Many file formats have compression built in, including,
 - images - JPG, GIF
 - video - MPEG, AVI
 - programs - installation programs are normally compressed
- Typical compression formats include,
 - zip - zip, medium range compression
 - gz - g-zip - good compression
 - Z - unix compression
 - Stuffit - A Mac compression format
- Some files, such as text, will become 1/10 of their original size.

4.1.4.7 Java

- This is a programming language that is supported on most Internet based computers.
- These programs will run on any computer - there is no need for a Mac, PC and Unix version.
- Most users don't need to program in Java, but the results can be used in your web pages

EXERCISE: Go to 'www.javasoft.com' and look at some sample java programs.

4.1.4.8 Javascript

- Simple programs can be written as part of an html file that will add abilities to the HTML page.

4.1.4.9 ActiveX

- This is a programming method proposed by Microsoft to reduce the success of Java - It has been part of the antitrust suit against Microsoft by the Justice Department.
- It will only work on IBM PC computers running the 'Internet Explorer' browser from Microsoft.
- One major advantage of ActiveX is that it allows users to take advantage of programs written for Windows machines.
- Note: Unless there is no choice avoid this technique. If similar capabilities are needed, use Java instead.

4.1.4.10 Graphics

- Two good formats are,
 - GIF - well suited to limited color images - no loss in compression. Use these for line images, technical drawings, etc
 - JPG - well suited to photographs - image can be highly compressed with minimal distortion. Use these for photographs.
- Digital cameras will permit image capture and storage - images in JPG format are best.
- Scanners will capture images, but this is a poor alternative as the image sizes are larger and image quality is poorer
 - Photographs tend to become grainy when scanned.

- Line drawings become blurred.
- Screen captures are also possible, but do these with a lower color resolution on the screen (256 color mode).

4.1.4.11 Animation

- These are not video, but moving drawings/cartoons.
- Animations are limited, but are best done with animated gif files.
- Other options include,
 - java programs
 - special plug-ins such as shockwave

EXERCISE: Find an animation on a student page at 'claymore.engineer.gvsu.edu/students.html'.

4.1.4.12 Video

- Streaming built into Netscape for real-time video.

EXERCISE: Point Netscape to 'www.aml.gvsu.edu'. Select and watch the video stream.

- We can also get special plug-ins that will allow us to see video files,
 - MPEG very popular, good compression, and fault tolerant.
 - AVI popular on PC platforms, but limited drivers elsewhere.
 - Apple Quicktime.
- Real-time video conferencing is possible, but not yet practical.

EXERCISE: Start the Netscape Conference software.

4.1.4.13 Sounds

- Sound files are poorly supported, and most require special plugins,
 - real audio
 - wav audio

- etc

EXERCISE: (If there is a sound card in the computer) Go to www.mtv.com and try to play a

4.1.4.14 Other Program Files

- We can connect any type of non-standard computer file to our pages, such as a Microsoft Excel file '.xls'.
- To do this we need to,
 1. Put the file in our web directory
 2. Link it to one or more HTML pages
 3. Have the system administrator add this as a MIME type to the system.
 4. When you click on the link to the file in Netscape it will ask you to choose an application. For an excel file you would want to choose '\program files\office97\excel'. This will automatically start when you choose the file next time. If you did not do step 3, or did not choose an application you would be asked to save the file.
- This is an excellent way to extend the capabilities of a web browser.

EXERCISE: Create a Word or Powerpoint file, and put it in your web directory using FTP. Add a link to it in your home page, and test to see if the link works.

4.2 PULLING ALL THE PROTOCOLS AND FORMATS TOGETHER WITH BROWSERS

- As you have already seen, the browser (ie, Netscape) helps pull these resources together.
- When we want to do things that are not part of the standard browser, we can use plug-ins.
- Plug-ins are small programs that can be used by Browsers to deal with different Protocols and Formats.

EXERCISE: Go to the Netscape home page, and call up the plug-in directory. Look for a plug-in you would be able to use. You may also want to try (www.autodesk.com) to find a DWF viewer plugin.

REFERENCES

1. www.presentersonline.com/training

AA:1. ENGINEERING JOKES

- These jokes poke a bit of fun at those disciplines that have provided some many of the fundamentals of engineering. If anybody know of jokes from other disciplines, please let me know and I will add/link them.
- These have come from a variety of sources, all shamelessly reproduced, but if I knew more of the sources I would list them.
- Contributions by,
Frank Currie, who got them from Khawlah Zaki Munshi
Colin Moore
Rukshan Jayawardene

AA:1.1 AN ENGINEER, A LAWYER AND A.....

- A physicist, an engineer, and a mathematician are all locked in separate buildings. The physicist runs to a chalkboard, calculates exactly how much water he will need to put out the fire, runs and finds that amount, puts out the fire and survives. The engineer pulls out a calculator, calculates exactly how much water he need to put out the fire, runs and finds 10 times that amount, puts out the fire and survives. The mathematician runs to a chalkboard, calculates exactly how much water he will need to put out the fire, declares, "There IS a solution!", and then burns to death.
- Three engineers and three accountants are traveling by train to a conference. At the station, the three accountants each buy tickets and watch as the three engineers buy only a single ticket. "How are three people going to travel on only one ticket?" asks an accountant. "Watch and you'll see," answers an engineer. They all board the train. The accountants take their respective seats but all three engineers cram into a restroom and close the door behind them. Shortly after the train has departed, the conductor comes around collecting tickets. He knocks on the restroom door and says, "Ticket, please." The door opens just a crack and a single arm emerges with a ticket in hand. The conductor takes it and moves on. The accountants saw this, and agreed it was quite a clever idea. So after the conference, the accountants decide to copy the engineers on the return trip and save some money (being clever with money, and all that). When they get to the station, they buy a single ticket for the return trip. To their astonishment, the engineers don't buy a ticket at all. "How are you going to travel without a ticket?" says one perplexed accountant. "Watch and you'll see," answers an engineer. When they board the train, the three accountants cram into a restroom and the three engineers cram into another one nearby. The train departs. Shortly afterward, one of the engineers leaves his restroom and walks over to the restroom where the accountants are hiding. He knocks on the door and says, "Ticket, please."

- A priest, a doctor, and an engineer were waiting one morning for a particularly slow group of golfers.
 - Engineer: What's with these guys? We must have been waiting for 15 minutes!
 - Doctor: I don't know but I've never seen such ineptitude!
 - Priest: Hey, here comes the greenskeeper. Let's have a word with him. ...Hi George. Say George, what's with that group ahead of us? They're rather slow aren't they?
 - George: Oh yes. That's a group of blind fire fighters. They lost their sight while saving our club house last year. So we let them play here anytime free of charge!
 - (silence)
 - Priest: That's so sad. I think I will say a special prayer for them tonight.
 - Doctor: Good idea. And I'm going to contact my ophthalmologist buddy and see if there's anything he can do for them.
 - Engineer: Why can't these guys play at night?
- A lead hardware engineer, a lead software engineer, and their program manager are taking a walk outdoors during their lunch break when they come upon an old brass lamp. They pick it up and dust it off. Poof -- out pops a genie. "Thank you for releasing me from my lamp-prison. I can grant you 3 wishes. Since there are 3 of you I will grant one wish to each of you." The hardware engineer thinks a moment and says, "I'd like to be sailing a yacht across the Pacific, racing before the wind, with an all-girl crew." "It is done", said the Genie, and poof, the hardware engineer disappears. The software engineer thinks a moment and says, "I'd like to be riding my Harley with a gang of beautiful women throughout the American Southwest." "It is done", said the Genie, and poof, the software engineer disappears. The program manager looks at where the other two had been standing and rubs his chin in thought. Then he tells the Genie, "I'd like those two back in the office after lunch." Sheldon Rosen and Anne Hamon
- Source: anonymous
 - An engineer dies and reports to the pearly gates. St. Peter checks his dossier and says, "Ah; you're an engineer - you're in the wrong place."
 - So the engineer reports to the gates of hell and is let in. Pretty soon, the engineer gets dissatisfied with the level of discomfort in hell, and starts designing and building improvements. After a while, they've got air conditioning and flush toilets and escalators, and the engineer is a pretty popular guy.
 - One day God calls Satan up on the telephone and says with a sneer, "So, how's it going down there in hell?"
 - Satan replies, "Hey, things are going great! We've got ait conditioning and flush toilets and escalators, and there's no telling what the engineer is going to come up with next."
 - God replies, "What? You've got an engineer? That's a mistake - he should never have gotten down there; send him up here." Satan says, "No way. I like having an engineer on the staff, and I'm keeping him."
 - God says, "Send him back up here or I'll sue."
 - Satan laughs uproariously and answers, "Yeah, right. And just where are YOU going to get a lawyer?"
- Three engineers and three accountants are traveling by train to a conference. At the station, the

three accountants each buy tickets and watch as the three engineers buy only a single ticket. "How are three people going to travel on only one ticket?" asks an accountant. "Watch and you'll see," answers an engineer.

They all board the train. The accountants take their respective seats but all three engineers cram into a restroom and close the door behind them. Shortly after the train has departed, the conductor comes around collecting tickets. He knocks on the restroom door and says, "Ticket, please." The door opens just a crack and a single arm emerges with a ticket in hand. The conductor takes it and moves on.

The accountants saw this and agreed it was quite a clever idea. So after the conference, the accountants decide to copy the engineers on the return trip and save some money (being clever with money, and all!). When they get to the station, they buy a single ticket for the return trip. To their astonishment, the engineers buy no tickets at all.

"How are you going to travel without a ticket?" says one perplexed accountant. "Watch and you'll see," answers an engineer. When they board the train the three accountants cram into a restroom and the three engineers cram into another one nearby. The train departs. Shortly afterward, one of the engineers leaves his restroom and walks over to the restroom where the accountants are hiding. He knocks on the door and says, "ticket please."

• THE CASTAWAY ENGINEER

A rather inhibited engineer finally splurged on a luxury cruise to the Caribbean. It was the "craziest" thing he had ever done in his life. Just as he was beginning to enjoy himself, a hurricane roared upon the huge ship, capsizing it like a child's toy. Somehow the engineer, desperately hanging on to a life preserver, managed to wash ashore on a secluded island.

Outside of beautiful scenery, a spring-fed pool, bananas and coconuts, there was little else.

He lost all hope and for hours on end, sat under same palm tree. One day, after several months had passed, a gorgeous woman in a small rowboat appeared.

"I'm from the other side of the island," she said. "Were you on the cruise ship, too?"

"Yes, I was," he answered. "But where did you get that rowboat?"

"Well, I whittled the oars from gum tree branches, wove the reinforced gunnel from palm branches, and made the keel and stern from a Eucalyptus tree."

"But, what did you use for tools?" asked the man.

"There was a very unusual strata of alluvial rock exposed on the south side of the island. I discovered that if I fired it to a certain temperature in my kiln, it melted into forgeable ductile iron. Anyhow, that's how I got the tools. But, enough of that," she said. "Where have you been living all this time? I don't see any shelter."

"To be honest, I've just been sleeping on the beach," he said.

"Would you like to come to my place?" the woman asked. The engineer nodded dumbly.

She expertly rowed them around to her side of the island, and tied up the boat with a handsome strand of hand-woven hemp topped with a neat back splice. They walked up a winding stone walk she had laid and around a palm tree. There stood an exquisite bungalow painted in blue and white.

"It's not much, but I call it home." Inside, she said, "Sit down please; would you like to have a drink?"

"No, thanks," said the man. "One more coconut juice and I'll throw up!"

"It won't be coconut juice," the woman replied. "I have a crude still out back, so we can have authentic Pina Coladas."

Trying to hide his amazement, the man accepted the drink, and they sat down on her couch to talk. After they had exchanged stories, the woman asked, "Tell me, have you always had a beard?"

"No," the man replied, "I was clean shaven all of my life until I ended up on this island."

"Well if you'd like to shave, there's a razor upstairs in the bathroom cabinet."

The man, no longer questioning anything, went upstairs to the bathroom and shaved with an intricate bone-and-shell device honed razor sharp. Next he showered -- not even attempting to fathom a guess as to how she managed to get warm water into the bathroom -- and went back downstairs. He couldn't help but admire the masterfully carved banister as he walked.

"You look great," said the woman. "I think I'll go up and slip into something more comfortable."

As she did, the man continued to sip his Pina Colada. After a short time, the woman, smelling faintly of gardenias, returned wearing a revealing gown fashioned out of pounded palm fronds.

"Tell me," she asked, "we've both been out here for a very long time with no companionship. You know what I mean. Haven't you been lonely, too...isn't there something that you really, really miss? Something that all men and woman need? Something that would be really nice to have right now!"

"Yes there is!" the man replied, shucking off his shyness. "There is something I've wanted to do for so long. But on this island all alone, it was just...well, it was impossible." "Well, it's not impossible any more," the woman said.

The man, practically panting in excitement, said breathlessly: "You mean . . .

.you actually figured out some way we can CHECK OUR EMAIL HERE!?!?!?"

- Programmers Joke

A programmer, a mechanical engineer, and a manager were on their way to a meeting. They were driving down a steep, winding, mountain road with many switchbacks, when suddenly the brakes failed. The car ran off the road, bounced down the mountain, rolled over a few times, bumped into some rocks, and finally stopped only a few feet from the road below. No one was hurt, but they had a problem. They were on a mountain with no brakes in the car.

"Let's have a meeting," the manager said. "Propose a Vision, formulate a Mission Statement, define some Goals, and by a process of Continuous Mission Improvement focusing on our Core Business, find a solution to this Critical Problem."

"No, no," said the mechanical engineer. "That's never solved anything. I've got two shirt hangars, half a roll of duct tape, and a Swiss army knife. Give me a few minutes, and the brakes will be good enough to get us to the nearest gas station."

"Wait," said the programmer. "Before we do anything, I think we should push the car back up the road and see if it happens again."

AA:1.2 GEEKY REFERENCES

- What do you get when you cross a goat and a mountain climber? Nothing..... a mountain climber is a scalar.
- Why does steak have more energy than hamburger? Because, hamburger is in its ground state.
- Heisenberg was driving down the Autobahn whereupon he was pulled over by a policeman. The policeman asked, "Do you know how fast you were going back there? Heisenberg replied, "No, but I know where I am."

AA:1.3 QUIPS

If it's green and wiggles, it's biology. If it stinks, it's chemistry. If it doesn't work, it's physics.

AA:1.4 ACADEMIA

AA:1.4.1 Other Disciplines

- One day a mathematician decides that he is sick of math and wants to become a fireman. So the mathematician walks down to the fire department and announces that he wants to become a fireman. The fire chief says, "Well, you look like a good guy. I'd be glad to hire you, but first I have to give you a little test." The firechief takes the mathematician to the alley behind the fire department which contains a dumpster, a spicket, and a hose. The chief then says, "OK, you're walking in the alley and you see the dumpster here is on fire. What do you do?" The mathematician replies, "Well, I hook up the hose to the spicket, turn the water on, and put out the fire." The chief says, "That's great...perfect. Now I have to ask you just one more question. What do you do if you're walking down the alley and you see the dumpster is not on fire?" The mathematician puzzles over the question for awhile and the finally says, "I light the dumpster on fire." The chief yells, "What? That's horrible! Why would you light the dumpster on fire?" The mathematician replies, "Well, that way I reduce the problem to one I've already solved."
- A biologist, a chemist, and a physicist are taking a walk through the country when they come upon a cow. For some odd reason, none of them knew what it was. The biologist thinks for a second and then declares, "I know what that is. That's Bos Bovine." The chemist looks for a second and then says, "It's just a carbon-based life form, approximately 75% water, and the remaining 25% trace elements." The physicist stares blankly for a second and then says, "Well.... I guess we could approximate it as a sphere."

- A theoretical physicist, an experimental physicist, and a mathematician are all locked in separate rooms for a week with only 1 can of soup and are told that if they want to eat they must find a way to open the can. After 1 week the rooms were opened:
- The experimental physicist's room had multiple dents in the walls, with a few soup stains, showing how he had thrown his can against the wall until he figured out the exact angle needed to open the can. He then threw the can at the wall at that angle, opened the can, and ate the soup. The theoretical physicist's room was covered in equations and 1 dent in the wall, showing he calculated the exact angle needed to open the can, threw the can at the wall at that angle, opened the can, and ate the soup. The mathematician was found in his room seated on the floor with the unopened can repeating, "I define this can to be open!"
- MURPHY'S LAWS OF COMPUTING
 1. When computing, whatever happens, behave as though you meant it to happen.
 2. When you get to the point where you really understand your computer, it's probably obsolete.
 3. The first place to look for information is in the section of the manual where you least expect to find it.
 4. When the going gets tough, upgrade.
 5. For every action, there is an equal and opposite malfunction.
 6. To err is human . . . to blame your computer for your mistakes is even more human, it is downright natural.
 7. He who laughs last probably made a back-up.
 8. If at first you do not succeed, blame your computer.
 9. A complex system that does not work is invariably found to have evolved from a simpler system that worked just fine.
 10. The number one cause of computer problems is computer solutions.
 11. A computer program will always do what you tell it to do, but rarely what you want to do.

AA:1.4.2 Faculty

- Here is a list of the ways professors here at the American University grade their final exams:
 - Dept Of Statistics: All grades are plotted along the normal bell curve.
 - Dept Of Psychology: Students are asked to blot ink in their exam books, close them and turn them in. The professor opens the books and assigns the first grade that comes to mind.
 - Dept Of History: All students get the same grade they got last year.
 - Dept Of Religion: Grade is determined by God.
 - Dept Of Philosophy: What is a grade?
 - Law School: Students are asked to defend their position of why they should receive an A.
 - Dept Of Mathematics: Grades are variable.
 - Dept Of Logic: If and only if the student is present for the final and the student has accu-

mulated a passing grade then the student will receive an A else the student will not receive an A.

Dept Of Computer Science: Random number generator determines grade.

Music Department: Each student must figure out his grade by listening to the instructor play the corresponding note (+ and - would be sharp and flat respectively).

AA:1.4.3 Students

- Top ten ways the Bible would have been different if written by college students.
 - 10). Loaves and Fishes replaced by Pizza and Chips
 - 9). Ten Commandments are actually only five, but because they are double-spaced and written in a large font, they look like ten.
 - 8). Forbidden fruit would have been eaten because it wasn't dorm food.
 - 7). Paul's Letters to the Romans become Paul's E-Mail to the Romans.
 - 6). Reason Cain killed Abel: They were roommates.
 - 5). The place where the end of the world occurs, not the Plains of Armageddon, rather Finals.
 - 4). Book of Armaments would be in there somewhere.
 - 3). Reason why Moses and followers walked in desert for 40 years: They didn't want to ask directions and look like a Freshman.
 - 2). Tower of Babel blamed for Foreign Language requirement.
 - 1). Instead of God creating the world in six days and resting on the seventh, He would have put it off until the night before it was due and then pulled an all-nighter and hoped no one noticed.
- Chani Silverberg and Seth Pertain

- THESE ARE ACTUAL EXCERPTS FROM STUDENT SCIENCE EXAM PAPERS:

Charles Darwin was a naturalist who wrote the organ of the species.

Benjamin Franklin produced electricity by rubbing cats backwards.

The theory of evolution was greatly objected to because it made man think.

Three kinds of blood vessels are arteries, vanes and caterpillars.

The dodo is a bird that is almost decent by now.

To remove air from a flask, fill it with water, tip the water out, and put the cork in quick before the air can get back in.

The process of turning steam back into water again is called conversation.

A magnet is something you find crawling all over a dead cat.

The Earth makes one resolution every 24 hours.

The cuckoo bird does not lay his own eggs.

To prevent conception when having intercourse, the male wears a condominium.

To collect fumes of sulfur, hold a deacon over a flame in a test tube.

Parallel lines never meet, unless you bend one or both of them.

Algebraical symbols are used when you do not know what you are talking about.

Geometry teaches us to bisex angles.

A circle is a line which meets its other end without ending.
The pistol of a flower is its only protection against insects.
The moon is a planet just like the Earth, only it is even deader.
Artificial insemination is when the farmer does it to the cow instead of the bull.
An example of animal breeding is the farmer who mated a bull that gave a great deal of milk with a bull with good meat.
We believe that the reptiles came from the amphibians by spontaneous generation and study of rocks.
English sparrows and starlings eat the farmers grain and soil his corpse.
By self-pollination, the farmer may get a flock of long-haired sheep.
If conditions are not favorable, bacteria go into a period of adolescence.
Dew is formed on leaves when the sun shines down on them and makes them perspire.
Vegetative propagation is the process by which one individual manufactures another individual by accident.
A super-saturated solution is one that holds more than it can hold.
A triangle which has an angle of 135 degrees is called an obscene triangle.
Blood flows down one leg and up the other.
A person should take a bath once in the summer, and not quite so often in the winter.
The hookworm larvae enters the human body through the soul.
When you haven't got enough iodine in your blood you get a glacier.
It is a well-known fact that a deceased body harms the mind.
Humans are more intelligent than beasts because the human branes have more convulsions.
For fainting: rub the person's chest, or if a lady, rub her arm above the hand instead.
For fractures: to see if the limb is broken, wiggle it gently back and forth.
For dog bite: put the dog away for several days. If he has not recovered, then kill it.
For nosebleed: put the nose much lower than the body.
For drowning: climb on top of the person and move up and down to make artificial perspiration.
To remove dust from the eye, pull the eye down over the nose.
For head colds: use an agonizer to spray the nose until it drops in your throat.
For snakebites: bleed the wound and rape the victim in a blanket for shock.
For asphyxiation: apply artificial respiration until the patient is dead.
Before giving a blood transfusion, find out if the blood is affirmative or negative.
Bar magnets have north and south poles, horseshoe magnets have east and west poles.
When water freezes you can walk on it. That is what Christ did long ago in wintertime.
When you smell an odorless gas, it is probably carbon monoxide.

- Source: Rukshan Jayawardene - The First Realizations That You're Not In College Anymore
You're waking up at 6 am instead of going to bed.
Beers at lunch get you reprimanded.
College sweatshirts are 'casual' instead of dress-up.
Your parents charge rent.
Your parents walk in on you having sex, not your roommate.
The 4 food groups are no longer beer, pizza, ramen and cereal.
It's 'getting late' when it's 9:30 p.m.

Three Words: School Loan Payments.

You make thousands of dollars a year - and still can't afford that dream Porsche.

You start eyeing the Light Beer section appreciatively.

Pickup football games mean that at least one person will be in the hospital by game's end.

Your friends are discussing,

THEN: GPA's, phone rates and tonsil hockey;

NOW: IRA's, Interest rates and their kid's orthodontia.

Sleeping on the couch is a no-no.

Naps are no longer available between noon and 6 p.m.

Sneakers are now 'weekend shoes'.

The letter from mom and dad now have portraits of their 'other' grandchildren instead of cash.

Dinner and a movie - The whole date instead of the beginning of one.

Your girlfriend being pregnant brings thought of tax deductions instead of coronaries.

Jack and Cokes become Dewers on the Rocks.

The only drugs you take are Tums and Tylenol.

That weak single you hit in the intramural softball game is now remembered as a Varsity dinger for the League Championship.

You get your news from sources other than USA Today, ESPN Sportscenter and MTV News.

Random hook-ups are no longer acceptable.

You wear more ties/skirts in a week than you owned while taking classes.

You find yourself reminiscing fondly of 2-hour Calculus exams.

You empathize with the characters from 'Friends'.

METABOLISM SLOWDOWN.

Football season tickets go from \$75 for the season with dozens of friends to \$750 for the season with the three other guys who get away from the family.

Wine appreciation expands beyond Boone's and Mad Dog.

You actually eat breakfast foods at breakfast time.

Grocery lists actually contain relatively healthy food.

When drinking, you say at least once per night, 'I just can't put it down the same as I used to'.

You are the only person over the age of 16 in your neighborhood with a Sega.

AA:1.5 COMPUTERS

- MORE COMPUTER VIRUSES... Militia Virus: wipes out your operating system claiming it has no right to control your PC.

Pro-Choice Virus: Although it presents the standard "Abort, Retry, Fail" prompt, it pressures you to choose "Abort", telling you the process being terminated is just "a blob of bits" which has no value.

Lyle And Eric Menendez Virus: wipes out your motherboard, claiming it was done in self-defense.

Bill Clinton Virus: causes your PC to behave unpredictably, working as expected one moment, then suddenly doing the exact opposite the next moment.
Politically Correct Virus: rephrases the "Abort, Retry, Fail" prompt as "Choice, Retry, Success-Impaired".
National Organization of Women (NOW) Virus: forces your PC to recognize its female connections as male connections.
Republican Virus: sells off your system resources to the highest bidder.
Democrat Virus: doesn't allow you to delete inefficient programs or wasted disc space - if you try, it accuses you of being a "mean-spirited extremist".
National Education Assoc. (NEA) Virus: although cleverly disguised as educational software intended to improve your system, in reality it "dumbs down" your 486DX into an 8086.
Jocelyn Elders Virus: teaches your computer to turn itself on.
LAPD Virus: attempts to stop your CPU. If your CPU resists, it is pummeled into hamburger.
Jack Kevorkian Virus: assists your CPU in destroying itself.
Ross Perot Virus: This erratic virus doesn't do much of anything, except surfacing occasionally to threaten to disrupt your system.

- Source: John DeJong

Once upon a midnight dreary, fingers cramped and vision bleary,
System manuals piled high and wasted paper on the floor,
Longing for the warmth of bed sheets, still I sat there doing spreadsheets.
Having reached the bottom line I took a floppy from the drawer,
I then invoked the SAVE command and waited for the disk to store,
Only this and nothing more.
Deep into the monitor peering, long I sat there wond'ring, fearing,
Doubting, while the disk kept churning, turning yet to churn some more.
But the silence was unbroken, and the stillness gave no token.
"Save!" I said, "You cursed mother! Save my data from before!"
One thing did the phosphors answer, only this and nothing more,
Just, "Abort, Retry, Ignore?"
Was this some occult illusion, some maniacal intrusion?
These were choices undesired, ones I'd never faced before.
Carefully I weighed the choices as the disk made impish noises.
The cursor flashed, insistent, waiting, baiting me to type some more.
Clearly I must press a key, choosing one and nothing more,
>From "Abort, Retry, Ignore?"
With fingers pale and trembling, slowly toward the keyboard bending,
Longing for a happy ending, hoping all would be restored,
Praying for some guarantee, timidly, I pressed a key.
But on the screen there still persisted words appearing as before.
Ghastly grim they blinked and taunted, haunted, as my patience wore,
Saying "Abort, Retry, Ignore?"
I tried to catch the chips off guard, and pressed again, but twice as hard.
I pleaded with the cursed machine: I begged and cried and then I swore.

Now in mighty desperation, trying random combinations,
Still there came the incantation, just as senseless as before.
Cursor blinking, angrily winking, blinking nonsense as before.
Reading, "Abort, Retry, Ignore?"
There I sat, distraught, exhausted, by my own machine accosted.
Getting up I turned away and paced across the office floor.
And then I saw a dreadful sight: a lightning bolt cut through the night.
A gasp of horror overtook me, shook me to my very core.
The lightning zapped my previous data, lost and gone forevermore.
Not even, "Abort, Retry, Ignore?"
To this day I do not know the place to which lost data go.
What demonic nether world us wrought where lost data will be stored,
Beyond the reach of mortal souls, beyond the ether, into black holes?
But sure as there's C, Pascal, Lotus, Ashton-Tate and more,
You will be one day be left to wander, lost on some Plutonian shore,
Pleading, "Abort, Retry, Ignore?"

• Suuport issues ...

1. Compaq is considering changing the command "Press Any Key" to "Press Return Key" because of the flood of calls asking where the "Any" key is.
2. AST technical support had a caller complaining that her mouse was hard to control with the dust cover on. The cover turned out to be the plastic bag the mouse was packaged in.
3. Another Compaq technician received a call from a man complaining that the system wouldn't read word processing files from his old diskettes. After trouble-shooting for magnets and heat failed to diagnose the problem, it was found that the customer had labeled the diskettes, then rolled them into the typewriter to type the labels.
4. Another AST customer was asked to send a copy of her defective diskettes. A few days later a letter arrived from the customer along with photocopies of the floppies.
5. A Dell technician advised his customer to put his troubled floppy back in the drive and close the door. The customer asked the tech to hold on, and was heard putting the phone down, getting up and crossing the room to close the door to his room.
6. Another Dell customer called to say he couldn't get his computer to fax anything. After 40 minutes of trouble-shooting, the technician discovered the man was trying to fax a piece of paper by holding it in front of the monitor screen and hitting the "send" key.
7. Yet another Dell customer called to complain that his keyboard no longer worked. He had cleaned it by filling up his tub with soap and water and soaking the keyboard for a day, then removing all the keys and washing them individually.
8. A Dell technician received a call from a customer who was enraged because his computer had told him he was "bad and an invalid". The tech explained that the computer's "bad command" and "invalid" responses shouldn't be taken personally.
9. A confused caller to IBM was having troubles printing documents. He told the technician that the computer had said it "couldn't find printer". The user had also tried turning the computer screen to face the printer - but that his computer still couldn't

“see” the printer.

10. An exasperated caller to Dell Computer Tech Support couldn't get her new Dell Computer to turn on. After ensuring the computer was plugged in, the technician asked her what happened when she pushed the power button. Her response, “I pushed and pushed on this foot pedal and nothing happens.” The “foot pedal” turned out to be the computer's mouse.
 11. Another customer called Compaq tech support to say her brand-new computer wouldn't work. She said she unpacked the unit, plugged it in and sat there for 20 minutes waiting for something to happen. When asked what happened when she pressed the power switch, she asked “What power switch?”
 12. True story from a Novell NetWire SysOp:
Caller: “Hello, is this Tech Support?”
Tech: “Yes, it is. How may I help you?”
Caller: “The cup holder on my PC is broken and I am within my warranty period. How do I go about getting that fixed?”
Tech: “I'm sorry, but did you say a cup holder?”
Caller: “Yes, it's attached to the front of my computer.”
Tech: “Please excuse me if I seem a bit stumped, It's because I am. Did you receive this as part of a promotional, at a trade show? How did you get this cup holder? Does it have any trademark on it?”
Caller: “It came with my computer, I don't know anything about a promotional. It just has '4X' on it.”
At this point the Tech Rep had to mute the caller, because he couldn't stand it. He was laughing too hard. The caller had been using the load drawer of the CD-ROM drive as a cup holder, and snapped it off the drive!
 13. Another IBM customer had troubles installing software and rang for support. “I put in the first disk, and that was OK. It said to put in the second disk, and had some problems with the disk. When it said to put in the third disk - I couldn't even fit it in...” The user hadn't realized that “Insert Disk 2” meant to remove Disk 1 first.
 14. In a similar incident, a customer had followed the instructions for installing software. The instructions said to remove the disk from it's cover and insert into the drive. The user had physically removed the casing of the disk and wondered why there were problems.
- At a recent Sacramento PC User's Group meeting, a company was demonstrating its latest speech-recognition software. A representative from the company was just about ready to start the demonstration and asked everyone in the room to quiet down. Just then someone in the back of the room yelled, “Format C: Return.” Someone else chimed in: “Yes, Return” Unfortunately, the software worked.

AA:1.5.1 Bill

- Dear Abby - I am writing to your advice-column because of a serious problem I am facing. You

see, I am a Vietnam-era deserter from the U. S. Marines, and I have a cousin who works for Microsoft's Customer Service Division. My mother peddles Nazi literature to Girl Scouts and my father (a former dentist) is in jail for 30 years for raping most of his patients while they were under anesthesia. The sole supports of our large family, including myself and my \$500-a-week heroin habit, are my uncle (master pick-pocket Benny "The Fingers") and my aunt and kid sisters, who are well-known streetwalkers. My problem is this: I have just gotten engaged to the most beautiful, sweetest girl in the world. She is just sweet sixteen, and we are going to marry as soon as she can escape from reform school. To support ourselves, we are going to move to Mexico and start a fake-Aztec-souvenir factory staffed by child labor. We look forward to bringing our kids into the family business. But -- I am worried that my family will not make a good impression on hers, once she has a chance to meet them. In your opinion, Abby: Should I -- or shouldn't I -- let her know about my cousin who works for Microsoft Customer Service?

Benjie Wolicki

- There are three engineers in a car; an electrical engineer, a chemical engineer, and a Microsoft engineer. Suddenly the car just stops by the side of the road, and the three engineers look at each other wondering what could be wrong. The electrical engineer suggests stripping down the electronics of the car and trying to trace where a fault might have occurred. The chemical engineer, not knowing much about cars suggests that maybe the fuel is becoming emulsified and getting blocked somewhere. Then, the Microsoft engineer, not knowing much about anything, comes up with a suggestion. "Why don't we close all the windows, get out, get back in, open the windows again, and maybe it'll work!?"

Chani Savet
- Subject: DIARY OF A DIGITAL HOMEOWNER
The (Future) Diary of a Digital Homeowner:
Nov 28: Moved in to my new digitally-maxed out Hermosa Beach house at last. Finally, we live in the smartest house in the neighborhood. Everything's networked. The cable TV is connected to our phone, which is connected to my personal computer, which is connected to the power lines, all the appliances and the security system. Everything runs off a universal remote with the friendliest interface I've ever used. Programming is a snap. I'm like, totally wired.
Nov 30: Hot Stuff! Programmed my VCR from the office, turned up the thermostat and switched on the lights with the car phone, remotely tweaked the oven a few degrees for my pizza. Everything nice & cozy when I arrived. Maybe I should get the universal remote surgically attached.
Dec 1: Had to call the SmartHouse people today about bandwidth problems. The TV drops to about 2 frames/second when I'm talking on the phone. They insist it's a problem with the cable company's compression algorithms. How do they expect me to order things from the Home Shopping Channel?
Dec 8: Got my first SmartHouse invoice today and was unpleasantly surprised. I suspect the cleaning woman of reading Usenet from the washing machine interface when I'm not here. She must be downloading one hell of a lot of GIFs from the binary groups, because packet charges were through the roof on the invoice.
Dec 3: Yesterday, the kitchen CRASHED. Freak event. As I opened the refrigerator door, the light bulb blew. Immediately, everything else electrical shut down -- lights, microwave, coffee maker -- everything. Carefully unplugged and replugged all the

appliances. Nothing. Call the cable company (but not from the kitchen phone). They refer me to the utility. The utility insists that the problem is in the software. So the software company runs some remote tele-diagnostics via my house processor. Their expert system claims it has to be the utility's fault. I don't care, I just want my kitchen back. More phone calls; more remote diag's. Turns out the problem was "unanticipated failure mode": The network had never seen a refrigerator bulb failure while the door was open. So the fuzzy logic interpreted the burnout as a power surge and shut down the entire kitchen. But because sensor memory confirmed that there hadn't actually been a power surge, the kitchen logic sequence was confused and it couldn't do a standard restart. The utility guy swears this was the first time this has ever happened. Rebooting the kitchen took over an hour.

Dec 7: The police are not happy. Our house keeps calling them for help. We discover that whenever we play the TV or stereo above 25 decibels, it creates patterns of micro-vibrations that get amplified when they hit the window. When these vibrations mix with a gust of wind, the security sensors are actuated, and the police computer concludes that someone is trying to break in. Go figure. Another glitch: Whenever the basement is in self-diagnostic mode, the universal remote won't let me change the channels on my TV. That means I actually have to get up off the couch and change the channels by hand. The software and the utility people say this flaw will be fixed in the next upgrade -- SmartHouse 2.1. But it's not ready yet. Finally, I'm starting to suspect that the microwave is secretly tuning into the cable system to watch Bay Watch. The unit is completely inoperable during that same hour. I guess I can live with that. At least the blender is not tuning in to old I Love Lucy episodes.

Dec 9: I just bought the new Microsoft Home. Took 93 gigabytes of storage, but it will be worth it, I think. The house should be much easier to use and should really do everything. I had to sign a second mortgage over to Microsoft, but I don't mind: I don't really own my house now--it's really the bank. Let them deal with Microsoft.

Dec 10: I'm beginning to have doubts about Microsoft House. I keep getting an hour glass symbol showing up when I want to run the dishwasher.

Dec 12: This is a nightmare. There's a virus in the house. My personal computer caught it while browsing on the public access network. I come home and the living room is a sauna, the bedroom windows are covered with ice, the refrigerator has defrosted, the washing machine has flooded the basement, the garage door is cycling up and down and the TV is stuck on the home shopping channel. Throughout the house, lights flicker like stroboscopes until they explode from the strain. Broken glass is everywhere. Of course, the security sensors detect nothing. I look at a message slowly throbbing on my personal computer screen: WELCOME TO Home-Wrecker!!! NOW THE FUN BEGINS ... (Be it ever so humble, there's no virus like the HomeWrecker...).

Dec 18: They think they've digitally disinfected the house, but the place is a shambles. Pipes have burst and we're not completely sure we've got the part of the virus that attacks toilets. Nevertheless, the Exorcists (as the anti-virus SWATteam members like to call themselves) are confident the worst is over. "HomeWrecker is pretty bad" one he tells me, "but consider yourself lucky you didn't get PolterGeist. That one is really evil."

Dec 19: Apparently, our house isn't insured for viruses. "Fires and mud slides, yes," says the claims adjuster. "Viruses, no." My agreement with the SmartHouse people explicitly states that all claims and warranties are null and void if any appliance or computer in my house networks in any way, shape or form with a non-certified on-line service. Everybody's very, very, sorry, but they can't be expected to anticipate every virus that might be created. We call our lawyer. He laughs. He's excited!

Dec 21: I get a call from a SmartHouse sales rep. As a special holiday offer, we get the free opportunity to become a beta site for the company's new SmartHouse 2.1 upgrade. He says I'll be able to meet the programmers personally. "Sure," I tell him.

Yakov Horenstein

- Toddler Property Laws

1. If I like it, it's mine.
2. If it's in my hand, it's mine.
3. If I can take it from you, it's mine.
4. If I had it a little while ago, it's mine.
5. If it's mine, it must never appear to be yours in any way.
6. If I'm doing or building something, all the pieces are mine.
7. If it looks just like mine, it's mine.
8. If I think it's mine, it's mine.
9. If I . . .Oops! I'm sorry, I goofed!

Instead of typing in the Toddler Property Laws, I've been typing in Bill Gates' Business Plan.

- Windows 95: n.

32 bit extensions and a graphical shell for a 16 bit patch to an 8 bit operating system originally coded for a 4 bit microprocessor, stolen by a 2 bit company, that can't stand 1 bit of competition.

- Is Windows a Virus? No, Windows is not a virus. Here's what viruses do:

- 1.They replicate quickly - okay, Windows does that.
- 2.Viruses use up valuable system resources, slowing down the system as they do so - okay, Windows does that.
- 3.Viruses will, from time to time, trash your hard disk - okay, Windows does that too.
- 4.Viruses are usually carried, unknown to the user, along with valuable programs and systems. Sigh... Windows does that, too.
- 5.Viruses will occasionally make the user suspect their system is too slow (see 2) and the user will buy new hardware. Yup, that's with Windows, too.

Until now it seems Windows is a virus but there are fundamental differences: Viruses are well supported by their authors, are running on most systems, their program code is fast, compact and efficient and they tend to become more sophisticated as they mature.

So .

It's a bug.

[And that's the truth ...]

AA:1.5.2 Internet

- Have you been spending more and more time using the Internet? Have your cheeks taken on that pasty white glow from over-exposure to your computer monitor? How do you know if you're addicted to the Net and losing touch with reality? Take the Net Addict's Reality Test. Answer the following multiple choice questions and check out your score to see if you should be concerned:

1. What do you think are good names for children?
 - a) Scott and Jenny.
 - b) Bill Gates IV.
 - c) Mozilla and Dotcom.
2. What's a telephone?
 - a) A thing with a round dial you use to talk to others.
 - b) A telecommunications device with 12 keys.
 - c) Something you plug into a modem.
3. Which punctuation is most correct?
 - a) I had a wonderful day!
 - b) I had a ****wonderful**** day!!!
 - c) I had a wonderful day :-)
4. You wake up at 4:00 a.m. and decide to:
 - a) Visit the washroom.
 - b) Raid the fridge.
 - c) Check your E-mail.
5. What are RAM and ROM?
 - a) A male sheep and a city in Italy.
 - b) Hulking stars of the WWF.
 - c) I need more of the former and should upgrade the latter.
6. To avoid a virus you should:
 - a) Stay away from people who sneeze and cough.
 - b) Never read E-mail titled "Good Times".
 - c) Use virus scanning software every time you boot up.
7. When you want to buy something hard-to-find you:
 - a) Ask friends where to purchase it.
 - b) Check out the Yellow Pages.
 - c) Go to Yahoo!
8. When you don't understand how to use a new appliance you:
 - a) Call the retailer.
 - b) Call the manufacturer's toll-free number.
 - c) Visit the manufacturer's Web site and look for the FAQ.
9. When you want to see all the beautiful people you:
 - a) Visit a club on a Saturday night.
 - b) Turn on the TV and tune in to Baywatch.

- c) Check out the alt.binary newsgroups.
- 10. How do you introduce yourself at a party?
 - a) Hi, I'm Jane!
 - b) Hi, I'm a Taurus on the cusp.
 - c) Hi, I'm a 5'10" hot blonde with a super bod.
- 11. When you're interested in someone at a party you say:
 - a) Tell me more about yourself.
 - b) What's your star sign?
 - c) What's your Profile?
- 12. If you really like the person, you say:
 - a) Could you tell me your phone number?
 - b) What's your E-mail address?
 - c) Let's chat Private.
- 13. When I say spam, you think:
 - a) Ham in a can.
 - b) Unsolicited advertising E-mail.
 - c) I mailbomb all spammers!
- 14. When you receive an AOL trial diskette, you say:
 - a) I don't need another mug coaster.
 - b) Great! I'll reformat and use it for backups.
 - c) Great! I'll sign up under a fake ID and use up the 50 hours.
- 15. When you want to research a reference you:
 - a) Open up a volume of your encyclopedia.
 - b) Slip Encarta in your CD-ROM drive.
 - c) Go to www.altavista.digital.com.
- 16. When you write a letter you:
 - a) Put pencil to paper.
 - b) Open Eudora.
 - c) Ask: What's a letter? Is it like E-mail?
- 17. Different types of text formatting include:
 - a) Writing and printing.
 - b) Underline and double-strike.
 - c) Bold and italic.
- 18. You correct errors using:
 - a) An eraser.
 - b) White-out.
 - c) Backspace or delete.
- 19. You sign your name:
 - a) Best regards, John Smith.
 - b) See you in IRC, John_Smith.
 - c) Check out my home page for the cool links, johnsmith@aol.com.
- 20. To keep a copy of your letter you:
 - a) Insert a carbon and a second sheet.
 - b) Take it to the photocopier.
 - c) Check your Sent Mail folder.

SCORING: Give yourself zero points for each "a" response, five for each

"b" and 10 for each "c".

If you scored 150 or higher, unplug your computer and log more hours in real life.

If you scored between 50 and 145, you're living a good mix of Net and reality.

If you scored under 50, you probably didn't read this far.

AA:1.6 OTHER STUFF

- Some legal oddities,

A Louisiana law upholds your right to grow as tall as you like.

Singing out of tune in North Carolina is against the law.

The laws of Portland, Me., do not allow one to tickle a girl under the chin with a feather duster.

A New York judge ruled that if two people behind you in a movie house are discussing the probable outcome of a film, you can give them a Bronx cheer.

It is illegal to lasso a fish in Knoxville, Tenn.

It is illegal to put tomatoes in clam chowder in Massachusetts.

In Nebraska, sneezing in public is prohibited by law.

The law prohibits unrestrained giggling in Helena, Mont.

A kiss can last no longer than one second in Halethorpe, Md.

In Baltimore, Md., it is against the law to mistreat an oyster.

In Denver, Colo., the law insists that dogcatchers notify dogs of impounding by posting a notice on a tree in the park.

The law forbids women in Oxford, Ohio, to undress in front of a photograph of a man.

In Cold Spring, Pa., liquor can be sold to a married man only if he has his wife's permission.

In Connecticut, the law states that if you are a beaver, you have the right to build a dam.

In Gary, Ind., it is illegal to attend a theater within four hours of eating garlic.

In Owensboro, Ky., if a woman wants to buy a new hat, her husband must try it on first.

The legal punishment in Minneapolis, Minn., for double parking is being put on a chain gang and fed only on a diet of bread and water.

In Roderfield, W. Va., only babies are allowed to ride in baby carriages.

You are not permitted to swim on dry land in Santa Ana, Calif.

An old ordinance of New London, Conn., forbids actresses from appearing in public.

In Saco, Mo., hats which may frighten people are outlawed.

A Virginia law makes it illegal to have a bathtub in one's house.

It is illegal to mispronounce the name of the city of Joliet, Ill.

A 17th-century Massachusetts law forbade the selling of cakes or buns except on special occasions-contrived to keep women from gossiping over tea and cake. The crafty females circumvented the law and gave rise to New England's famous pies and doughnuts.

- Today's scientific question is: What in the world is electricity? And where does it go after it leaves the toaster? Here is a simple experiment that will teach you an important electrical les-

son: On a cool, dry day, scuff your feet along a carpet, then reach your hand into a friend's mouth and touch one of his dental fillings. Did you notice how your friend twitched violently and cried out in pain? This teaches us that electricity can be a very powerful force, but we must never use it to hurt others unless we need to learn an important electrical lesson. It also teaches us how an electrical circuit works. When you scuffed your feet, you picked up batches of "electrons," which are very small objects that carpet manufacturers weave into carpet so that they will attract dirt. The electrons travel through your bloodstream and collect in your finger, where they form a spark that leaps to your friend's filling, then travel down to his feet and back into the carpet, thus completing the circuit. AMAZING ELECTRONIC FACT: If you scuffed your feet long enough without touching anything, you would build up so many electrons that your finger would explode! But this is nothing to worry about... unless you have carpeting. Although we modern persons tend to take our electric lights, radios, mixers, etc. for granted. Hundreds of years ago people did not have any of these things, which is just as well because there was no place to plug them in. Then along came the first Electrical Pioneer, Benjamin Franklin, who flew a kite in a lightning storm and received a serious electrical shock. This proved that lightning was powered by the same force as carpets, but it also damaged Franklin's brain so severely that he started speaking only in incomprehensible maxims, such as, "A penny saved is a penny earned." Eventually he had to be given a job running the post office. After Franklin came a herd of Electrical Pioneers whose names have become part of our electrical terminology: Myron Volt, Mary Louise Amp, James Watt, Bob Transformer, etc. These pioneers conducted many important electrical experiments - - Among them, Galvani discovered (this is the truth) that when he attached two different kinds of metal to the leg of a frog, an electrical current developed and the frog's leg kicked, even though it was no longer attached to the frog, which was dead anyway. Galvani's discovery led to enormous advances in the field of amphibian medicine. Today, skilled veterinary surgeons can take a frog that has been seriously injured or killed, implant pieces of metal in its muscles, and watch it hop back into the pond just like a normal frog, except for the fact that it sinks like a stone. But the greatest Electrical Pioneer of them all was Thomas Edison, who was a brilliant inventor despite the fact that he had little formal education and lived in New Jersey. Edison's first major invention in 1877 was the phonograph, which could soon be found in thousands of American homes, where it basically sat until 1923, when the record was invented. But Edison's greatest achievement came in 1879 when he invented the electric company. Edison's design was a brilliant adaptation of the simple electrical circuit: the electric company sends electricity through a wire to a customer, then immediately gets the electricity back through another wire, then (this is the brilliant part) sends it right back to the customer again. This means that an electric company can sell a customer the same batch of electricity thousands of times a day and never get caught, since very few customers take the time to examine their electricity closely. In fact, the last year any new electricity was generated was 1937; the electric companies have been merely re-selling it ever since, which is why they have so much time to apply for rate increases. Today, thanks to men like Edison and Franklin, and frogs like Galvani's, we receive almost unlimited benefits from electricity. For example, in the past decade scientists have developed the laser, an electronic appliance so powerful that it can vaporize a bulldozer 2000 yards away, yet so precise that doctors can use it to perform delicate operations on the human eyeball, provided they remember to change the power setting from "Vaporize Bulldozer" to "Delicate." A N Agrawal

- From Colin Moore - GUINNESS, AND HERITAGE

One day an Englishman, a Scotsman, and an Irishman walked into a pub together. They each bought a pint of Guinness. Just as they were about to enjoy their creamy beverage, three flies landed in each of their pints, and were stuck in the thick head. The Englishman pushed his beer away in disgust. The Scotsman fished the fly out of his beer, and continued drinking it, as if nothing had happened. The Irishman, too, picked the fly out of his drink, held it out over the beer, and started yelling, "SPIT IT OUT, SPIT IT OUT YOU BASTARD!!!!"

- From Colin Moore

A man has been in business for many, many years and the business is going down the drain. He is seriously contemplating suicide and he doesn't know what to do. He goes to the Rabbi to seek his advice. He tells the Rabbi about all of his problems in the business and asks the Rabbi what he should do. The Rabbi says "Take a beach chair and a bible and put them in your car and drive down to the edge of the ocean. Go to the water's edge. Take the beach chair out of the car, sit on it and take the bible out and open it up. The wind will rifle the pages for a while and eventually the bible will stay open at a particular page. Read the bible and it will tell you what to do." The man does as he is told. He places a beach chair and a bible in his car and drives down to the beach. He sits on the chair at the water's edge and opens the bible. The wind rifles the pages of the bible and then stops at a particular page. He looks down at the bible and sees what he has to do. Three months later the man and his family come back to see the Rabbi. The man is wearing a \$1,000 Italian suit, The wife is all decked out with a full-length mink coat and the child is dressed in beautiful silk. The man hands the Rabbi a thick envelope full of money and tells him that he wants to donate this money to the synagogue in order to thank the Rabbi for his wonderful advice. The Rabbi is delighted. He recognizes the man and asks him what advice in the bible brought this good fortune to him. The man replies: "Chapter 11."

- Source: Colin Moore

A guy walks into a bar and sits down. He starts dialing numbers...like a telephone...on his hand and talking into his hand. The bar tender walks over and tells him that this is a very tough neighborhood and he doesn't need any trouble here. The guy says, "You don't understand; I'm very hi-tech. I had a phone installed in my hand because I was tired of carrying the cellular." The bar tender says, "Prove it." The guy dials up a number and `hands' his hand to the bar tender. The bartender talks into the hand and carries on a conversation. "That's incredible", says the bartender ... "I would never believe it!" "Yeah", said the guy, "I'm really very hi-tech. I can keep in touch with my broker, my wife, you name it! By the way, where is the men's room?" The bar tender directs him to the men's room. The guy goes in and doesn't come out for the longest time. Fearing the worst given the tough neighborhood, the bar tender goes into the men's room. There is the guy... he is spread-eagle on the wall...his pants are pulled down and he has a roll of toilet paper up his butt. "Oh my god", said the bar tender, "Did they rob you? How much did they get?" The guy turns and says: "No, no,... I'm just waiting for a fax!"

- Source Ruckshan Jayawardene - Here's to all you `process thinkers'..... Enjoy. The answers are at the end of this message, but don't cheat.

- 1) The Elder Twin - One day Kerry celebrated her birthday. Two days later her older twin brother, Terry, celebrated his birthday. How come?
- 2) Manhole Covers - Why is it better to have round manhole covers than square ones?
This is logical rather than lateral, but it is a good puzzle which can be solved by lateral thinking techniques. It is supposedly used by a very well-known software company as an interview question for prospective employees.
- 3) The Deadly Party - A man went to a party and drank some of the punch. He then left early. Everyone else at the party who drank the punch subsequently died of poisoning. Why did the man not die?
- 4) Heaven - A man died and went to Heaven. There were thousands of other people there. They were all naked and all looked as they died at the age of 21. He looked around to see if there was anyone he recognized. He saw a couple and he knew immediately that they were Adam and Eve. How did he know?
- 5) Trouble with Sons - A woman had two sons who were born on the same hour of the same day of the same year. But they were not twins. How could this be so?
- 6) The Man in the Bar - A man walks into a bar and asks the barman for a glass of water. The barman pulls out a gun and points it at the man. The man says `Thank you' and walks out. This puzzle has claims to be the best of the genre. It is simple in its statement, absolutely baffling and yet with a completely satisfying solution. Most people struggle very hard to solve this one yet they like the answer when they hear it or have the satisfaction of figuring it out.

SOLUTIONS:

- 1) At the time she went into labor, the mother of the twins was traveling by boat. The older twin, Terry, was born first early on March 1st. The boat then crossed the International Date line (or anytime zone line) and Kerry, the younger twin, was born on February the 28th. In a leap year the younger twin celebrates her birthday two days before her older brother.
- 2) A square manhole cover can be turned and dropped down the diagonal of the manhole. A round manhole cannot be dropped down the manhole. So for safety and practicality, all manhole covers should be round.
- 3) The poison in the punch came from the ice cubes. When the man drank the punch the ice was fully frozen. Gradually it melted, poisoning the punch.
- 4) He recognized Adam and Eve as the only people without navels. Because they were not born of women, they had never had umbilical cords and therefore they never had navels. This one seems perfectly logical but it can sometimes spark fierce theological arguments!
- 5) They were two of a set of triplets (or quadruplets etc.) This simple little puzzle stumps many people. They try outlandish solutions involving test-tube babies or surrogate mothers. Why does the brain search for complex solutions when there is a much simpler one available?
- 6) The man had hiccups. The barman recognized this from his speech and drew the gun in order to give him a shock. It worked and cured the hiccups - so the man no longer needed the water. This is a simple puzzle to state but a difficult one to solve. It is a

perfect example of a seemingly irrational and incongruous situation having a simple and complete explanation. Amazingly this classic puzzle seems to work in different cultures and languages.

- A Canadian is walking down the street with a case of beer under his arm. His friend Doug stops him and asks, "Hey Bob! Whacha get the case of beer for?" "I got it for my wife, eh." answers Bob. "Oh!" exclaims Doug, "Good trade."

- CLASSIFIED ADS

Illiterate? Write today for free help.

Auto Repair Service. Free pick-up and delivery. Try us once, you'll never go anywhere again.

Our experienced Mom will care for your child. Fenced yard, meals, and smacks included.

Dog for sale: eats anything and is fond of children.

Man wanted to work in dynamite factory. Must be willing to travel.

Stock up and save. Limit: one.

Semi-Annual after-Christmas Sale.

3-year old teacher needed for pre-school. Experience preferred.

Mixing bowl set designed to please a cook with round bottom for efficient beating.

Girl wanted to assist magician in cutting-off-head illusion. Blue Cross and salary.

Dinner Special -- Turkey \$2.35; Chicken or Beef \$2.25; Children \$2.00

For sale: antique desk suitable for lady with thick legs and large drawers.

Now is your chance to have your ears pierced and get an extra pair to take home, too.

We do not tear your clothing with machinery. We do it carefully by hand.

For sale. Three canaries of undermined sex.

Great Dames for sale.

Have several very old dresses from grandmother in beautiful condition.

Tired of cleaning yourself. Let me do it.

Vacation Special: have your home exterminated.

Get rid of aunts. Zap does the job in 24 hours.

Toaster: A gift that every member of the family appreciates. Automatically burns toast.

For Rent: 6-room hated apartment.

Man, honest. Will take anything.

Used Cars: Why go elsewhere to be cheated. Come here first.

Christmas tag-sale. Handmade gifts for the hard-to-find person.

Wanted: Hair cutter. Excellent growth potential.

Wanted. Man to take care of cow that does not smoke or drink.

Our bikinis are exciting. They are simply the tops.

Wanted. Widower with school age children requires person to assume general housekeeping duties. Must be capable of contributing to growth of family.

And now, the Superstore-unequaled in size, unmatched in variety, unrivaled inconvenience.

We will oil your sewing machine and adjust tension in your home for \$1.00.

- READ this sentence:

FINISHED FILES ARE THE RE-

SULT OF YEARS OF SCIENTIF-
IC STUDY COMBINED WITH
THE EXPERIENCE OF YEARS.

Now count aloud the F's in that sentence. Count them ONLY ONCE; do NOT go back and count them again.

ANSWER:

There are six F's in the sentence. One of average intelligence finds three of them. If you spotted four, you're above average. If you got five, You can turn your nose at most anybody. If you caught six, you are a genius. There is no catch. Many people forget the OFs. The human brain tends to see them as "V's" instead of "F's".

- How to keep a healthy level of insanity in the workplace
 1. Page yourself over the intercom. (don't disguise your voice)
 2. Find out where your boss shops and buy exactly the same outfits. Always wear them one day after your boss does. (this is especially effective if your boss is a different gender then you are)
 3. Make up nicknames for all your coworkers and refer to them only by these names. ("That's a good point, Sparky." "No I'm sorry I'm going to have to disagree with you there, Baby Cakes.")
 4. Send e-mail to the rest of the company telling them what you're doing. For example, "If anyone needs me, I'll be in the bathroom."
 5. "Hi-lite" your shoes. Tell people that you haven't lost your shoes since you did this.
 6. Ask people to call you "Captain"
 7. Put up mosquito netting around your cubicle
 8. Put a chair facing the printer, sit there all day and tell people you're waiting for your document.
 9. Arrive at a meeting late, say you're sorry, but you didn't have time for lunch, and you're going to be nibbling during the meeting. During the meeting eat five entire raw potatoes.
 10. Insist that your e-mail address be "zena_goddess_of_fire@companyname.com"
 11. Every time someone asks you to do something, ask them if they want fries with that.
 12. Send e-mail to yourself engaging yourself in an intelligent debate about the direction of one of your company's products. Forward the mail to a co-worker and ask her to settle the disagreement.
 13. Suggest that beer be put in the soda machine.
 14. Encourage your colleagues to join you in a little synchronized chair dancing.
 15. Put your garbage can on your desk. Label it "IN".
 16. Determine how many cups of coffee is "too many."
 17. Develop an unnatural fear of staplers.
 18. Decorate your office with pictures of Cindy Brady and Danny Partridge. Try to pass them off as your children.
 19. Send e-mail messages saying "free pizza, free donuts etc..." in the lunchroom; when people complain that there was none, just lean back, pat your stomach, and say, "Oh, you've got to be faster than that."
 20. Put decaf in the coffeemaker for 3 weeks. Once everyone has gotten over their caffeine

addictions, switch to espresso.

2. PUZZLES

- These puzzles have been collected from a number of sources to help build problem solving skills, primarily for design.
- In most/all cases the problems have been rewritten to appeal to the engineering approach to problems.

2.1 MATH

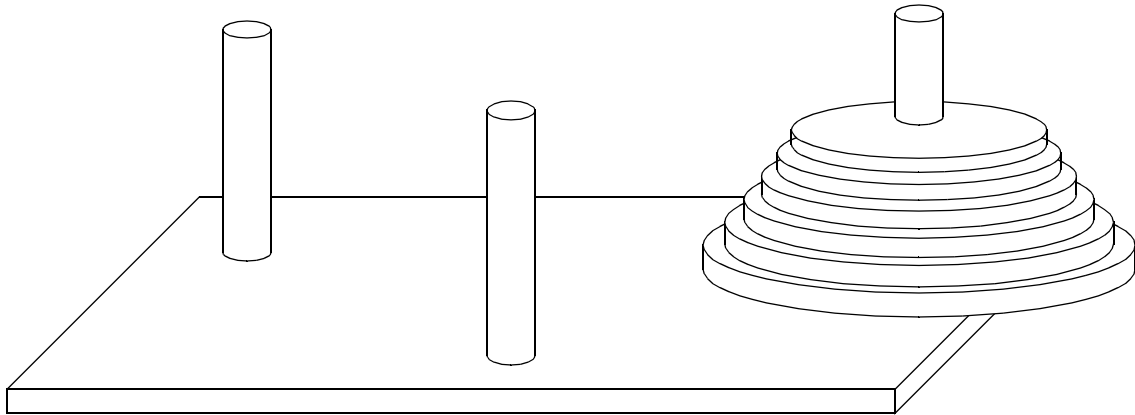
- We are planning to build a new autoparts factory to supply stores along a straight section of highway. It doesn't matter where we build the factory, except the total driving distance will vary. We want to choose a location that minimizes the total driving time. Each grid space below represents 10 miles. [Carter & Russell, 1995]



Historical demand for deliveries

Deliver to A 8 times per day
 Deliver to B 4 times per day
 Deliver to C 7 times per day
 Deliver to D 6 times per day

- A customer has indicated that 10 years from now four inventory items will be a total of 100 years old. What will their total age be 7 years from now? [Carter & Russell, 1995]
- The Towers of Hanoi is a classic puzzle that requires that discs of smaller sizes be moved one piece at a time to the other posts, while never putting a larger disc over a smaller one. In the case below the discs are on one post. How many moves are required to move all of the discs to another post?



- Nellie the pig can eat a trough of slop in 2 hours. Billie the pig can each a trough of slop in 1.5 hours. How long would it take both of the pigs together to eat one trough of slop?
- A total of 35,555 marbles were dropped in a tank, and sorted into four sizes. We missed the count of the first bin, but the second bin was 2,384 marbles less, the third was 5,285 less, and the last was 8,923 less. How many were in the first bin?
- A lineup was used to check three automobile safety systems. The tally sheets indicated the numbers below. How many cars had only one safety system working?

20 had adequate brakes
23 had working headlights
40 had seatbelts
6 had brakes and headlights
7 had brakes and seatbelts
5 had headlights and seatbelts
2 had all systems functioning

- If a train is half a mile long, and enters a tunnel that is 3 miles long, how long will some part of the train be in the tunnel if the train is travelling at 55 m.p.h.?
- Using one stroke of a pen, make the following equation true,

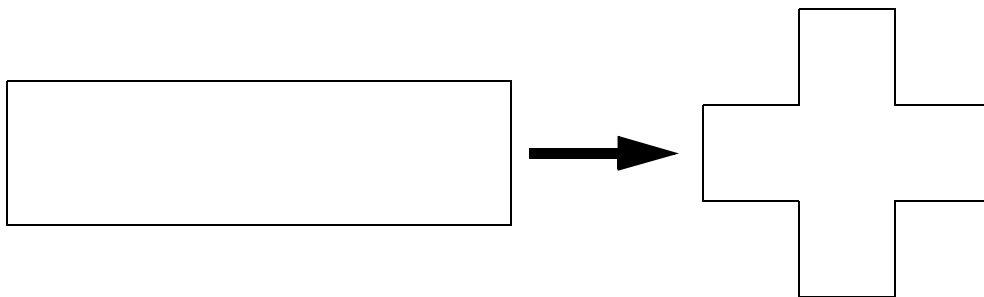
$$5 + 5 + 5 = 550$$

- Add plus/minus signs to the left hand side of the equation below to balance the equation,

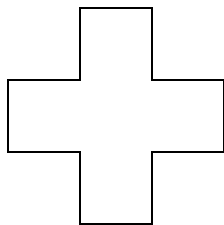
$$1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9 = 100$$

2.2 STRATEGY

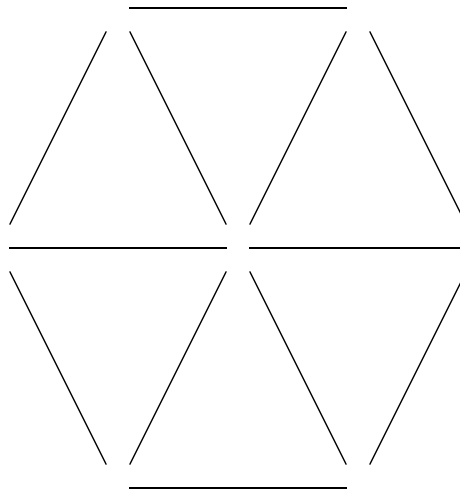
- Using six equal size sticks, create three squares of equal size.
- Divide the rectangle into three pieces that will make a cross,



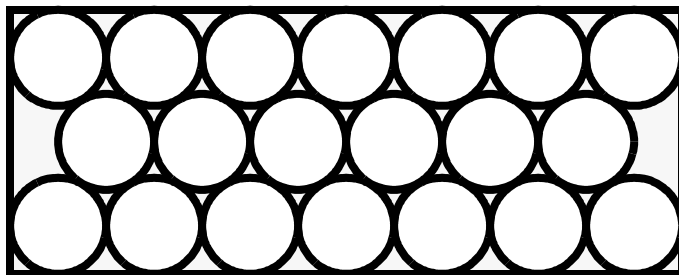
- We have four sticks of equal length, and four more sticks that are twice as long. Move the sticks to make four squares of the same size.
- Cut the cross below into four identical pieces that can be rearranged into a square [Carter & Russell, 1995]



- Move four matches to make three equilateral triangles

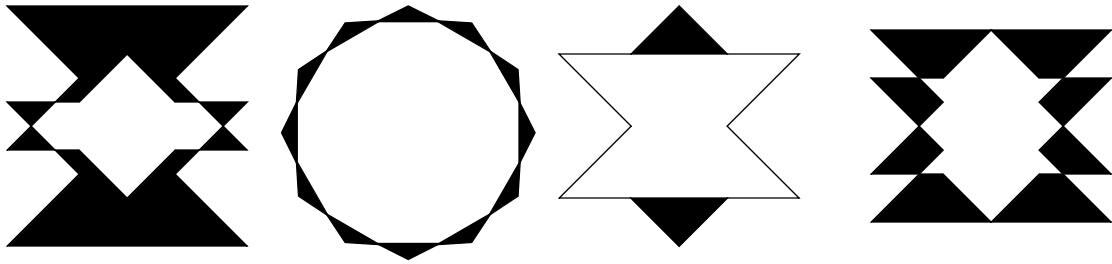
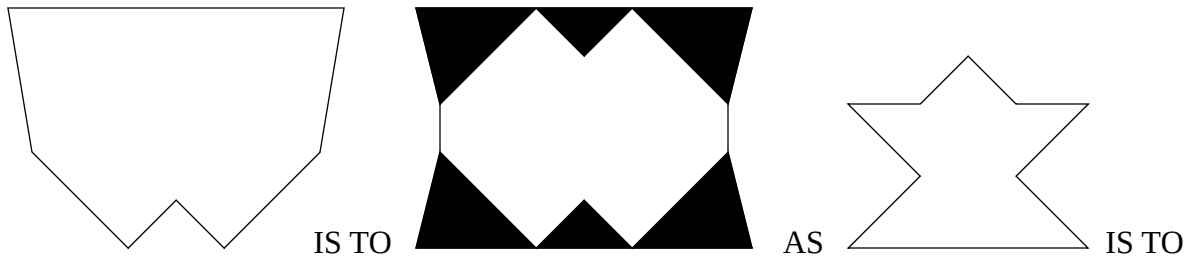


- Chess Stuff: Move a knight about a chessboard so that in 16 moves it touches all of the squares on the board. Move a bishop around a board to touch all black squares in 17 moves. Move the queen about the board to touch all the squares in 14 moves. [pentagram]
- How many balls can be removed from the box below, while still leaving the others locked in place? [Pentagram]

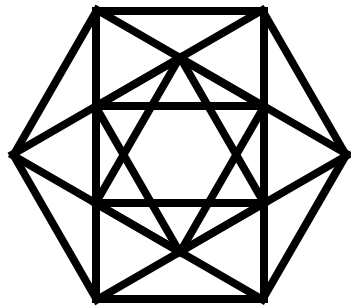


2.3 GEOMETRY

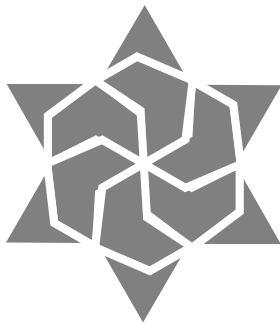
- Consider the associations below,



- Find all of the triangles of different sizes in the figure below. [Pentagram]

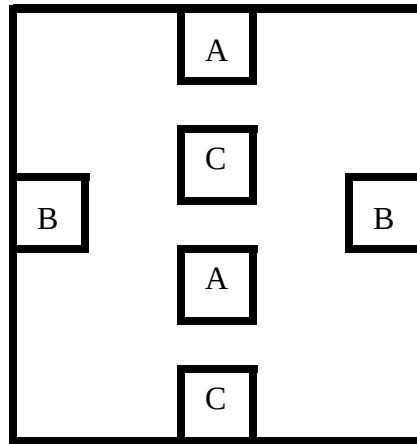


- Rearrange the shapes below into three equal shaped smaller six pointed stars,



2.4 PLANNING/DESIGN

- How can a brick be suspended with a single sheet of paper.
- Why are manhole covers round.
- Put a marble on a table. Using only a glass remove the marble from the table (without touching it).
- Draw lines that do not cross between boxes containing the same letters.



2.5 REFERENCES

Carter, P., and Russell, K., IQ Puzzles; Beat The Mensa Puzzlemasters, Robinson Publishing, London, England, 1995.

Pentagram, More Puzzlegrams, Simon & Schuster Inc., New York, 1994.

3. ATOMIC MATERIAL DATA

Element	Atomic #	Atomic weight	density g/cm ³	Valances	H _w
copper	29	63.57	8.96	1/2	
iron	26	55.85	7.86	2/3	
aluminum	13	26.97	2.67	3	
tungsten	74	183.85	19.3	6/8	
zinc	30	65.37	7.13	2	
silicon	14	28.09	2.33	4	
carbon	6	12.01	2.1	2/4	
titanium	22	47.9	4.51	3/4	

F = 96,500 coulombs

4. MECHANICAL MATERIAL PROPERTIES

Table 1:

Material	Type	Density	E	G	poisson	ten - yield stress	ten - ult. stress	ten - yield strain	yield stress - comp.	ult. stress - comp.	ult. stress - shear	thermal expansion
		(Mg/m ³)	(GPa)	(GPa)		(MPa)		(%)	(MPa)			10 ⁻⁶ /deg C
aluminum	typical	2.6-2.8	70-79	26-30	0.33	35-500	100-550	1-45				23
	1100-h14											
	2024-T6											
	5456-h116											
	2014-T6	2.8	73	36-41	0.33	410	480	13				
	6061-T6	2.8	70	26	0.33	270	310	17				
	7075-T6	2.7	72	26	0.33	480	550	11				

Table 1:

Material	Type	Density	E	G	poisson	ten - yield stress	ten - ult. stress	ten - yield strain	yield stress - comp.	ult. stress - comp.	ult. stress - shear	thermal expansion
		(Mg/m ³)	(GPa)	(GPa)		(MPa)		(%)	(MPa)			10 ⁻⁶ /deg C
Brass	typical	8.4-8.6	96-110	36-41	0.34	70-550	200-620	4-60				19.1-21.2
Yellow Brass												
	cold rolled											
	annealed											
Red Brass												
	cold rolled											
	annealed											
Bronze	typical	8.2-8.8	96-120	36-44	0.34	82-690	200-830	5-60				18-21
Cast Iron	typical	7.0-7.4	83-170	32-69	0.2-0.3	120	69-480	0-1		340-1400		9.9-12
Concrete	typical	2.3	17-31	na	0.1-0.2					10-70		7-14
	reinforced	2.4	17-31	na	0.1-0.2							
	lightweight	1.1-1.8	17-31	na	0.1-0.2							
Copper	typical	8.9	110-120	40-47	0.33-0.36	55-760	230-830	4-50				16.6-17.6
	annealed											
	hard drawn											
Glass	typical	2.4-2.8	48-83	19-35	0.17-0.27		30-1000	0				5-11
	Plate						70					
	Fibers						7000-20000					
Magnesium	typ. alloys	1.76-1.83	41-45	15-17	0.35	80-280	140-340	2-20				26.1-28.8
Monel	67%Ni,30% Cu	8.8	170	66	0.32	170-1100	450-1200	2-50				14
Nickel		8.8	210	80	0.31	100-620	310-760	2-50				13

Table 1:[illegible]

Table 1:

Material	Type	Density	E	G	poisson	ten - yield stress	ten - ult. stress	ten - yield strain	yield stress - comp.	ult. stress - comp.	ult. stress - shear	thermal expansion
		(Mg/m ³)	(GPa)	(GPa)		(MPa)		(%)	(MPa)			10 ⁻⁶ /deg C
	astm-a47											
	astm-a36					250	400	30				12
	astm-a5242											
	astm-a441											
	astm-a572					340	500	20				12
	astm-a514					700	830	15				12
	wire					280-1000	550-1400	5-40				
Stainless Steel	aisi 302											17
Titanium	typ. alloys	4.5	100-120	39-44	0.33	760-1000	900-1200	10				8.1-11
Tungsten		1.9	340-380	140-160	0.2		1400-4000	0-4				4.3
Water	fresh	1.0										
	sea	1.02										
Wood (dry)												
	Douglas fir	.48-.56	11-13			30-50	50-80		30-50	40-70		
	Oak	.64-.72	11-12			40-60	50-100		30-40	30-50		
	Southern pine	.56-.64	11-14			40-60	50-100		30-50	40-70		

4.1 FORMULA SHEET

- A collection of the essential mechanics of materials formulas are given below

Axial/Normal Stress/Strain

$$\sigma = \epsilon E = \frac{P}{A} \quad \delta = \epsilon L = \frac{PL}{AE}$$

Shear Stress/Strain

$$\tau = \gamma G \quad \gamma_{xy} = \frac{\tau_{xy}}{G}$$

Poisson's ratio

$$\nu = -\frac{\epsilon_{lateral}}{\epsilon_{longitudinal}} \quad E = 2G(1 + \nu)$$

$$\epsilon_x = \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E} - \nu \frac{\sigma_z}{E}$$

Torsion

$$\tau_{max} = \frac{Tc}{J} \quad \gamma_{max} = \frac{\tau_{max}}{G}$$

$$\phi = \frac{TL}{JG} \quad J = \frac{\pi r^4}{2} \quad (\text{cylinder})$$

$$\sigma_{ten} = -\sigma_{comp} = \tau \quad J = \frac{\pi}{2}(r_o^4 - r_i^4) \quad (\text{hollow tube})$$

$$\epsilon = \frac{\gamma}{2} \quad P = 2\pi fT$$

Beams

$$\sigma = -\frac{My}{I} \quad I = \frac{bh^3}{12} \quad (\text{for rectangle})$$

$$\tau = \frac{VQ}{Ib} \quad I = \frac{\pi r^4}{2} \quad (\text{for circle})$$

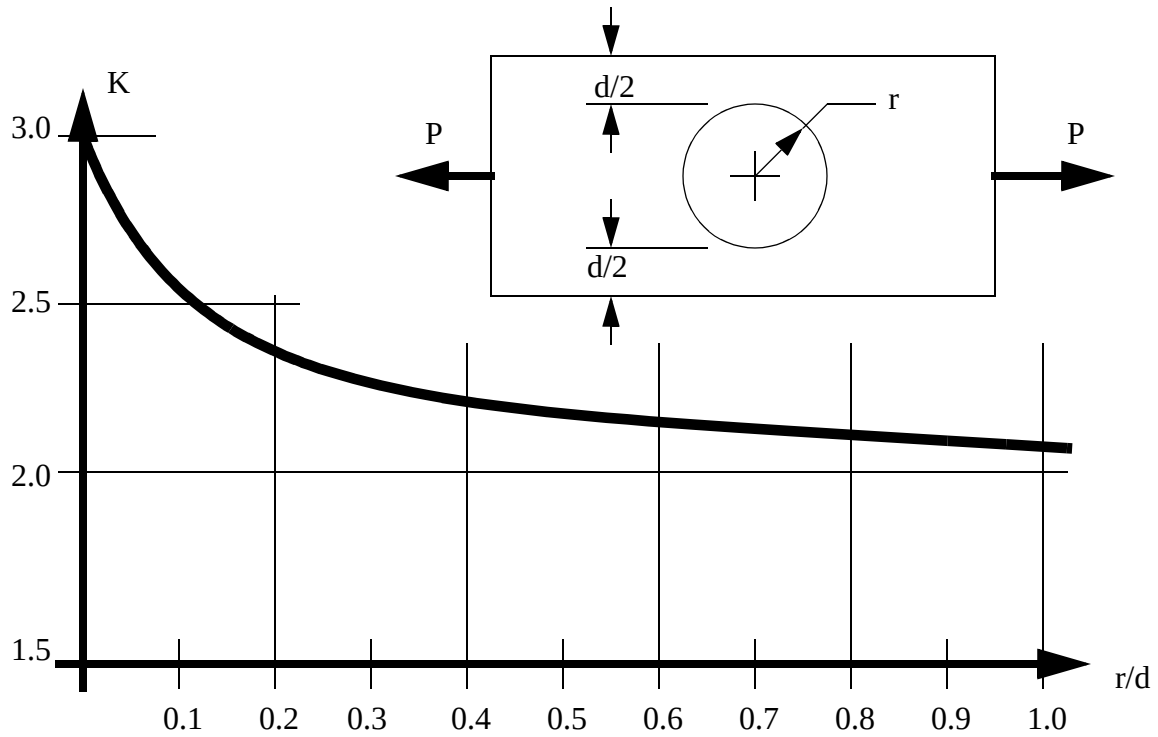
$$L = \rho \theta \quad \epsilon_{max} = \frac{c}{\rho}$$

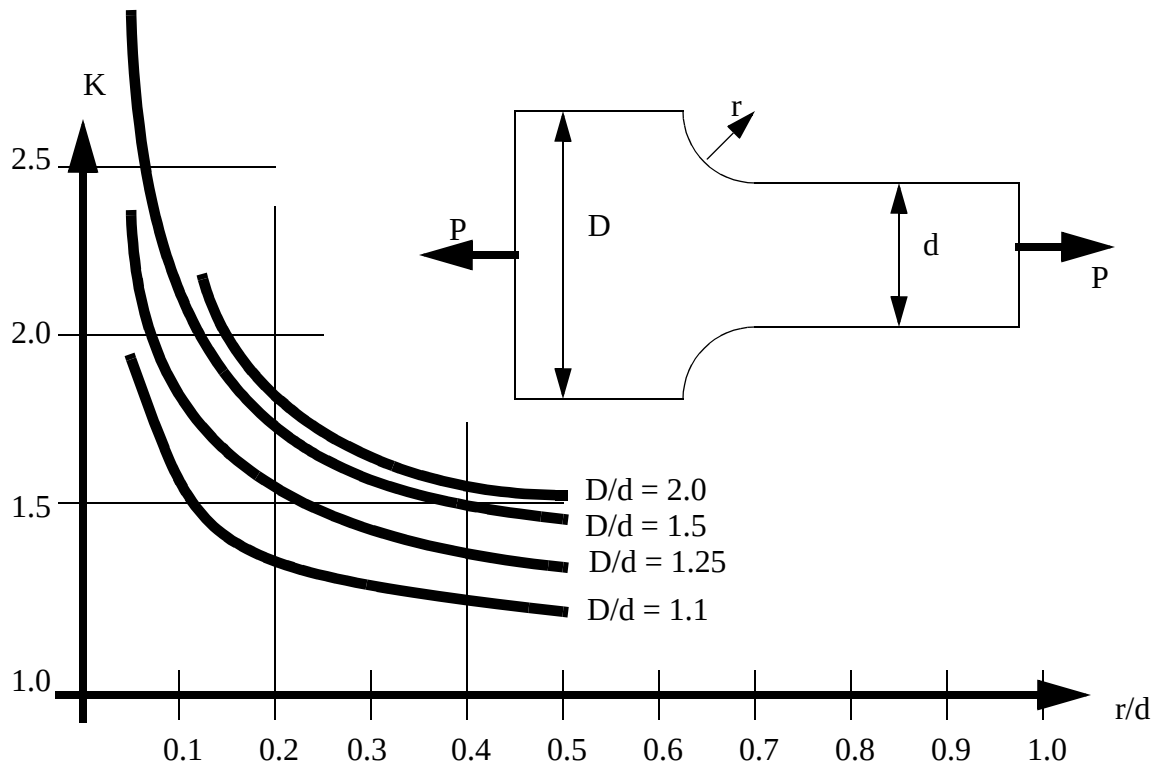
$$\sigma = -\frac{My}{I} \quad \rho = \frac{EI}{M}$$

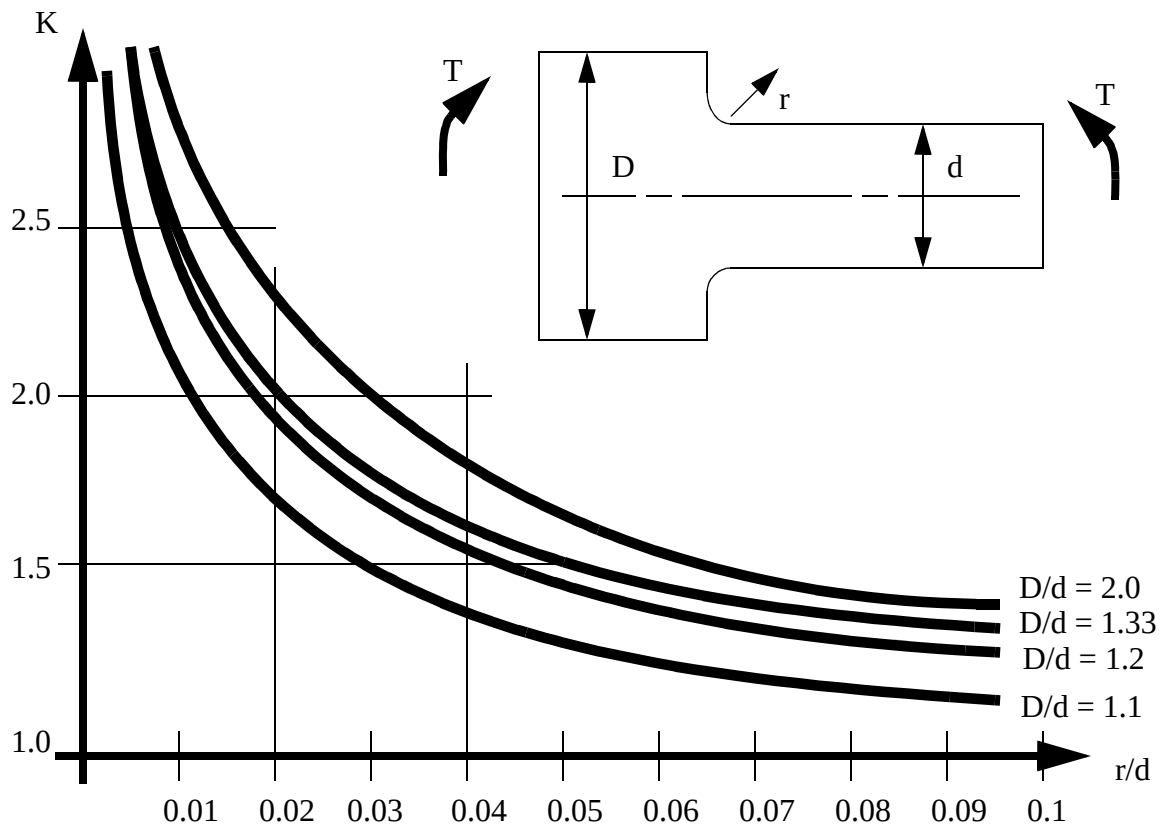
Buckling

$$P < \frac{\pi^2 EI}{k^2 L^2}$$

$k=0.5$ (both ends fixed)
 $k=0.7$ (one end fixed, one pinned)
 $k=1.0$ (both ends pinned)
 $k=2.0$ (one end pinned, one free)







5. UNITS AND CONVERSIONS

- Units are essential when describing real things.
- Good engineering practice demands that each number should always be accompanied with a unit.

5.1 HOW TO USE UNITS

- This table does not give an exhaustive list of conversion factors, but instead a minimal (but fairly complete) set is given. From the values below any conversion value can be derived. If you are not sure about this, ask the instructor to show you how.
- A simple example of unit conversion is given below,

****a simple unit conversion example:**

Given,

$$d_x = 10m \quad d_y = 5ft$$

Find the distance 'd',

$$d = \sqrt{d_x^2 + d_y^2}$$

$$\therefore d = \sqrt{(10m)^2 + (5ft)^2}$$

$$\therefore d = \sqrt{100m^2 + 25ft^2}$$

$$\therefore d = \sqrt{100m^2 + 25ft^2 \left(\frac{0.3048m}{1ft} \right)^2}$$

$$\therefore d = \sqrt{100m^2 + 25ft^2 (0.092903) \frac{m^2}{ft^2}}$$

$$\therefore d = \sqrt{100m^2 + 25(0.092903)m^2}$$

$$\therefore d = \sqrt{102.32m^2} = 10.12m$$

keep the units in the equation

multiply by 1

From the tables

$$1ft = 0.3048m$$

$$\therefore 1 = \frac{0.3048m}{1ft}$$

cancel out units

5.2 HOW TO USE SI UNITS

1. Beware upper/lower case letter in many cases they can change meanings.
e.g. m = milli, mega
 2. Try to move prefixes out of the denominator of the units.
e.g., N/cm or KN/m
 3. Use a slash or exponents.
e.g., (kg•m/s²) or (kg•m•s⁻²)
 4. Use a dot in compound units.
e.g., N•m
 5. Use spaces to divide digits when there are more than 5 figures, commas are avoided because their use is equivalent to decimal points in some cultures.
- In some cases units are non-standard. There are two major variations US units are marked with 'US' and Imperial units are marked with 'IMP'.

5.3 THE TABLE

Major Division

Distance

1 ft. (feet) = 12 in. (inches) = 0.3048 m (meter)
 1 mile = 1760 yards = 5280 ft = 1.609km
 1 in.(inch) = 2.540 cm
 1 yd (yard) = 3 ft.
 1 nautical mile = 6080 ft. = 1852 m = 1.150782 mi
 1 micron = 10⁻⁶ m
 1 angstrom = 10⁻¹⁰ m
 1 mil = 10⁻³ in
 1 acre = 43,560 ft. = 0.4047 hectares
 1 furlong = 660 ft
 1 lightyear = 9.460528e15 m
 1 parsec = 3.085678e16 m

Area

1 acre = 43,559.66 ft²
 1 Hectare (ha) = 10,000 m²
 1 Hectare (ha) = 10,000 m²
 1 Hectare (ha) = 10,000 m²
 1 Hectare (ha) = 10,000 m²

Velocity

1 mph = 0.8689762 knot

Angle

1 rev = 2PI radians = 360 degrees = 400 gradians

1 degree = 60 minutes

1 minute = 60 seconds

Volume

1 US gallon = 231 in³

1 CC = 1 cm³

1 IMP gallon = 277.274 in³

1 barrel = 31 IMP gal. = 31.5 US gal.

1 US gal. = 3.785 l = 4 quarts = 8 pints = 16 cups

1 liter (l) = 0.001 m³ = 2.1 pints (pt) = 1.06 quarts (qt) = 0.26 gallons (gal)

1 qt (quart) = 0.9464 l

1 cup (c) = 0.2365882 l = 8 USoz

1 US oz = 1 dram = 456.0129 drops = 480 US minim = 1.040842 IMP oz
= 2 tablespoons = 6 teaspoons

1 IMP gal. = 1.201 U.S. gal.

1 US pint = 16 US oz

1 IMP pint = 20 IMP oz

1 tablespoon = 0.5 oz.

1 bushel = 32 quarts

1 peck = 8 quarts

Force/Mass

1 N (newton) = 1 kg•m/s² = 100,000 dyne

1 dyne = 2.248*10⁻⁶ lb. (pound)

1 kg = 9.81 N (on earth surface) = 2.2046 lb

1 lb = 16 oz. (ounce) = 4.448N

1 oz. = 28.35 g (gram) = 0.2780N

1 lb = 0.03108 slug

1 kip = 1000 lb.

1 slug = 14.59 kg

1 imperial ton = 2000 lb = 907.2 kg

1 metric tonne = 1000 kg

1 troy oz = 480 grain (gr)

1 g = 5 carat

1 pennyweight = 24 grain

1 stone = 14 lb

1 long ton = 2240 lb

1 short ton = 2000 lb

Pressure

$$1 \text{ Pascal (Pa)} = 1 \text{ N/m}^2 = 6.895 \text{ kPa}$$

$$1 \text{ atm (metric atmos.)} = 760 \text{ mmHg at } 0^\circ\text{C} = 14.223 \text{ lb/in}^2 = 1.0132 \times 10^5 \text{ N/m}^2$$

$$1 \text{ psi} = 2.0355 \text{ in. Hg at } 32^\circ\text{F} = 2.0416 \text{ in. Hg at } 62^\circ\text{F}$$

$$1 \text{ microbar} = 0.1 \text{ N/m}^2$$

Scale/Magnitude

$$\text{atto (a)} = 10^{-18}$$

$$\text{femto (f)} = 10^{-15}$$

$$\text{pico (p)} = 10^{-12}$$

$$\text{nano (n)} = 10^{-9}$$

$$\text{micro } (\mu) = 10^{-6}$$

$$\text{milli (m)} = 10^{-3}$$

$$\text{centi (c)} = 10^{-2}$$

$$\text{deci (d)} = 10^{-1}$$

$$\text{deka (da)} = 10$$

$$\text{hecto (H)} = 10^2$$

$$\text{kilo (K)} = 10^3$$

$$\text{mega (M)} = 10^6$$

$$\text{giga (G)} = 10^9$$

$$\text{tera (T)} = 10^{12}$$

$$\text{peta (P)} = 10^{15}$$

$$\text{exa (E)} = 10^{18}$$

Power

$$1 \text{ h.p. (horsepower)} = 745.7 \text{ W (watts)} = 2.545 \text{ BTU/hr.} = 550 \text{ ft.lb./sec.}$$

$$1 \text{ ft} \cdot \text{lb/s} = 1.356 \text{ W}$$

$$1 \text{ J (joule)} = 1 \text{ N} \cdot \text{m} = 10^7 \text{ ergs} = 0.2389 \text{ cal.}$$

$$1 \text{ W} = 1 \text{ J/s}$$

$$1 \text{ ev} = 1.60219 \times 10^{-19} \text{ J}$$

$$1 \text{ erg} = 10^{-7} \text{ J}$$

Temperature

$$^\circ\text{F} = [(^\circ\text{C} \times 9)/5] + 32, \text{ } ^\circ\text{C} = \text{Celsius (Centigrade)}, \text{ F} = \text{Fahrenheit}$$

$$\text{K} = \text{Kelvin}$$

$$\text{Rankine (R)} = \text{F} - 459.666$$

$$0.252 \text{ calories} = 1 \text{ BTU (British Thermal Unit)}$$

$$-273.2 \text{ } ^\circ\text{C} = -459.7 \text{ } ^\circ\text{F} = 0 \text{ K} = 0 \text{ R} = \text{absolute zero}$$

$$0 \text{ } ^\circ\text{C} = 32 \text{ } ^\circ\text{F} = 273.3 \text{ K} = 491.7 \text{ R} = \text{Water Freezes}$$

$$100 \text{ } ^\circ\text{C} = 212 \text{ } ^\circ\text{F} = 373.3 \text{ K} = 671.7 \text{ R} = \text{Water Boils (1 atm. pressure)}$$

$$1 \text{ therm} = 100,000 \text{ BTU}$$

Mathematical

π radians = 3.1416 radians = 180 degrees = 0.5 cycles
 1 Hz = 1 cycle/sec.
 1 rpm (revolutions per minute) = 60 RPS (Revolutions per second) = 60Hz
 1 fps (foot per second) = 1 ft/sec
 1 mph (miles per hour) = 1 mi./hr.
 1 cfm (cubic foot per minute) = 1 ft³/min.
 e = 2.718

Time

1 Hz (hertz) = 1 s⁻¹
 1 year = 365 days = 52 weeks = 12 months
 1 leap year = 366 days
 1 day = 24 hours
 1 fortnight = 14 days
 1 hour = 60 min.
 1 min = 60 seconds
 1 millenium = 1000 years
 1 century = 100 years
 1 decade = 10 years

Physical Constants

R = 1.987 cal/mole K = ideal gas law constant
 K = Boltzmann's constant = 1.3×10^{-16} erg/K = 1.3×10^{-23} J/K
 h = Planck's constant = 6.62×10^{-27} erg-sec = 6.62×10^{-34} J.sec
 Avagadro's number = 6.02×10^{23} atoms/atomic weight
 density of water = 1 g/cm³
 electron charge = 1.60×10^{-19} coul.
 electron rest mass = 9.11×10^{-31} Kg
 proton rest mass = 1.67×10^{-27} Kg
 speed of light (c) = 3.00×10^{10} cm/sec
 speed of sound in dry air 25 C = 331 m/s
 gravitational constant = 6.67×10^{-11} Nm²/Kg²
 permittivity of free space = 8.85×10^{-12} farad/m
 permeability of free space = 1.26×10^{-6} henry/m
 mean radius of earth = 6370 Km
 mass of earth = 5.98×10^{24} Kg

Electromagnetic

magnetic flux = weber (We) = 10^8 maxwell
 inductance = henry
 magnetic flux density = tesla (T) = 10^4 gauss
 magnetic intensity = ampere/m = 0.004*PI oersted

electric flux density = coulomb/m**2
capacitance = farad
permeability = henry/m
electric field strength = V/m
luminous flux = lumen
luminance = candela/m**2
1 flame = 4 foot candles = 43.05564 lux = 43.05564 meter-candles
illumination = lux
resistance = ohm

5.4 ASCII, HEX, BINARY CONVERSION

- The table below will allow conversions between decimal, binary, hexadecimal, and ASCII values. The values shown only go up to 127. ASCII values above this are not commonly used in robust applications.

decimal	hexadecimal	binary	ASCII	decimal	hexadecimal	binary	ASCII
0	0	00000000	NUL	32	20	00100000	space
1	1	00000001	SOH	33	21	00100001	!
2	2	00000010	STX	34	22	00100010	“
3	3	00000011	ETX	35	23	00100011	#
4	4	00000100	EOT	36	24	00100100	\$
5	5	00000101	ENQ	37	25	00100101	%
6	6	00000110	ACK	38	26	00100110	&
7	7	00000111	BEL	39	27	00100111	‘
8	8	00001000	BS	40	28	00101000	(
9	9	00001001	HT	41	29	00101001	)
10	A	00001010	LF	42	2A	00101010	*
11	B	00001011	VT	43	2B	00101011	+
12	C	00001100	FF	44	2C	00101100	,
13	D	00001101	CR	45	2D	00101101	-
14	E	00001110	S0	46	2E	00101110	.
15	F	00001111	S1	47	2F	00101111	/
16	10	00010000	DLE	48	30	00110000	0
17	11	00010001	DC1	49	31	00110001	1
18	12	00010010	DC2	50	32	00110010	2
19	13	00010011	DC3	51	33	00110011	3
20	14	00010100	DC4	52	34	00110100	4
21	15	00010101	NAK	53	35	00110101	5
22	16	00010110	SYN	54	36	00110110	6
23	17	00010111	ETB	55	37	00110111	7
24	18	00011000	CAN	56	38	00111000	8
25	19	00011001	EM	57	39	00111001	9
26	1A	00011010	SUB	58	3A	00111010	:
27	1B	00011011	ESC	59	3B	00111011	;
28	1C	00011100	FS	60	3C	00111100	<
29	1D	00011101	GS	61	3D	00111101	=
30	1E	00011110	RS	62	3E	00111110	>
31	1F	00011111	US	63	3F	00111111	?

decimal	hexadecimal	binary	ASCII	decimal	hexadecimal	binary	ASCII
64	40	01000000	@	96	60	01100000	`
65	41	01000001	A	97	61	01100001	a
66	42	01000010	B	98	62	01100010	b
67	43	01000011	C	99	63	01100011	c
68	44	01000100	D	100	64	01100100	d
69	45	01000101	E	101	65	01100101	e
70	46	01000110	F	102	66	01100110	f
71	47	01000111	G	103	67	01100111	g
72	48	01001000	H	104	68	01101000	h
73	49	01001001	I	105	69	01101001	i
74	4A	01001010	J	106	6A	01101010	j
75	4B	01001011	K	107	6B	01101011	k
76	4C	01001100	L	108	6C	01101100	l
77	4D	01001101	M	109	6D	01101101	m
78	4E	01001110	N	110	6E	01101110	n
79	4F	01001111	O	111	6F	01101111	o
80	50	01010000	P	112	70	01110000	p
81	51	01010001	Q	113	71	01110001	q
82	52	01010010	R	114	72	01110010	r
83	53	01010011	S	115	73	01110011	s
84	54	01010100	T	116	74	01110100	t
85	55	01010101	U	117	75	01110101	u
86	56	01010110	V	118	76	01110110	v
87	57	01010111	W	119	77	01110111	w
88	58	01011000	X	120	78	01111000	x
89	59	01011001	Y	121	79	01111001	y
90	5A	01011010	Z	122	7A	01111010	z
91	5B	01011011	[	123	7B	01111011	{
92	5C	01011100	yen	124	7C	01111100	
93	5D	01011101	]	125	7D	01111101	}
94	5E	01011110	^	126	7E	01111110	r arr.
95	5F	01011111	_	127	7F	01111111	l arr.

5.5 G-CODES

- A basic list of 'G' operation codes is given below. These direct motion of the tool.
G00 - Rapid move (not cutting)

G01 - Linear move
G02 - Clockwise circular motion
G03 - Counterclockwise circular motion
G04 - Dwell
G05 - Pause (for operator intervention)
G08 - Acceleration
G09 - Deceleration
G17 - x-y plane for circular interpolation
G18 - z-x plane for circular interpolation
G19 - y-z plane for circular interpolation
G20 - turning cycle or inch data specification
G21 - thread cutting cycle or metric data specification
G24 - face turning cycle
G25 - wait for input #1 to go low (Prolight Mill)
G26 - wait for input #1 to go high (Prolight Mill)
G28 - return to reference point
G29 - return from reference point
G31 - Stop on input (INROB1 is high) (Prolight Mill)
G33-35 - thread cutting functions (Emco Lathe)
G35 - wait for input #2 to go low (Prolight Mill)
G36 - wait for input #2 to go high (Prolight Mill)
G40 - cutter compensation cancel
G41 - cutter compensation to the left
G42 - cutter compensation to the right
G43 - tool length compensation, positive
G44 - tool length compensation, negative
G50 - Preset position
G70 - set inch based units or finishing cycle
G71 - set metric units or stock removal
G72 - indicate finishing cycle (EMCO Lathe)
G72 - 3D circular interpolation clockwise (Prolight Mill)
G73 - turning cycle contour (EMCO Lathe)
G73 - 3D circular interpolation counter clockwise (Prolight Mill)
G74 - facing cycle contour (Emco Lathe)
G74.1 - disable 360 deg arcs (Prolight Mill)
G75 - pattern repeating (Emco Lathe)
G75.1 - enable 360 degree arcs (Prolight Mill)
G76 - deep hole drilling, cut cycle in z-axis
G77 - cut-in cycle in x-axis
G78 - multiple threading cycle
G80 - fixed cycle cancel
G81-89 - fixed cycles specified by machine tool manufacturers
G81 - drilling cycle (Prolight Mill)
G82 - straight drilling cycle with dwell (Prolight Mill)
G83 - drilling cycle (EMCO Lathe)
G83 - peck drilling cycle (Prolight Mill)

G84 - tapping cycle (EMCO Lathe)
G85 - reaming cycle (EMCO Lathe)
G85 - boring cycle (Prolight mill)
G86 - boring with spindle off and dwell cycle (Prolight Mill)
G89 - boring cycle with dwell (Prolight Mill)
G90 - absolute dimension program
G91 - incremental dimensions
G92 - Spindle speed limit
G93 - Coordinate system setting
G94 - Feed rate in ipm (EMCO Lathe)
G95 - Feed rate in ipr (EMCO Lathe)
G96 - Surface cutting speed (EMCO Lathe)
G97 - Rotational speed rpm (EMCO Lathe)
G98 - withdraw the tool to the starting point or feed per minute
G99 - withdraw the tool to a safe plane or feed per revolution
G101 - Spline interpolation (Prolight Mill)

• M-Codes control machine functions and these include,

M00 - program stop
M01 - optional stop using stop button
M02 - end of program
M03 - spindle on CW
M04 - spindle on CCW
M05 - spindle off
M06 - tool change
M07 - flood with coolant
M08 - mist with coolant
M08 - turn on accessory #1 (120VAC outlet) (Prolight Mill)
M09 - coolant off
M09 - turn off accessory #1 (120VAC outlet) (Prolight Mill)
M10 - turn on accessory #2 (120VAC outlet) (Prolight Mill)
M11 - turn off accessory #2 (120VAC outlet) (Prolight Mill) or tool change
M17 - subroutine end
M20 - tailstock back (EMCO Lathe)
M20 - Chain to next program (Prolight Mill)
M21 - tailstock forward (EMCO Lathe)
M22 - Write current position to data file (Prolight Mill)
M25 - open chuck (EMCO Lathe)
M25 - set output #1 off (Prolight Mill)
M26 - close chuck (EMCO Lathe)
M26 - set output #1 on (Prolight Mill)
M30 - end of tape (rewind)
M35 - set output #2 off (Prolight Mill)
M36 - set output #2 on (Prolight Mill)
M38 - put stepper motors on low power standby (Prolight Mill)
M47 - restart a program continuously, or a fixed number of times (Prolight Mill)

M71 - puff blowing on (EMCO Lathe)
M72 - puff blowing off (EMCO Lathe)
M96 - compensate for rounded external curves
M97 - compensate for sharp external curves
M98 - subprogram call
M99 - return from subprogram, jump instruction
M101 - move x-axis home (Prolight Mill)
M102 - move y-axis home (Prolight Mill)
M103 - move z-axis home (Prolight Mill)

- Other codes and keywords include,

A_{nnn} - an orientation, or second x-axis spline control point
B_{nnn} - an orientation, or second y-axis spline control point
C_{nnn} - an orientation, or second z-axis spline control point, or chamfer
F_{nnn} - a feed value (in ipm or m/s, not ipr), or thread pitch
I_{nnn} - x-axis center for circular interpolation, or first x-axis spline control point
J_{nnn} - y-axis center for circular interpolation, or first y-axis spline control point
K_{nnn} - z-axis center for circular interpolation, or first z-axis spline control point
L_{nnn} - arc angle, loop counter and program cycle counter
N_{nnn} - a sequence/line number
O_{nnn} - subprogram block number
P_{nnn} - subprogram reference number
R_{nnn} - a clearance plane for tool movement, or arc radius, or taper value
Q_{nnn} - peck depth for pecking cycle
S_{nnn} - cutting speed (rpm), spindle speed
T_{nnn} - a tool number
U_{nnn} - relative motion in x
V_{nnn} - relative motion in y
W_{nnn} - relative motion in z
X_{nnn} - an x-axis value
Y_{nnn} - a y-axis value
Z_{nnn} - a z-axis value
; - starts a comment (proLight Mill), or end of block (EMCO Lathe)

6. COMBINED GLOSSARY OF TERMS

6.1 A

abort - the disruption of normal operation.

absolute pressure - a pressure measured relative to zero pressure.

absorption Loss - when sound or vibration energy is lost in a transmitting or reflecting medium. This is the result of generation of other forms of energy such as heat.

absorbtive law - a special case of Boolean algebra where $A(A+B)$ becomes A .

AC (Alternating Current) - most commonly an electrical current and voltage that changes in a sinusoidal pattern as a function of time. It is also used for voltages and currents that are not steady (DC). Electrical power is normally distributed at 60Hz or 50Hz.

AC contactor - a contactor designed for AC power.

acceptance Test - a test for evaluating a newly purchased system's performance, capabilities, and conformity to specifications, before accepting, and paying the supplier.

accumulator - 1. a temporary data register in a computer CPU 2.

accuracy - the difference between an ideal value and a physically realizable value. The companion to accuracy is repeatability.

acidity - a solution that has an excessive number of hydrogen atoms. Acids are normally corrosive.

acoustic - another term for sound.

acknowledgement (ACK) - a response that indicates that data has been transmitted correctly.

actuator - a device that when activated will result in a mechanical motion. For example a motor, a solenoid valve, etc.

A/D - Analog to digital converter (see ADC).

ADC (Analog to Digital Converter) - a circuit that will convert an analog voltage to a digital value, also referred to as A/D.

ADCCP (Advanced Data Communications Procedure) - ANSI standard for synchronous commu-

nication links with primary and secondary functions.

address - a code (often a number) that specifies a location in a computers memory.

address register - a pointer to memory locations.

adsorption - the ability of a material or apparatus to adsorb energy.

agitator - causes fluids or gases to mix.

AI (Artificial Intelligence) - the use of computer software to mimic some of the cognitive human processes.

algorithms - a software procedure to solve a particular problem.

aliasing - in digital systems there are natural limits to resolution and time that can be exceeded, thus aliasing the data. For example. an event may happen too fast to be noticed, or a point may be too small to be displayed on a monitor.

alkaline - a solution that has an excess of HO pairs will be a base. This is the compliment to an acid.

alpha rays - ions that are emitted as the result of atomic fission or fusion.

alphanumeric - a sequence of characters that contains both numbers and letters.

ALU (Arithmetic Logic Unit) - a part of a computer that is dedicated to mathematical operations.

AM (Amplitude Modulation) - a fixed frequency carrier signal that is changed in amplitude to encode a change in a signal.

ambient - normal or current environmental conditions.

ambient Noise - a sort of background noise that is difficult to isolate, and tends to be present throughout the volume of interest.

ambient temperature - the normal temperature of the design environment.

analog signal - a signal that has continuous values, typically voltage.

analysis - the process of review to measure some quality.

and - a Boolean operation that requires all arguments to be true before the result is true.

annealing - heating of metal to relieve internal stresses. In many cases this may soften the material.

annotation - a special note added to a design for explanatory purposes.

ANSI (American National Standards Organization) - a developer of standards, and a member of ISO.

APF (All Plastic Fibre Cable) -

API (Application Program Interface) - a set of functions, and procedures that describes how a program will use another service/library/program/etc.

APPC (Advanced Program to Program Communication) -

APT (Automatically Programmed Tools) - a language used for directing computer controlled machine tools.

application - the task which a tool is put to, This normally suggests some level of user or real world interaction.

application layer - the top layer in the OSI model that includes programs the user would run, such as a mail reader.

arc - when the electric field strength exceeds the dielectric breakdown voltage, electrons will flow.

architecture - the general layout or design at a higher level.

armature - the central rotating portion of a DC motor or generator, or a moving part of a relay.

ARPA (Advanced Research Projects Agency) - now DARPA. Originally funded ARPANET.

ARPANET - originally sponsored by ARPA. A packet switching network that was in service from the early 1970s, until 1990.

ASCII (American Standard Code for Information Interchange) - a set of numerical codes that correspond to numbers, letters, special characters, and control codes. The most popular standard

ASIC (Application Specific Integrated Circuit) - a specially designed and programmed logic circuit. Used for medium to low level production of complex functions.

aspirator - a device that moves materials with suction.

assembler - converts assembly language into machine code.

assembly language - a mnemonic set of commands that can be directly converted into commands for a CPU.

associative dimensioning - a method for linking dimension elements to elements in a drawing.

associative laws - Boolean algebra laws $A+(B+C) = (A+B)+C$ or $A(BC) = (AB)C$

asynchronous - events that happen on an irregular basis, and are not predictable.

asynchronous communications (serial) - strings of characters (often ASCII) are broken down into a series of on/off bits. These are framed with start/stop bits, and parity checks for error detection, and then send out one character at a time. The use of start bits allows the characters to be sent out at irregular times.

attenuation - to decrease the magnitude of a signal.

attenuation - as the sound/vibration energy propagates, it will undergo losses. The losses are known as attenuation, and are often measured in dB. For general specifications, the attenuation may be tied to units of dB/ft.

attribute - a nongraphical feature of a part, such as color.

audible Range - the range of frequencies that the human ear can normally detect from 16 to 20,000 Hz.

automatic control - a feedback of a system state is compared to a desired value and the control value for the system is adjusted by electronics, mechanics and/or computer to compensate for differences.

automated - a process that operates without human intervention.

auxiliary power - secondary power supplies for remote or isolated systems.

AWG (American Wire Gauge) - specifies conductor size. As the number gets larger, the conductors get smaller.

6.2 B

B-spline - a fitted curve/surface that is commonly used in CAD and graphic systems.

backbone - a central network line that ties together distributed networks.

background - in multitasking systems, processes may be running in the background while the user is working in the foreground, giving the user the impression that they are the only user of the machine (except when the background job is computationally intensive).

background suppression - the ability of a sensing system to discriminate between the signal of

interest, and background noise or signals.

backplane - a circuit board located at the back of a circuit board cabinet. The backplane has connectors that boards are plugged into as they are added.

backup - a redundant system to replace a system that has failed.

backward chaining - an expert system looks at the results and looks at the rules to see logically how to get there.

band pressure Level - when measuring the spectrum of a sound, it is generally done by looking at frequencies in a certain bandwidth. This bandwidth will have a certain pressure value that is an aggregate for whatever frequencies are in the bandwidth.

base - 1. a substance that will have an excess of HO ions in solution form. This will react with an acid. 2. the base numbering system used. For example base 10 is decimal, base 2 is binary

baseband - a network strategy in which there is a single carrier frequency, that all connected machines must watch continually, and participate in each transaction.

BASIC (Beginner's All-purpose Basic Instruction Code) - a computer language designed to allow easy use of the computer.

batch processing - an outdated method involving running only one program on a computer at once, sequentially. The only practical use is for very intensive jobs on a supercomputer.

battery backup - a battery based power supply that keeps a computer (or only memory) on when the master power is off.

BAUD () - The maximum number of bits that may be transmitted through a serial line in one second

baudot code - an old code similar to ASCII for teleprinter machines.

BCC (Block Check Character) - a character that can check the validity of the data in a block.

BCD (Binary Coded Decimal) - numerical digits (0 to 9) are encoded using 4 bits. This allows two numerical digits to each byte.

BCP (Byte Controlled Protocol) -

beam - a wave of energy waves such as light or sound. A beam implies that it is not radiating in all directions, and covers an arc or cone of a few degrees.

bearing - a mechanical support between two moving surfaces. Common types are ball bearings (light weight) and roller bearings (heavy weight), journal bearings (rotating shafts).

beats - if two different sound frequencies are mixed, they will generate other frequencies. if a 1000Hz and 1001Hz sound are heard, a 1Hz ($=1000-1001$) sound will be perceived.

benchmark - a figure to compare with. If talking about computers, these are often some numbers that can be use to do relative rankings of speeds, etc. If talking about design, we can benchmark our products against our competitors to determine our weaknesses.

Bernoulli's principle - a higher fluid flow rate will result in a lower pressure.

beta ratio - a ratio of pipe diameter to orifice diameter.

beta rays - electrons are emitted from a fission or fusion reaction.

beta site - a software tester who is actually using the software for practical applications, while looking for bugs. After this stage, software will be released commercially.

big-endian - a strategy for storing or transmitting the most significant byte first.

BIOS (Basic Input Output System) - a set of basic system calls for accessing hardware, or software services in a computer. This is typically a level lower than the operating system.

binary - a base 2 numbering system with the digits 0 and 1.

bit - a single binary digit. Typically the symbols 0 and 1 are used to represent the bit value.

bit/nibble/byte/word - binary numbers use a 2 value number system (as opposed to the decimal 0-9, binary uses 0-1). A bit refers to a single binary digit, and as we add digits we get larger numbers. A bit is 1 digit, a nibble is 4 digits, a byte is 8 digits, and a word is 16 digits.

decimal(base 10)	binary(base 2)	octal(base 8)
0	0	0
1	1	1
2	10	2
3	11	3
4	100	4
5	101	5
6	110	6
7	111	7
8	1000	10
9	1001	11
10	1010	12
11	1011	13
.	.	.
.	.	.
.	.	.

e.g. differences

decimal	15 ... tens 3,052 ... thousands 1,000,365 ... millions
binary	1 ... bit 0110 nibble (up to 16 values) 10011101 ... byte (up to 256 values) 0101000110101011 ... word (up to 64,256 values)
Most significant bit	least significant bit

BITNET (Because It's Time NET) - An academic network that has been merged with CSNET.

blackboard - a computer architecture when different computers share a common memory area (each has its own private area) for sharing/passing information.

block - a group of bytes or words.

block diagrams - a special diagram for illustrating a control system design.

binary - specifies a number system that has 2 digits, or two states.

binary number - a collection of binary values that allows numbers to be constructed. A binary number is base 2, whereas normal numbering systems are base 10.

blast furnace - a furnace that generates high temperatures by blowing air into the combustion.

bleed nozzle - a valve or nozzle for releasing pressure from a system.

block diagram - a symbolic diagram that illustrates a system layout and connection. This can be used for analysis, planning and/or programming.

BOC (Bell Operating Company) - there are a total of 7 regional telephone companies in the U.S.A.

boiler - a device that will boil water into steam by burning fuel.

BOM (Bills Of Materials) - list of materials needed in the production of parts, assemblies, etc. These lists are used to ensure all required materials are available before starting an operation.

Boolean - a system of numbers based on logic, instead of real numbers. There are many similarities to normal mathematics and algebra, but a separate set of operators, axioms, etc. are used.

bottom-up design - the opposite of top-down design. In this methodology the most simple/basic functions are designed first. These simple elements are then combined into more complex elements. This continues until all of the hierarchical design elements are complete.

bounce - switch contacts may not make absolute contact when switching. They make and break contact a few times as they are coming into contact.

Bourdon tube - a pressure tube that converts pressure to displacement.

BPS (Bits Per Second) - the total number of bits that can be passed between a sender and listener in one second. This is also known as the BAUD rate.

branch - a command in a program that can cause it to start running elsewhere.

bread board - a term used to describe a temporary electronic mounting board. This is used to prototype a circuit before doing final construction. The main purpose is to verify the basic design.

breadth first search - an AI search technique that examines all possible decisions before making the next move.

breakaway torque - the start-up torque. The value is typically high, and is a function of friction, inertia, deflection, etc.

breakdown torque - the maximum torque that an AC motor can produce at the rated voltage and frequency.

bridge - 1. an arrangement of (typically 4) balanced resistors used for measurement. 2. A network

device that connects two different networks, and sorts out packets to pass across.

broadband networks - multiple frequencies are used with multiplexing to increase the transmission rates in networks.

Broad-Band Noise - the noise spectrum for a particular noise source is spread over a large range of frequencies.

broadcast - a network term that describes a general broadcast that should be delivered to all clients on a network. For example this is how Ethernet sends all of its packets.

brush - a sliding electrical conductor that conducts power to/from a rotor.

BSC (Binary Synchronous Communication) - a byte oriented synchronous communication protocol developed by IBM.

BSD (Berkeley Software Distribution) - one of the major versions of UNIX.

buffer - a temporary area in which data is stored on its way from one place to another. Used for communication bottlenecks and asynchronous connections.

bugs - hardware or software problems that prevent desired components operation.

burn-in - a high temperature pre-operation to expose system problems.

burner - a term often used for a device that programs EPROMs, PALs, etc. or a bad cook.

bus - a computer has buses (collections of conductors) to move data, addresses, and control signals between components. For example to get a memory value, the address value provided the binary memory address, the control bus instructs all the devices to read/write, and to examine the address. If the address is valid for one part of the computer, it will put a value on the data bus that the CPU can then read.

byte - an 8 bit binary number. The most common unit for modern computers.

6.3 C

C - A programming language that followed B (which followed A). It has been widely used in software development in the 80s and 90s. It has grown up to become C++ and Java.

CAA (Computer Aided Analysis) - allows the user to input the definition of a part and calculate the performance variables.

cable - a communication wire with electrical and mechanical shielding for harsh environments.

CAD (Computer Aided Design) - is the creation and optimization of the design itself using the computer as a productivity tool. Components of CAD include computer graphics, a user interface, and geometric modelling.

CAD (Computer Aided Drafting) - is one component of CAD which allows the user to input engineering drawings on the computer screen and print them out to a plotter or other device.

CADD (Computer Aided Design Drafting) - the earliest forms of CAD systems were simple electronic versions of manual drafting, and thus are called CADD.

CAE (Computer Aided Engineering) - the use of computers to assist in engineering. One example is the use of Finite Element Analysis (FEA) to verify the strength of a design.

CALS (Computer Aided Acquisition and Logistics Support) -

CAM (Computer Aided Engineering) -

capacitor - a device for storing energy or mass.

capacitance - referring to the ability of a device to store energy. This is used for electrical capacitors, thermal masses, gas cylinders, etc.

capacity - the ability to absorb something else.

carrier - a high/low frequency signal that is used to transmit another signal.

carry flag - an indication when a mathematical operator has gone past the limitations of the hardware/software.

cascade - a method for connecting devices to increase their range, or connecting things so that they operate in sequence. This is also called chaining.

CASE (Computer Aided Software Engineering) - software tools are used by the developer/programmer to generate code, track changes, perform testing, and a number of other possible functions.

cassette - a holder for tapes normally.

CCITT (Consultative Committee for International Telegraph and Telephone) - recommended X25. A member of the ITU of the United Nations.

CD-ROM (Compact Disc Read Only Memory) - originally developed for home entertainment, these have turned out to be high density storage media available for all platforms at very low prices (< \$100 at the bottom end). The storage of these drives is well over 500 MB.

CE (Concurrent Engineering) -

CE - a mark placed on products to indicate that they conform to the standards set by the European Common Union.

Celsius - a temperature scale the uses 0 as the freezing point of water and 100 as the boiling point

centrifugal force - the force on an orbiting object the would cause it to accelerate outwards.

centripetal force - the force that must be applied to an orbiting object so that it will not fly outwards.

channel - an independent signal pathway.

character - a single byte, that when displayed is some recognizable form, such as a letter in the alphabet, or a punctuation mark.

checksum - when many bytes of data are transmitted, a checksum can be used to check the validity of the data. It is commonly the numerical sum of all of the bytes transmitted.

chip - a loose term for an integrated circuit.

chromatography - gases or liquids can be analyzed by how far their constituent parts can migrate through a porous material.

CIM (Computer Integrated Manufacturing) - computers can be used at a higher level to track and guide products as they move through the facility. CIM may or may not include CAD/CAM.

CL (Cutter Location) - an APT program is converted into a set of x-y-z locations stored in a CL file. In turn these are sent to the NC machine via tapes, etc.

clear - a signal or operation to reset data and status values.

client-server - a networking model that describes network services, and user programs.

clipping - the automatic cutting of lines that project outside the viewing area on a computer screen.

clock - a signal from a digital oscillator. This is used to make all of the devices in a digital system work synchronously.

clock speed - the rate at which a computers main time clock works at. The CPU instruction speed is usually some multiple or fraction of this number, but true program execution speeds are loosely related at best.

closed loop - a system that measures system performance and trims the operation. This is also

known as feedback. If there is no feedback the system is called open loop.

CMOS (Complimentary Metal Oxide Semi-conductor) - a low power microchip technology that has high noise immunity.

CNC (Computer Numerical Control) - machine tools are equipped with a control computer, and will perform a task. The most popular is milling.

coalescing - a process for filtering liquids suspended in air. The liquid condenses on glass fibers.

coaxial cable - a central wire contains a signal conductor, and an outer shield provides noise immunity. This configuration is limited by its coaxial geometry, but it provides very high noise immunity.

coax - see coaxial cable.

cogging - a machine steps through motions in a jerking manner. The result may be low frequency vibration.

coil - wire wound into a coil (tightly packed helix) used to create electromagnetic attraction. Used in relays, motors, solenoids, etc. These are also used alone as inductors.

collisions - when more than one network client tries to send a packet at any one time, they will collide. Both of the packets will be corrupted, and as a result special algorithms and hardware are used to abort the write, wait for a random time, and retry the transmission. Collisions are a good measure of network overuse.

colorimetry - a method for identifying chemicals using their colors.

combustion - a burning process generating heat and light when certain chemicals are added.

command - a computer term for a function that has an immediate effect, such as listing the files in a directory.

communication - the transfer of data between computing systems.

commutative laws - Booleans algebra laws $A+B = B+A$ and $AB=BA$.

compare - a computer program element that examines one or more variables, determines equality/inequality, and then performs some action, sometimes a branch.

compatibility - a measure of the similarity of a design to a standard. This is often expressed as a percentage for software. Anything less than 100% is not desirable.

compiler - a tool to change a high level language such as C into assembler.

complement - to take the logical negative. TRUE becomes false and vice versa.

component - an interchangeable part of a larger system. Components can be used to cut down manufacturing and maintenance difficulties.

compressor - a device that will decrease the volume of a gas - and increase the pressure.

computer - a device constructed about a central instruction processor. In general the computer can be reconfigured (software/firmware/hardware) to perform alternate tasks.

Computer Graphics - is the use of the computer to draw pictures using an input device to specify geometry and other attributes and an output device to display a picture. It allows engineers to communicate with the computer through geometry.

concentric - a shared center between two or more objects.

concurrent - two or more activities occur at the same time, but are not necessarily the same.

concurrent engineering - all phases of the products life are considered during design, and not later during design review stages.

condenser - a system component that will convert steam to water. Typically used in power generators.

conduction - the transfer of energy through some medium.

configuration - a numbers of multifunction components can be connected in a variety of configurations.

connection - a network term for communication that involves first establishing a connection, second data transmission, and third closing the connection. Connectionless networking does not require connection.

constant - a number with a value that should not vary.

constraints - are performance variables with limits. Constraints are used to specify when a design is feasible. If constraints are not met, the design is not feasible.

contact - 1. metal pieces that when touched will allow current to pass, when separated will stop the flow of current. 2. in PLCs contacts are two vertical lines that represent an input, or internal memory location.

contactor - a high current relay.

continuous Noise - a noise that is ongoing, and present. This differentiates from instantaneous, or intermittent noise sources.

continuous Spectrum - a noise has a set of components that are evenly distributed on a spectral graph.

control relay - a relay that does not control any external devices directly. It is used like a variable in a high level programming language.

control variable - a system parameter that we can set to change the system operation.

controls - a system that is attached to a process. Its purpose is to direct the process to some set value.

convection - the transfer of heat energy to liquid or gas that is moving past the surface of an object.

cook's constant - another name for the fudge factor

core memory - an outdated term describing memory made using small torii that could be polarized magnetically to store data bits. The term lives on when describing some concepts, for example a 'core dump' in UNIX. Believe it or not this has not been used for decades but still appears in many new textbooks.

coriolis force - a force that tends to cause spinning in moving frames of reference. Consider the direction of the water swirl down a drain pipe, it changes from the north to the south of the earth.

correction factor - a formal version of the 'fudge factor'. Typically a value used to multiply or add another value to account for hard to quantify values. This is the friend of the factor of safety.

counter - a system to count events. Tis can be either software or hardware.

cps (characters per second) - This can be a good measure of printing or data transmission speed, but it is not commonly used, instead the more confusing 'baud' is preferred.

CPU (Central Processing Unit) - the main computer element that examines machine code instructions and executes results.

CRC (Cyclic Redundancy Check) - used to check transmitted blocks of data for validity.

criteria - are performance variables used to measure the quality of a design. Criteria are usually defined in terms of degree - for example, lowest cost or smallest volume or lowest stress. Criteria are used to optimize a design.

crosstalk - signals in one conductor induce signals in other conductors, possibly creating false signals.

CRT (Cathode Ray Tubes) - are the display device of choice today. A CRT consists of a phosphor-coated screen and one or more electron guns to draw the screen image.

crucible - 1. a vessel for holding high temperature materials 2.

CSA (Canadian Standards Association) -

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) - a protocol that causes computers to use the same communication line by waiting for turns. This is used in networks such as Ethernet.

CSNET (Computer+Science NETwork) - a large network that was merged with BITNET.

CTS (Clear To Send) - used to prevent collisions in asynchronous serial communications.

current loop - communications that use a full electronic loop to reduce the effects of induced noise. RS-422 uses this.

current rating - this is typically the maximum current that a designer should expect from a system, or the maximum current that an input will draw. Although some devices will continue to work outside rated values, not all will, and thus this limit should be observed in a robust system. Note: exceeding these limits is unsafe, and should be done only under proper engineering conditions.

current sink - a device that allow current to flow through to ground when activated.

current source - a device that provides current from another source when activated.

cursors - are movable trackers on a computer screen which indicate the currently addressed screen position, or the focus of user input. The cursor is usually represented by an arrow, a flashing character or cross-hair.

customer requirements - the qualitative and quantitative minimums and maximums specified by a customer. These drive the product design process.

cycle - one period of a periodic function.

cylinder - a piston will be driven in a cylinder for a variety of purposes. The cylinder guides the piston, and provides a seal between the front and rear of the piston.

6.4 D

daisy chain - allows serial communication of devices to transfer data through each (and every) device between two points.

darlington coupled - two transistors are ganged together by connecting collectors to bases to increase the gain. These increase the input impedance, and reduce the back propagation of noise from loads.

DARPA (Defense Advanced Research Projects Agency) - replaced ARPA. This is a branch of the US department of defence that has participated in a large number of research projects.

data acquisition - refers to the automated collection of information collected from a process or system.

data highway - a term for a communication bus between two separated computers, or peripherals. This term is mainly used for PLC's.

data link layer - an OSI model layer

data logger - a dedicated system for data acquisition.

data register - stores data values temporarily in a CPU.

database - a software program that stores and recalls data in an organized way.

DARPA (Defense Advanced Research Projects Agency) -

DC (Direct Current) -

DCA (Defense Communications Agency) - developed DDN.

DCA (Document Content Architecture) -

DCD (Data Carrier Detect) - used as a handshake in asynchronous communication.

DCE (Data Communications Equipment) -

DCE (Distributed Computing Environment) - applications can be distributed over a number of computers because of the use of standards interfaces, functions, and procedures.

DDCMP (Digital Data Communication Message Protocol) -

DDN (Defense Data Network) - a group of DoD networks, including MILNET.

dead band - a region for a device when it no longer operates.

dead time - a delay between an event occurring and the resulting action.

debounce - a switch may not make sudden and complete contact as it is closes, circuitry can be

added to remove a few on-off transitions as the switch mechanically bounces.

debug - after a program has been written it undergoes a testing stage called debugging that involves trying to locate and eliminate logic and other errors. This is also a time when most engineers deeply regret not spending more time on the initial design.

decibel (dB) - a logarithmic compression of values that makes them more suited to human perception (for both scalability and reference)

decision support - the use of on-line data, and decision analysis tools are used when making decisions. One example is the selection of electronic components based on specifications, projected costs, etc.

DECnet (Digital Equipment Corporation net) - a proprietary network architecture developed by DEC.

decrement - to decrease a numeric value.

dedicated computer - a computer with only one task.

default - a standard condition.

demorgan's laws - Boolean laws great for simplifying equations $\sim(AB) = \sim A + \sim B$, or $\sim(A+B) = \sim A \sim B$.

density - a mass per unit volume.

depth first search - an artificial intelligence technique that follows a single line of reasoning first.

derivative control - a control technique that uses changes in the system of setpoint to drive the system. This control approach gives fast response to change.

design - creation of a new part/product based on perceived needs. Design implies a few steps that are ill defined, but generally include, rough conceptual design, detailed design, analysis, redesign, and testing.

design capture - the process of formally describing a design, either through drafted drawings, schematic drawings, etc.

design cycle - the steps of the design. The use of the word cycle implies that it never ends, although we must at some point decide to release a design.

design Variables - are the parameters in the design that describe the part. Design variables usually include geometric dimensions, material type, tolerances, and engineering notes.

detector - a device to determine when a certain condition has been met.

device driver - controls a hardware device with a piece of modular software.

DFA (Design For Assembly) - a method that guides product design/redesign to ease assembly times and difficulties.

DFT (Design for Testability) - a set of design axioms that generally calls for the reduction of test steps, with the greatest coverage for failure modes in each test step.

DIA (Document Interchange Architecture) -

diagnostic - a system or set of procedures that may be followed to identify where systems may have failed. These are most often done for mission critical systems, or industrial machines where the user may not have the technical capability to evaluate the system.

diaphragm - used to separate two materials, while allowing pressure to be transmitted.

differential -

differential amplifier - an amplifier that will subtract two or more input voltages.

diffuse field - multiple reflections result in a uniform and high sound pressure level.

digital - a system based on binary on-off values.

diode - a semiconductor device that will allow current to flow in one direction.

DIP switches - small banks of switches designed to have the same footprint as an integrated circuit.

DISOSS (DIStributed Office Support System) -

distributed - suggests that computer programs are split into parts or functions and run on different computers

distributed system - a system can be split into parts. Typical components split are mechanical, computer, sensors, software, etc.

DLE (Data Link Escape) - An RS-232 communications interface line.

DMA (Direct Memory Access) - used as a method of transferring memory in and out of a computer without slowing down the CPU.

DMD (Directory Management Domain) -

DNS (Domain Name System) - an internet method for name and address tracking.

documentation - (don't buy equipment without it) - one or more documents that instruct in the use, installation, setup, maintenance, troubleshooting, etc. for software or machinery. A poor design supported by good documentation can often be more useful than a good design unsupported by poor documentation.

domain - the basic name for a small or large network. For example (unc.edu) is the general extension for the University on North Carolina.

doppler shift - as objects move relative to each other, a frequency generated by one will be perceived at another frequency by the other.

DOS (Disk Operating System) - the portion of an operating system that handles basic I/O operations. The most common example is Microsoft MS-DOS for IBM PCs.

dotted decimal notation - the method for addressing computers on the internet with IP numbers such as '129.100.100.13'.

double pole - a double pole switch will allow connection between two contacts. These are useful when making motor reversers. see also single pole.

double precision - a real number is represented with 8 bytes (single precision is 4) to give more precision for calculations.

double throw - a switch or relay that has two sets of contacts.

download - to retrieve a program from a server or higher level computer.

downtime - a system is removed from production for a given amount of downtime.

drag - a force that is the result of a motion of an object in a viscous fluid.

drop - a term describing a short connection to peripheral I/O.

drum sequencer - a drum has raised/lowered sections and as it rotates it opens/closes contacts and will give sequential operation.

dry contact - an isolated output, often a relay switched output.

DSP (Digital Signal Processor) - a medium complexity microcontroller that has a built in floating point unit. These are very common in devices such as modems.

DSR (Data Set Ready) - used as a data handshake in asynchronous communications.

DTE (Data Terminal Equipment) - a serial communication line used in RS-232

DTR (Data Terminal Ready) - used as a data handshake in asynchronous communications to indicate a listener is ready to receive data.

dump - a large block of memory is moved at once (as a sort of system snapshot).

duplex - serial communication that is in both directions between computers at the same time.

dynamic braking - a motor is used as a brake by connecting the windings to resistors. In effect the motor becomes a generator, and the resistors dissipate the energy as heat.

dynamic variable - a variable with a value that is constantly changing.

dyne - a unit of force

6.5 E

EAROM (Electrically Alterable Read Only Memory) -

EBCDIC (Extended Binary-Coded Decimal Information Code) - a code for representing keyboard and control characters.

ECC (Error Correction Code) -

eccentric - two or more objects do not have a common center.

echo - a reflected sound wave.

ECMA (European Computer Manufacturer's Association) -

eddy currents - small currents that circulate in metals as currents flow in nearby conductors. Generally unwanted.

EDIF (Electronic Design Interchange Format) - a standard to allow the interchange of graphics and data between computers so that it may be changed, and modifications tracked.

EEPROM (Electrically Erasable Programmable Read Only Memory) -

effective sound pressure - the RMS pressure value gives the effective sound value for fluctuating pressure values. This value is some fraction of the peak pressure value.

EIA (Electronic Industries Association) -

electro-optic isolator - uses optical emitter, and photo sensitive switches for electrical isolation.

electromagnetic - a broad range term referring to magnetic waves. This goes from low frequency signals such as AM radio, up to very high frequency waves such as light and X-rays.

electrostatic - devices that use trapped charge to apply forces and cause distribution. An example is droplets of paint that have been electrically charged can be caused to disperse evenly over a surface that is oppositely charged.

electrostatic discharge - a sudden release of static electric charge (in nongrounded systems). This can lead to uncomfortable electrical shocks, or destruction of circuitry.

email (electronic mail) - refers to messages passed between computers on networks, that are sent from one user to another. Almost any modern computer will support some form of email.

EMI (ElectroMagnetic Interference) - transient magnetic fields cause noise in other systems.

emulsify - to mix two materials that would not normally mix. For example an emulsifier can cause oil and water to mix.

enable - a digital signal that allows a device to work.

encoding - a conversion between different data forms.

energize - to apply power to a circuit or component.

energy - the result of work. This concept underlies all of engineering. Energy is shaped, directed and focused to perform tasks.

engineering work stations - are self contained computer graphics systems with a local CPU which can be networked to larger computers if necessary. The engineering work station is capable of performing engineering synthesis, analysis, and optimization operations locally. Work stations typically have more than 1 MByte of RAM, and a high resolution screen greater than 512 by 512 pixels.

EOH (End of Header) -

EOT (End Of Transmission) - an ASCII code to indicate the end of a communications.

EPROM (Erasable Programmable Read Only Memory) -

EPS (Encapsulated PostScript) - a high quality graphics description language understood by high end printers. Originally developed by Adobe Systems Limited. This standard is becoming very popular.

error -

error signal - a control signal that is the difference between a desired and actual position.

ESD - see electrostatic discharge.

esters - a chemical that was formed by a reaction between alcohol and an acid.

ETX (End Of Text) -

even parity - a checksum bit used to verify data in other bits of a byte.

execution - when a computer is under the control of a program, the program is said to be executing.

expansion principle - when heat is applied a liquid will expand.

expert systems - is a branch of artificial intelligence designed to emulate human expertise with software. Expert systems are in use in many arenas and are beginning to be seen in CAD systems. These systems use rules derived from human experts.

6.6 F

fail safe - a design concept where system failure will bring the system to an idle or safe state.

false - a logical negative, or zero.

Faraday's electromagnetic induction law - if a conductor moves through a magnetic field a current will be induced. The angle between the motion and the magnetic field needs to be 90 deg for maximum current.

Fahrenheit - a temperature system that has 180 degrees between the freezing and boiling point of water.

fatal error - an error so significant that a software/hardware cannot continue to operate in a reliable manner.

fault - a small error that may be recoverable, or may result in a fatal error.

FAX (facsimile) - an image is scanned and transmitted over phone lines and reconstructed at the other end.

FCS (Frame Check Sequence) - data check flag for communications.

FDDI (Fibre Distributed Data Interface) - a fibre optic token ring network scheme in which the control tokens are counter rotating.

FDX (Full Duplex) - all characters that are transmitted are reflected back to the sender.

FEA (Finite Element Analysis) - is a numerical technique in which the analysis of a complex part is subdivided into the analysis of small simple subdivisions.

feedback - a common engineering term for a system that examines the output of a system and uses it to tune the system. Common forms are negative feedback to make systems stable, and positive feedback to make systems unstable (e.g. oscillators).

fetch - when the CPU gets a data value from memory.

fiberoptics - data can be transmitted by switching light on/off, and transmitting the signal through an optical fiber. This is becoming the method of choice for most long distance data lines because of the low losses and immunity to EMI.

FIFO (First In First Out) - items are pushed on a stack. The items can then be pulled back off last first.

file - a concept of a serial sequence of bytes that the computer can store information in, normally on the disk. This is a ubiquitous concept, but file is also used by Allen Bradley to describe an array of data.

filter - a device that will selectively pass matter or energy.

firmware - software stored on ROM (or equivalent).

flag - a single binary bit that indicates that an event has/has not happened.

flag - a single bit variable that is true or not. The concept is that if a flag is set, then some event has happened, or completed, and the flag should trigger some other event.

flame - an email, or netnews item that is overtly critical of another user, or an opinion. These are common because of the ad-hoc nature of the networks.

flange - a thick junction for joining two pipes.

floating point - uses integer math to represent real numbers.

flow chart - a schematic diagram for representing program flow. This can be used during design of software, or afterwards to explain its operation.

flow meter - a device for measuring the flow rate of fluid.

flow rate - the volume of fluid moving through an area in a fixed unit of time.

fluorescence - incoming UV light or X-ray strike a material and cause the emission of a different

frequency light.

FM (Frequency Modulation) - transmits a signal using a carrier of constant magnitude but changing frequency. The frequency shift is proportional to the signal strength.

force - a PLC output or input value can be set on artificially to test programs or hardware. This method is not suggested.

format - 1. a physical and/or data structure that makes data rereadable, 2. the process of putting a structure on a disk or other media.

forward chaining - an expert system approach to examine a set of facts and reason about the probable outcome.

fragmentation - the splitting of an network data packet into smaller fragments to ease transmission.

frame buffers - store the raster image in memory locations for each pixel. The number of colors or shades of gray for each pixel is determined by the number of bits of information for each pixel in the frame buffer.

free field - a sound field where none of the sound energy is reflected. Generally there aren't any nearby walls, or they are covered with sound absorbing materials.

frequency - the number of cycles per second for a sinusoidally oscillating vibration/sound.

friction - the force resulting from the mechanical contact between two masses.

FSK (Frequency Shift Keying) - uses two different frequencies, shifting back and forth to transmit bits serially.

FTP (File Transfer Protocol) - a popular internet protocol for moving files between computers.

fudge factor - a number that is used to multiply or add to other values to make the experimental and theoretical values agree.

full duplex - a two way serial communication channel can carry information both ways, and each character that is sent is reflected back to the sender for verification.

fuse - a device that will destruct when excessive current flows. It is used to protect the electrical device, humans, and other devices when abnormally high currents are drawn. Note: fuses are essential devices and should never be bypassed, or replaced with fuses having higher current rating.

6.7 G

galvanometer - a simple device used to measure currents. This device is similar to a simple DC motor.

gamma rays - high energy electromagnetic waves resulting from atomic fission or fusion.

gate - 1. a circuit that performs one of the Boolean algebra functions (i.e., and, or, not, etc.) 2. a connection between a runner and a part, this can be seen on most injection molded parts as a small bump where the material entered the main mold cavity.

gateway - translates and routes packets between dissimilar networks.

Geiger-Mueller tube - a device that can detect ionizing particles (eg, atomic radiation) using a gas filled tube.

global optimum - the absolute best solution to a problem. When found mathematically, the maximum or minimum cost/utility has been obtained.

gpm (gallons per minute) - a flow rate.

grafcet - a method for programming PLCs that is based on Petri nets. This is now known as SFCs and is part of the IEC 1131-3 standard.

gray code - a modified binary code used for noisy environments. It is devised to only have one bit change at any time. Errors then become extremely obvious when counting up or down.

ground - a buried conductor that acts to pull system neutral voltage values to a safe and common level. All electrical equipment should be connected to ground for safety purposes.

GUI (Graphical User Interface) - the user interacts with a program through a graphical display, often using a mouse. This technology replaces the older systems that use menus to allow the user to select actions.

6.8 H

half cell - a probe that will generate a voltage proportional to the hydrogen content in a solution.

half duplex - see HDX

handshake - electrical lines used to establish and control communications.

hard copy - a paper based printout.

hardware - a mechanical or electrical system. The 'functionality' is 'frozen' in hardware, and often difficult to change.

HDLC (High-level Data Link Control) - an ISO standard for communications.

HDX (Half Duplex) - a two way serial connection between two computer. Unlike FDX, characters that are sent are not reflected back to the sender.

head - pressure in a liquid that is the result of gravity.

hermetic seal - an airtight seal.

hertz - a measure of frequency in cycles per second. The unit is Hz.

hex - see hexadecimal.

hexadecimal - a base 16 number system where the digits are 0 to 9 then A to F, to give a total of 16 digits. This is commonly used when providing numbers to computers.

high - another term used to describe a Boolean true, logical positive, or one.

high level language - a language that uses very powerful commands to increase programming productivity. These days almost all applications use some form of high level language (i.e., basic, fortran, pascal, C, C++, etc.).

horsepower - a unit for measuring power

host - a networked (fully functional) computer.

hot backup - a system on-line that can quickly replace a failed system.

hydraulic - 1. a study of water 2. systems that use fluids to transmit power.

hydrocarbon - a class of molecules that contain carbon and hydrogen. Examples are propane, octane.

hysteresis - a sticking or lagging phenomenon that occurs in many systems. For example, in magnetic systems this is a small amount of magnetic repolarization in a reversing field, and in friction this is an effect based on coulomb friction that reverses sticking force.

Hz - see hertz

6.9 I

IAB (internet Activities Board) - the developer of internet standards.

IC (Integrated Circuit) - a microscopic circuit placed on a thin wafer of semiconductor.

IEC (International Electrical Commission) -

IEEE (Institute of Electrical and Electronics Engineers) -

IEEE802 - a set of standards for LANs and MANs.

IGES (Initial Graphics Exchange Specification) - a standard for moving data between various CAD systems. In particular the format can handle basic geometric entities, such as NURBS, but it is expected to be replaced by PDES/STEP in the near future.

impact instrument - measurements are made based by striking an object. This generally creates an impulse function.

impedance - In electrical systems this is both reactive and real resistance combined. This also applies to power transmission and flows in other types of systems.

impulse Noise - a short duration, high intensity noise. This type of noise is often associated with explosions.

increment - increase a numeric value.

inductance - current flowing through a coil will store energy in a magnetic field.

inductive heating - a metal part is placed inside a coil. A high frequency AC signal is passed through the coil and the resulting magnetic field melts the metal.

infrared - light that has a frequency below the visible spectrum.

inertia - a property where stored energy will keep something in motion unless there is energy added or released.

inference - to make a decision using indirect logic. For example if you are wearing shoes, we can infer that you had to put them on. Deduction is the complementary concept.

inference engine - the part of an expert system that processes rules and facts using forward or backward chaining.

Insertion Loss - barriers, hoods, enclosures, etc. can be placed between a sound source, and listener, their presence increases reverberant sound levels and decreases direct sound energy. The

increase in the reverberant sound is the insertion loss.

instruction set - a list of all of the commands that available in a programmable system. This could be a list of PLC programming mnemonics, or a list of all of the commands in BASIC.

instrument - a device that will read values from external sensors or probes, and might make control decision.

intake stroke - in a piston cylinder arrangement this is the cycle where gas or liquid is drawn into the cylinder.

integral control - a control method that looks at the system error over a long period of time. These controllers are relatively immune to noise and reduce the steady state error, but the do not respond quickly.

integrate - to combine two components with clearly separable functions to obtain a new single component capable of more complex functions.

intelligence - systems will often be able to do simple reasoning or adapt. This can mimic some aspects of human intelligence. These techniques are known as artificial intelligence.

intelligent device - a device that contains some ability to control itself. This reduces the number of tasks that a main computer must perform. This is a form of distributed system.

interface - a connection between a computer and another electrical device, or the real world.

interlock - a device that will inhibit system operation until certain conditions are met. These are often required for safety on industrial equipment to protect workers.

intermittent noise - when sounds change level fluctuate significantly over a measurement time period.

internet - an ad-hoc collection of networks that has evolved over a number of years to now include millions of computers in every continent, and by now every country. This network will continue to be the defacto standard for personal users. (commentary: The information revolution has begun already, and the internet has played a role previously unheard of by overcoming censorship and misinformation, such as that of Intel about the Pentium bug, a military coup in Russia failed because they were not able to cut off the flow of information via the internet, the Tiananmen square massacre and related events were widely reported via internet, etc. The last stage to a popular acceptance of the internet will be the World Wide Web accessed via Mosaic/ Netscape.)

internet address - the unique identifier assigned to each machine on the internet. The address is a 32 bit binary identifier commonly described with the dotted decimal notation.

interlacing - is a technique for saving memory and time in displaying a raster image. Each pass

alternately displays the odd and then the even raster lines. In order to save memory, the odd and even lines may also contain the same information.

interlock - a flag that ensures that concurrent streams of execution do not conflict, or that they cooperate.

interpreter - programs that are not converted to machine language, but slowly examined one instruction at a time as they are executed.

interrupt - a computer mechanism for temporarily stopping a program, and running another.

inverter - a logic gate that will reverse logic levels from TRUE to/from FALSE.

I/O (Input/Output) - a term describing anything that goes into or out of a computer.

IOR (Inclusive OR) - a normal OR that will be true when any of the inputs are true in any combinations. also see Exclusive OR (EOR).

ion - an atom, molecule or subatomic particle that has a positive or negative charge.

IP (internet Protocol) - the network layer (OSI model) definitions that allow internet use.

IP datagram - a standard unit of information on the internet.

ISDN (Integrated Services Digital Network) - a combined protocol to carry voice, data and video over 56KB lines.

ISO (International Standards Organization) -

isolation - electrically isolated systems have no direct connection between two halves of the isolating device. Sound isolation uses barriers to physically separate rooms.

isolation transformer - a transformer for isolating AC systems to reduce electrical noise.

6.10 J

JEC (Japanese Electrotechnical Committee) -

JIC (Joint International Congress) - drafted relay logic standards.

JIT (Just in Time) - a philosophy when setting up and operating a manufacturing system such that materials required arrive at the worksite just in time to be used. This cuts work in process, storage space, and a number of other logistical problems, but requires very dependable supplies

and methods.

jog - a mode where a motor will be advanced while a button is held, but not latched on. It is often used for clearing jams, and loading new material.

jump - a forced branch in a program

jumper - a short wire, or connector to make a permanent setting of hardware parameters.

6.11 K

k, K - specifies magnitudes. 1K = 1024, 1k = 1000 for computers, otherwise 1K = 1k = 1000.

Note - this is not universal, so double check the meanings when presented.

Kelvin - temperature units that place 0 degrees at absolute zero. The magnitude of one degree is the same as the Celsius scale.

KiloBaud, KBaud, KB, Baud - a transmission rate for serial communications (e.g. RS-232C, TTY, RS-422). A baud = 1bit/second, 1 Kilobaud = 1KBaud = 1KB = 1000 bits/second. In serial communication each byte typically requires 11 bits, so the transmission rate is about 1Kbaud/11 = 91 Bytes per second when using a 1KB transmission.

Karnaugh maps - a method of graphically simplifying logic.

kermit - a popular tool for transmitting binary and text files over text oriented connections, such as modems or telnet sessions.

keying - small tabs, prongs, or fillers are used to stop connectors from mating when they are improperly oriented.

kinematics/kinetics - is the measure of motion and forces of an object. This analysis is used to measure the performance of objects under load and/or in motion.

6.12 L

label - a name associated with some point in a program to be used by branch instructions.

ladder diagram - a form of circuit diagram normally used for electrical control systems.

ladder logic - a programming language for PLCs that has been developed to look like relay diagrams from the preceding technology of relay based controls.

laminar flow - all of the particles of a fluid or gas are travelling in parallel. The complement to this is turbulent flow.

laptop - a small computer that can be used on your lap. It contains a monitor and keyboard.

LAN (Local Area Network) - a network that is typically less than 1km in distance. Transmission rates tend to be high, and costs tend to be low.

latch - an element that can have a certain input or output lock in. In PLCs these can hold an output on after an initial pulse, such as a stop button.

LCD (Liquid Crystal Display) - a fluid between two sheets of light can be polarized to block light. These are commonly used in low power displays, but they require backlighting.

leakage current - a small amount of current that will be present when a device is off.

LED (Light Emitting Diode) - a semiconductor light that is based on a diode.

LIFO (Last In First Out) - similar to FIFO, but the last item pushed onto the stack is the first pulled off.

limit switch - a mechanical switch actuated by motion in a process.

line printer - an old printer style that prints single lines of text. Most people will be familiar with dot matrix style of line printers.

linear - describes a mathematical characteristic of a system where the differential equations are simple linear equations with coefficients.

little-endian - transmission or storage of data when the least significant byte/bit comes first.

load - In electrical system a load is an output that draws current and consumes power. In mechanical systems it is a mass, or a device that consumes power, such as a turbine.

load cell - a device for measuring large forces.

logic - 1. the ability to make decisions based on given values. 2. digital circuitry.

loop - part of a program that is executed repeatedly, or a cable that connects back to itself.

low - a logic negative, or zero.

LRC (Longitudinal Redundancy Check) -

LRC (Linear Redundancy Check) - a block check character

LSB (Least Significant Bit) -

LSD (Least Significant Digit) -

LSI (Large Scale Integration) - an integrated circuit that contains thousands of elements.

LVDT (Linear Variable Differential Transformer) - a device that can detect linear displacement of a central sliding core in the transformer.

6.13 M

machine language - CPU instructions in numerical form.

macro - a set of commands grouped for convenience.

magnetic field - a field near flowing electrons that will induce other electrons nearby to flow in the opposite direction.

MAN (Metropolitan Area Network) - a network designed for municipal scale connections.

manifold - 1. a connectors that splits the flow of fluid or gas. These are used commonly in hydraulic and pneumatic systems. 2. a description for a geometry that does not have any infinitely small points or lines of contact or separation. Most solid modelers deal only with manifold geometry.

MAP (Manufacturers Automation Protocol) -

mask - one binary word (or byte, etc) is used to block out, or add in digits to another binary number.

mass flow rate - instead of measuring flow in terms of volume per unit of time we use mass per unit time.

mass spectrometer - an instrument that identifies materials and relative proportions at the atomic level. This is done by observing their deflection as passed through a magnetic field.

master/slave - a control scheme where one computer will control one or more slaves. This scheme is used in interfaces such as GPIB, but is increasingly being replaced with peer-to-peer and client/server networks.

mathematical models - of an object or system predict the performance variable values based upon certain input conditions. Mathematical models are used during analysis and optimization pro-

cedures.

matrix - an array of numbers

MB MByte, KB, KByte - a unit of memory commonly used for computers. 1 KiloByte = 1 KByte = 1 KB = 1024 bytes. 1 MegaByte = 1 MByte = 1MB = 1024*1024 bytes.

MCR (Master Control Reset) - a relay that will shut down all power to a system.

memory - binary numbers are often stored in memory for fast recall by computers. Inexpensive memory can be purchased in a wide variety of configurations, and is often directly connected to the CPU.

memory - memory stores binary (0,1) patterns that a computer can read or write as program or data. Various types of memories can only be read, some memories lose their contents when power is off.

RAM (Random Access Memory) - can be written to and read from quickly. It requires power to preserve the contents, and is often coupled with a battery or capacitor when long term storage is required. Storage available is over 1MByte

ROM (Read Only Memory) - Programs and data are permanently written on this low cost ship. Storage available is over 1 MByte.

- EPROM (ELECTRICALLY Programmable Read Only Memory) - A program can be written to this memory using a special programmer, and erased with ultraviolet light. Storage available over 1MByte. After a program is written, it does not require power for storage. These chips have small windows for ultraviolet light.

- EEPROM/E2PROM (Electronically Erasable Programmable Read Only Memory) - These chips can be erased and programmed while in use with a computer, and store memory that is not sensitive to power. These can be slower, more expensive and with lower capacity (measured in Kbytes) than other memories. But, their permanent storage allows system configurations/data to be stored indefinitely after a computer is turned off.

memory map - a listing of the addresses of different locations in a computer memory. Very useful when programming.

menu - a multiple choice method of selecting program options.

message - a short sequence of data passed between processes.

microbar - a pressure unit (1 dyne per sq. cm)

microphone - an audio transducer (sensor) used for sound measurements.

microprocessor - the central control chip in a computer. This chip will execute program instructions to direct the computer.

MILNET (MILitary NETwork) - began as part of ARPANET.

MMI (Man Machine Interface) - a user interface terminal.

mnemonic - a few characters that describe an operation. These allow a user to write programs in an intuitive manner, and have them easily converted to CPU instructions.

MODEM (MODulator/DEModulator) - a device for bidirectional serial communications over phone lines, etc.

module - a part of a larger system that can be interchanged with others.

monitor - an operation mode where the computer can be watched in detail from step to step. This can also refer to a computer screen.

MOS (Metal Oxide Semiconductor) -

motion detect flow meter - a fluid flow induces measurement.

MRP (Material Requirements Planning) - a method for matching material required by jobs, to the equipment available in the factory.

MSD (Most Significant Digit) - the largest valued digit in a number (eg. 6 is the MSD in 63422). This is often used for binary numbers.

MTBF (Mean Time Between Failure) - the average time (hours usually) between the last repair of a product, and the next expected failure.

MTTR (Mean Time To Repair) -

multicast - a broadcast to some, but not necessarily all, hosts on a network.

multiplexing - a way to efficiently use transmission media by having many signals run through one conductor, or one signal split to run through multiple conductors and rejoined at the receiving end.

multiprocessor - a computer or system that uses more than one computer. Normally this term means a single computer with more than one CPU. This scheme can be used to increase processing speed, or increase reliability.

multivibrator - a digital oscillator producing square or rectangular waveforms.

6.14 N

NAK (Negative Acknowledgement) - an ASCII control code.

NAMUR - A European standards organization.

NAND (Not AND) - a Boolean AND operation with the result inverted.

narrowband - uses a small data transmission rate to reduce spectral requirements.

NBS (National Bureau of Standards) -

NC - see normally opened/closed

NC (Numerical Control) - a method for controlling machine tools, such as mills, using simple programs.

negative logic - a 0 is a high voltage, and 1 is a low voltage. In Boolean terms it is a duality.

NEMA (National Electrical Manufacturers Association) - this group publishes numerous standards for electrical equipment.

nephelometry - a technique for determining the amount of solids suspended in water using light.

nesting - a term that describes loops (such as FOR-NEXT loops) within loops in programs.

network - a connection of typically more than two computers so that data, email, messages, resources and files may be shared. The term network implies, software, hardware, wires, etc.

NFS (Network File System) - a protocol developed by Sun Microsystems to allow dissimilar computers to share files. The effect is that the various mounted remote disk drives act as a single local disk.

NIC (Network Interface Card) - a computer card that allows a computer to communicate on a network, such as ethernet.

NIH (Not Invented Here) - a short-lived and expensive corporate philosophy in which employees believe that if idea or technology was not developed in-house, it is somehow inferior.

NIST (National Institute of Standards and Technology) - formerly NBS.

NO - see normally opened

node - one computer connected to a network.

noise - 1. electrical noise is generated mainly by magnetic fields (also electric fields) that induce currents and voltages in other conductors, thereby decreasing the signals present. 2. a sound of high intensity that can be perceived by the human ear.

non-fatal error - a minor error that might indicate a problem, but it does not seriously interfere with the program execution.

nonpositive displacement pump - a pump that does not displace a fixed volume of fluid or gas.

nonretentive - when power is lost values will be set back to 0.

NOR (Not OR) - a Boolean function OR that has the results negated.

normally opened/closed - refers to switch types. when in their normal states (not actuated) the normally open (NO) switch will not conduct current. When not actuated the normally closed (NC) switch will conduct current.

NOT - a Boolean function that inverts values. A 1 will become a 0, and a 0 will become a 1.

NOVRAM (NOn Volatile Random Access Memory) - memory that does not lose its contents when turned off.

NPN - a bipolar junction transistor type. When referring to switching, these can be used to sink current to ground.

NPSM - American national standard straight pipe thread for mechanical parts.

NPT - American national standard taper pipe thread.

NSF (National Science Foundation) - a large funder of science projects in USA.

NSFNET (National Science Foundation NETwork) - funded a large network(s) in USA, including a high speed backbone, and connection to a number of super computers.

NTSC (National Television Standards Committee) - a Red-Green-Blue based transmission standard for video, and audio signals. Very popular in North America, Competes with other standards internationally, such as PAL.

null modem - a cable that connects two RS-232C devices.

6.15 O

OCR (Optical Character Recognition) - Images of text are scanned in, and the computer will try to interpret it, much as a human who is reading a page would. These systems are not perfect, and often rely on spell checkers, and other tricks to achieve reliabilities up to 99%

octal - a base 8 numbering system that uses the digits 0 to 7.

Octave - a doubling of frequency

odd parity - a bit is set during communication to indicate when the data should have an odd number of bits.

OEM (Original Equipment Manufacturer) -

off-line - two devices are connected, but not communicating.

offset - a value is shifted away or towards some target value.

one-shot - a switch that will turn on for one cycle.

on-line - two devices are put into communications, and will stay in constant contact to pass information as required.

opcode (operation code) - a single computer instruction. Typically followed by one or more operands.

open collector - this refers to using transistors for current sourcing or sinking.

open loop - a system that does monitor the result. open loop control systems are common when the process is well behaved.

open-system - a computer architecture designed to encourage interconnection between various vendors hardware and software.

operand - an operation has an argument (operand) with the mnemonic command.

operating system - software that existing on a computer to allow a user to load/execute/develop their own programs, to interact with peripherals, etc. Good examples of this is UNIX, MS-DOS, OS/2.

optimization - occurs after synthesis and after a satisfactory design is created. The design is optimized by iteratively proposing a design and using calculated design criteria to propose a better design.

optoisolators - devices that use a light emitter to control a photoswitch. The effect is that inputs and outputs are electrically separate, but connected. These are of particular interest when an interface between very noisy environments are required.

OR - the Boolean OR function.

orifice - a small hole. Typically this is places in a fluid/gas flow to create a pressure difference and slow the flow. It will increase the flow resistance in the system.

oscillator - a device that produces a sinusoidal output.

oscilloscope - a device that can read and display voltages as a function for time.

OSF (Open Software Foundation) - a consortium of large corporations (IBM, DEC, HP) that are promoting DCE. They have put forth a number of popular standards, such as the Motif Widget set for X-Windows programming.

OSHA (Occupational safety and Health Act) - these direct what is safe in industrial and commercial operations.

OSI (Open System Interconnect) - an international standards program to promote computer connectivity, regardless of computer type, or manufacturer.

overshoot - the inertia of a controlled system will cause it to pass a target value and then return.

overflow - the result of a mathematical operation passes by the numerical limitations of the hardware logic, or algorithm.

6.16 P

parallel communication - bits are passed in parallel conductors, thus increasing the transmission rates dramatically.

parallel design process - evaluates all aspects of the design simultaneously in each iteration. The design itself is sent to all analysis modules including manufacturability, inspectibility, and engineering analysis modules; redesign decisions are based on all results at once.

parallel programs -

parity - a parity bit is often added to bytes for error detection purposes. The two typical parity methods are even and odd. Even parity bits are set when an even number of bits are present in the transmitted data (often 1 byte = 8 bits).

particle velocity - the instantaneous velocity of a single molecule.

Pascal - a basic unit of pressure

Pascal's law - any force applied to a fluid will be transmitted through the fluid and act on all enclosing surfaces.

PC (Programmable Controller) - also called PLC.

PCB (Printed Circuit Board) - alternate layers of insulating materials, with wire layout patterns are built up (sometimes with several layers). Holes through the layers are used to connect the conductors to each other, and components inserted into the boards and soldered in place.

PDES (Product Data Exchange using Step) - a new product design method that has attempted to include all needed information for all stages of a product's life, including full solids modeling, tolerances, etc.

peak level - the maximum pressure level for a cyclic variation

peak-to-peak - the distance between the top and bottom of a sinusoidal variation.

peer-to-peer - a communications form where connected devices can both read and write messages at any time. This is opposed to a master slave arrangement.

performance variables - are parameters which define the operation of the part. Performance variables are used by the designer to measure whether the part will perform satisfactorily.

period - the time for a repeating pattern to go from beginning to end.

peripheral - devices added to computers for additional I/O.

permanent magnet - a magnet that retains a magnetic field when the original magnetizing force is removed.

petri-net - an enhanced state space diagram that allows concurrent execution flows.

pH - a scale for determining if a solution is an acid or a base. 0-7 is acid, 7-14 is a base.

photocell - a device that will convert photons to electrical energy.

photoconductive cell - a device that has a resistance that will change as the number of incident photons changes.

photoelectric cell - a device that will convert photons to electrical energy.

photon - a single unit of light. Light is electromagnetic energy emitted as an electron orbit decays.

physical layer - an OSI network model layer.

PID (Proportional Integral Derivative) - a linear feedback control scheme that has gained popularity because of its relative simplicity.

piezoelectric - a material (crystals/ceramics) that will generate a charge when a force is applied. A common transducer material.

ping - an internet utility that makes a simple connection to a remote machine to see if it is reachable, and if it is operating.

pink noise - noise that has the same amount of energy for each octave.

piston - it will move inside a cylinder to convert a pressure to a mechanical motion or vice versa.

pitch - a perceptual term for describing frequency. Low pitch means low frequency, high pitch means a higher frequency.

pitot tube - a tube that is placed in a flow stream to measure flow pressure.

pixels - are picture elements in a digitally generated and displayed picture. A pixel is the smallest addressable dot on the display device.

PLA (Programmable Logic Array) - an integrated circuit that can be programmed to perform different logic functions.

plane sound wave - the sound wave lies on a plane, not on a sphere.

PLC (Programmable Logic Controller) - A rugged computer designs for control on the factory floor.

pneumatics - a technique for control and actuation that uses air or gases.

PNP - a bipolar junction transistor type. When referring to switching, these can be used to source current from a voltage source.

poise - a unit of dynamic viscosity.

polling - various inputs are checked in sequence for waiting inputs.

port - 1. an undedicated connector that peripherals may be connected to. 2. a definable connection number for a machine, or a predefined value.

positive displacement pump - a pump that displaces a fixed volume of fluid.

positive logic - the normal method for logic implementation where 1 is a high voltage, and 0 is a low voltage.

potentiometer - displacement or rotation is measured by a change in resistance.

potting - a process where an area is filled with a material to seal it. An example is a sensor that is filled with epoxy to protect it from humidity.

power level - the power of a sound, relative to a reference level

power rating - this is generally the maximum power that a device can supply, or that it will require. Never exceed these values, as they may result in damaged equipment, fires, etc.

power supply - a device that converts power to a usable form. A typical type uses 115Vac and outputs a DC voltage to be used by circuitry.

PPP (Point-to-Point Protocol) - allows router to router or host to network connections over other synchronous and asynchronous connections. For example a modem connection can be used to connect to the internet using PPP.

presentation layer - an OSI network model layer.

pressure - a force that is distributed over some area. This can be applied to solids and gases.

pressure based flow meter - uses difference in fluid pressures to measure speeds.

pressure switch - activated above/below a preset pressure level.

prioritized control - control operations are chosen on the basis of priorities.

procedural language - a computer language where instructions happen one after the other in a clear sequence.

process - a purposeful set of steps for some purpose. In engineering a process is often a machine, but not necessarily.

processor - a loose term for the CPU.

program - a sequential set of computer instructions designed to perform some task.

programmable controller - another name for a PLC, it can also refer to a dedicated controller that uses a custom programming language.

PROM (Programmable Read Only Memory) -

protocol - conventions for communication to ensure compatibility between separated computers.

proximity sensor - a sensor that will detect the presence of a mass nearby without contact. These use a variety of physical techniques including capacitance and inductance.

pull-up resistor - this is used to normally pull a voltage on a line to a positive value. A switch/circuit can be used to pull it low. This is commonly needed in CMOS devices.

pulse - a brief change in a digital signal.

purge bubbling - a test to determine the pressure needed to force a gas into a liquid.

PVC - poly vinyl chloride - a tough plastic commonly used in electrical and other applications.

pyrometer - a device for measuring temperature

6.17 Q

QA (Quality Assurance) - a formal system that has been developed to improve the quality of a product.

QFD (Quality Functional Deployment) - a matrix based method that focuses the designers on the significant design problems.

quality - a measure of how well a product meets its specifications. Keep in mind that a product that exceeds its specifications may not be higher quality.

quality circles - a team from all levels of a company that meets to discuss quality improvement. Each members is expected to bring their own perspective to the meeting.

6.18 R

rack - a housing for holding electronics modules/cards.

rack fault - cards in racks often have error indicator lights that turn on when a fault has occurred. This allows fast replacement.

radar () - radio waves are transmitted and reflected. The time between emission and detection determines the distance to an object.

radiation - the transfer of energy or small particles (e.g., neutrons) directly through space.

radiation pyrometry - a technique for measuring temperature by detecting radiated heat.

radix - the base value of a numbering system. For example the radix of binary is 2.

RAID (Redundant Array of Inexpensive Disks) - a method for robust disk storage that would allow removal of any disk drive without the interruption of service, or loss of data.

RAM (Random Access Memory) -

random noise - there are no periodic waveforms, frequency and magnitude vary randomly.

random-scan devices - draw an image by refreshing one line or vector at a time; hence they are also called vector-scan or calligraphic devices. The image is subjected to flicker if there are more lines in the scene that can be refreshed at the refresh rate.

Rankine - A temperature system that uses absolute 0 as the base, and the scale is the same as the Fahrenheit scale.

raster devices - process pictures in parallel line scans. The picture is created by determining parts of the scene on each scan line and painting the picture in scan-line order, usually from top to bottom. Raster devices are not subject to flicker because they always scan the complete display on each refresh, independent of the number of lines in the scene.

rated - this will be used with other terms to indicate suggested target/maximum/minimum values for successful and safe operation.

RBOC (Regional Bell Operating Company) -

Read/Write (R/W) - a digital device that can store and retrieve data, such as RAM.

reagent - an chemical used in one or more chemical reactions. these are often used for identifying other chemicals.

real-time - suggests a system must be able to respond to events that are occurring outside the computer in a reasonable amount of time.

reciprocating - an oscillating linear motion.

redundancy - 1. added data for checking accuracy. 2. extra system components or mechanisms added to decrease the chance of total system failure.

refreshing - is required of a computer screen to maintain the screen image. Phosphors, which glow to show the image, decay at a fast rate, requiring the screen to be redrawn or refreshed several times a second to prevent the image from fading.

regenerative braking - the motor windings are reverse, and in effect return power to the power source. This is highly efficient when done properly.

register - a high speed storage area that can typically store a binary word for fast calculation. Registers are often part of the CPU.

regulator - a device to maintain power output conditions (such as voltage) regardless of the load.

relay - an electrical switch that comes in many different forms. The switch is activated by a magnetic coil that causes the switch to open or close.

relay - a magnetic coil driven switch. The input goes to a coil. When power is applied, the coil

generates a magnetic field, and pulls a metal contact, overcoming a spring, and making contact with a terminal. The contact and terminal are separately wired to provide an output that is isolated from the input.

reliability - the probability of failure of a device.

relief valve - designed to open when a pressure is exceeded. In a hydraulic system this will dump fluid back in the reservoir and keep the system pressure constant.

repeatability - the ability of a system to return to the same value time after time. This can be measured with a standard deviation.

repeater - added into networks to boost signals, or reduce noise problems. In effect one can be added to the end of one wire, and by repeating the signals into another network, the second network wire has a full strength signal.

reset - a signal to computers that restarts the processor.

resistance - this is a measurable resistance to energy or mass transfer.

resistance heating - heat is generated by passing a current through a resistive material.

resolution - the smallest division or feature size in a system.

resonant frequency - the frequency at which the material will have the greatest response to an applied vibration or signal. This will often be the most likely frequency of self destruction.

response time - the time required for a system to respond to a directed change.

return - at the end of a subroutine, or interrupt, the program execution will return to where it branched.

reverberation - when a sound wave hits a surface, part is reflected, and part is absorbed. The reflected part will add to the general (reverberant) sound levels in the room.

Reynolds number - a dimensionless flow value based on fluid density and viscosity, flow rate and pipe diameter.

RF (Radio Frequency) -

RFI (Radio Frequency Interference) -

RFS (Remote File System) - allows shared file systems (similar to NFS), and has been developed for System V UNIX.

RGB (Red Green Blue) - three additive colors that can be used to simulate the other colors of the

spectrum. This is the most popular scheme for specifying colors on computers. The alternate is to use Cyan-Magenta-Yellow for the subtractive color scheme.

ripple voltage - when an AC voltage is converted to DC it is passed through diodes that rectify it, and then through capacitors that smooth it out. A small ripple still remains.

RISC (Reduced Instruction Set Computer) - the more standard computer chips were CISC (Complete Instruction Set Computers) but these had architecture problems that limited speed. To overcome this the total number of instructions were reduced, allowing RISC computers to execute faster, but at the cost of larger programs.

rlogin - allows a text based connection to a remote computer system in UNIX.

robustness - the ability of a system to deal with and recover from unexpected input conditions.

ROM (Read Only Memory) -

rotameter - for measuring flow rate with a plug inside a tapered tube.

router - as network packets travel through a network, a router will direct them towards their destinations using algorithms.

RPC (Remote Procedure Call) - a connection to a specific port on a remote computer will request that a specific program be run. Typical examples are ping, mail, etc.

RS-232C - a serial communication standard for low speed voltage based signals, this is very common on most computers. But, it has a low noise immunity that suggests other standards in harsh environments.

RS-422 - a current loop based serial communication protocol that tends to perform well in noisy environments.

RS-485 - uses two current loops for serial communications.

RTC (Real-Time Clock) -

RTD (Resistance Temperature Detector) - as temperature is changed the resistance of many materials will also change. We can measure the resistance to determine the temperature.

RTS (Request To Send) -

rung - one level of logic in a ladder logic program or ladder diagram.

R/W (Read/Write) -

6.19 S

safety margin - a factor of safety between calculated maximums and rated maximums.

SCADA (Supervisory Control And Data Acquisition) - computer remote monitoring and control of processes.

scan-time - the time required for a PLC to perform one pass of the ladder logic.

schematic - an abstract drawing showing components in a design as simple figures. The figures drawn are often the essential functional elements that must be considered in engineering calculations.

scintillation - when some materials are hit by high energy particles visible light or electromagnetic radiation is produced

SCR (Silicon Controlled Rectifier) - a semiconductor that can switch AC loads.

SDLC (Synchronous Data-Link Control) - IBM oriented data flow protocol with error checking.

self-diagnosis - a self check sequence performed by many operation critical devices.

sensitivity - the ability of a system to detect a change.

sensor - a device that is externally connected to survey electrical or mechanical phenomena, and convert them to electrical or digital values for control or monitoring of systems.

serial communication - elements are sent one after another. This method reduces cabling costs, but typically also reduces speed, etc.

serial design - is the traditional design method. The steps in the design are performed in serial sequence. For example, first the geometry is specified, then the analysis is performed, and finally the manufacturability is evaluated.

servo - a device that will take a desired operation input and amplify the power.

session layer - an OSI network model layer.

setpoint - a desired value for a controlled system.

shield - a grounded conducting barrier that stops the propagation of electromagnetic waves.

Siemens - a measure of electrical conductivity.

signal conditioning - to prepare an input signal for use in a device through filtering, amplification,

integration, differentiation, etc.

simplex - single direction communication at any one time.

simulation - a model of the product/process/etc is used to estimate the performance. This step comes before the more costly implementation steps that must follow.

single-discipline team - a team assembled for a single purpose.

single pole - a switch or relay that can only be opened or closed. See also single pole.

single throw - a switch that will only switch one line. This is the simplest configuration.

sinking - using a device that when active will allow current to flow through it to ground. This is complimented by sourcing.

SLIP (Serial Line internet Protocol) - a method to run the internet Protocol (IP) over serial lines, such as modem connections.

slip-ring - a connector that allows indefinite rotations, but maintains electrical contacts for passing power and electrical signals.

slurry - a liquid with suspended particles.

SMTP (Simple Mail Transfer Protocol) - the basic connection protocol for passing mail on the internet.

snubber - a circuit that suppresses a sudden spike in voltage or current so that it will not damage other devices.

software - a program, often stored on non-permanent media.

solenoid - an actuator that uses a magnetic coil, and a lump of ferrous material. When the coil is energized a linear motion will occur.

solid state - circuitry constructed entirely of semiconductors, and passive devices. (i.e., no gas as in tubes)

sonar - sound waves are emitted and travel through gas/liquid. they are reflected by solid objects, and then detects back at the source. The travel time determines the distance to the object.

sound - vibrations in the air travel as waves. As these waves strike the human ear, or other surfaces, the compression, and rarefaction of the air induces vibrations. In humans these vibrations induce perceived sound, in mechanical devices they manifest as distributed forces.

sound absorption - as sound energy travels through, or reflects off a surface it must induce motion

of the propagating medium. This induced motion will result in losses, largely heat, that will reduce the amplitude of the sound.

sound analyzer - measurements can be made by setting the instrument for a certain bandwidth, and centre frequency. The measurement then encompasses the values over that range.

sound level - a legally useful measure of sound, weighted for the human ear. Use dBA, dBB, dBC values.

sound level meter - an instrument for measuring sound exposure values.

source - an element in a system that supplies energy.

sourcing - an output that when active will allow current to flow from a voltage source out to a device. It is complimented by sinking.

specific gravity - the ratio between the density of a liquid/solid and water or a gas and air.

spectrometer - determines the index of refraction of materials.

spectrophotometer - measures the intensities of light at different points in the spectrum.

spectrum - any periodic (and random) signal can be described as a collection of frequencies using a spectrum. The spectrum uses signal power, or intensity, plotted against frequency.

spherical wave - a wave travels outward as if on the surface of an expanding sphere, starting from a point source.

SQL (Structured Query Language) - a standard language for interrogating relational databases.

standing wave - if a wave travels from a source, and is reflected back such that it arrives back at the source in phase, it can undergo superposition, and effectively amplify the sound from the source.

static head - the hydrostatic pressure at the bottom of a water tank.

steady state - describes a system response after a long period of time. In other words the transient effects have had time to dissipate.

STEP (Standard for the Exchange of Product model data) - a standard that will allow transfer of solid model data (as well as others) between dissimilar CAD systems.

step response - a typical test of system behavior that uses a sudden step input change with a measured response.

stoichiometry - the general field that deals with balancing chemical equations.

strain gauge - a wire mounted on a surface that will be stretched as the surface is strained. As the wire is stretched, the cross section is reduced, and the proportional change in resistance can be measured to estimate strain.

strut - a two force structural member.

subroutine - a reusable segment of a program that is called repeatedly.

substrate - the base piece of a semiconductor that the layers are added to.

switching - refers to devices that are purely on or off. Clearly this calls for discrete state devices.

synchronous - two or more events happen at predictable times.

synchronous motor - an AC motor. These motors tend to keep a near constant speed regardless of load.

syntax error - an error that is fundamentally wrong in a language.

synthesis - is the specification of values for the design variables. The engineer synthesizes a design and then evaluates its performance using analysis.

system - a complex collection of components that performs a set of functions.

6.20 T

T1 - a 1.54 Mbps network data link.

T3 - a 45 Mbps network data link. This can be done with parallel T1 lines and packet switching.

tap - a connection to a power line.

tare - the ratio between unloaded and loaded weights.

TCP (Transmission Control Protocol) - a transport layer protocol that ensures reliable data communication when using IP communications. The protocol is connection oriented, with full duplex streams.

tee - a tap into a larger line that does not add any special compensation, or conditioning. These connectors often have a T-shape.

telnet - a standard method for logging into remote computers and having access if connect by a

dumb terminal.

temperature - the heat stored in an object. The relationship between temperature and energy content is specific to a material and is called the specific heat.

temperature dependence - as temperature varies, so do physical properties of materials. This makes many devices sensitive to temperatures.

thermal conductivity - the ability of a material to transfer heat energy.

thermal gradient - the change in temperature as we move through a material.

thermal lag - a delay between the time heat energy is applied and the time it arrives at the load.

thermistor - a resistance based temperature measurement device.

thermocouple - a device using joined metals that will generate a junction potential at different temperatures, used for temperature measurement.

thermopiles - a series of thermocouples in series.

thermoreistors - a category including RTDs and thermistors.

throughput - the speed that actual data is transmitted/processed, etc.

through beam - a beam is projected over an opening. If the beam is broken the sensor is activated.

thumbwheel - a mechanical switch with multiple positions that allow digits to be entered directly.

TIFF (Tagged Image File Format) - an image format best suited to scanned pictures, such as Fax transmissions.

time-division multiplex - a circuit is switched between different devices for communication.

time-proportional control - the amount of power delivered to an AC device is varied by changing the number of cycles delivered in a fixed period of time.

timer - a device that can be set to have events happen at predetermined times.

titration - a procedure for determining the strength of a solution using a reagent for detection. A chemical is added at a slow rate until the reagent detects a change.

toggle switch - a switch with a large lever used for easy reviews of switch settings, and easy grasping.

token - an indicator of control. Often when a process receives a token it can operate, when it is

done it gives it up.

TOP (Technical Office Protocol) -

top-down design - a design is done by first laying out the most abstract functions, and then filling in more of the details as they are required.

topology - 1. The layout of a network. 2. a mathematical topic describing the connection of geometric entities. This is used for B-Rep models.

torque - a moment or twisting action about an axis.

torus - a donut shape

toroidal core - a torus shaped magnetic core to increase magnetic conductivity.

TPDDI (Twisted Pair Distributed Data Interface) - counter rotating token ring network connected with twisted pair medium.

TQC (Total Quality Control) -

transceiver (transmitter receiver) - a device to electrically interface between the computer network card, and the physical network medium. Packet collision hardware is present in these devices.

transducer - a device that will convert energy from one form to another at proportional levels.

transformations - include translation, rotation, and scaling of objects mathematically using matrix algebra. Transformations are used to move objects around in a scene.

transformer - two separate coils wound about a common magnetic coil. Used for changing voltage, current and resistance levels.

transient - a system response that occurs because of a change. These effects dissipate quickly and we are left with a steady state response.

transmission path - a system component that is used for transmitting energy.

transport layer - an OSI network model layer.

TRIAC (TRIode Alternating Current) - a semiconductor switch suited to AC power.

true - a logic positive, high, or 1.

truth table - an exhaustive list of all possible logical input states, and the logical results.

TTL (Transistor Transistor Logic) - a high speed for of transistor logic.

TTY - a teletype terminal.

turbine - a device that generates a rotational motion using gas or fluid pressure on fan blades or vanes.

turbulent flow - fluids moving past an object, or changing direction will start to flow unevenly. This will occur when the Reynold's number exceeds 4000.

twisted pair - a scheme where wires are twisted to reduce the effects of EMI so that they may be used at higher frequencies. This is casually used to refer to 10b2 ethernet.

TXD (Transmitted Data) -

6.21 U

UART (Universal Asynchronous Receiver/Transmitter) -

UDP (User Datagram Protocol) - a connectionless method for transmitting packets to other hosts on the network. It is seen as a counterpart to TCP.

ultrasonic - sound or vibration at a frequency above that of the ear (> 16KHz typ.)

ultraviolet - light with a frequency above the visible spectrum.

UNIX - a very powerful operating system used on most high end and mid-range computers. The predecessor was Multics. This operating system was developed at AT & T, and grew up in the academic environment. As a result a wealth of public domain software has been developed, and the operating system is very well debugged.

UPS (Uninterruptable Power Supply) -

user friendly - a design scheme that simlifies interaction so that no knowledge is needed to operae a device and errors are easy to recover from. It is also a marketing term that is badly misused.

user interfaces - are the means of communicating with the computer. For CAD applications, a graphical interface is usually preferred. User friendliness is a measure of the ease of use of a program and implies a good user interface.

UUCP (Unix to Unix Copy Program) - a common communication method between UNIX systems.

6.22 V

Vac - a voltage that is AC.

vacuum - a pressure that is below another pressure.

vane - a blade that can be extended to provide a good mechanical contact and/or seal.

variable - a changeable location in memory.

varistor - voltage applied changes resistance.

valve - a system component for opening and closing mass/energy flow paths. An example is a water faucet or transistor.

vapor - a gas.

variable - it is typically a value that will change or can be changed. see also constant.

VDT (Video Display Terminal) - also known as a dumb terminal

velocity - a rate of change or speed.

Venturi - an effect that uses an orifice in a flow to generate a differential pressure. These devices can generate small vacuums.

viscosity - when moved a fluid will have some resistance proportional to internal friction. This determines how fast a liquid will flow.

viscosity index - when heated fluid viscosity will decrease, this number is the relative rate of change with respect to temperature.

VLSI (Very Large Scale Integration) -

volt - a unit of electrical potential.

voltage rating - the range or a maximum/minimum limit that is required to prevent damage, and ensure normal operation. Some devices will work outside these ranges, but not all will, so the limits should be observed for good designs.

volume - the size of a region of space or quantity of fluid.

volatile memory - most memory will lose its contents when power is removed, making it volatile.

vortex - a swirling pattern in fluid flow.

vortex shedding - a solid object in a flow stream might cause vortices. These vortices will travel with the flow and appear to be shed.

VRV (Vertical Redundancy Check) -

6.23 W

watchdog timer - a timer that expects to receive a pulse every fraction of a second. If a pulse is not received, it assumes the system is not operating normally, and a shutdown procedure is activated.

watt - a unit of power that is commonly used for electrical systems, but applies to all.

wavelength - the physical distance occupied by one cycle of a wave in a propagating medium.

word - 1. a unit of 16 bits or two bytes. 2. a term used to describe a binary number in a computer (not limited to 16 bits).

work - the transfer of energy.

write - a digital value is stored in a memory location.

WYSIWYG (What You See Is What You Get) - newer software allows users to review things on the screen before printing. In WYSIWYG mode, the layout on the screen matches the paper version exactly.

6.24 X

X.25- a packet switching standard by the CCITT.

X.400 - a message handling system standard by the CCITT.

X.500 - a directory services standard by the CCITT.

X rays - very high frequency electromagnetic waves.

X Windows - a window driven interface system that works over networks. The system was developed at MIT, and is quickly becoming the standard windowed interface. Personal computer manufacturers are slowly evolving their windowed operating systems towards X-Windows like standards. This standard only specifies low level details, higher level standards have been developed: Motif, and Openlook.

XFER - transfer.

XMIT - transmit.

xmodem - a popular protocol for transmitting files over text based connections. compression and error checking are included.

6.25 Y

ymodem - a popular protocol for transmitting files over text based connections. compression and error checking are included.

6.26 Z